VERY EASY & CUTE SECTION
WITH MANY MINI ANSWERS

VSAQ
SECTION-B

Baby Bullet Vs Baby Boss

Minimum Reading Maximum Scoring

VSAQ CHAPTERS

28 MARKS

★★☆	1.	ELECTRIC CHARGES & FIELDS (2 Marks)	62
★★	2.	ELECTROSTATIC POTENTIAL (2 Marks)	63-64
★★☆	3.	CURRENT ELECTRICITY (2 Marks)	65-66
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★★★	5.	MAGNETISM AND MATTER (2 Marks)	68
★★	6.	ELECTROMAGNETIC INDUCTION (2 Marks)	69-70
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★★★	13.	NUCLEI (2 Marks)	78
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Super Scoring Pages

★★★SSP★★★

Pages 62, 63, 65, 67, 68, 69, 71, 73, 74, 75, 76, 77, 78, 79

Dear Dude, Please note that VSAQs in each chapter are jumbled, and Easy VSAQ's are brought to the beginning and some VSAQs are pushed back to **STAR Questions plus.** To gain **maximum score**, read all the Qs given in the STAR Questions plus.

1 ELECTRIC CHARGES & FIELDS

2 Marks

78. The electric lines of force do not intersect. Why ? **[BMP]**

A : If electric lines of force intersect, the electric field at the point of intersection will have two directions simultaneously, which is physically impossible. So, the electric lines of force do not intersect.

---

79. What is meant by the statement charge is quantised ?

A : **Quantisation of charge:** The electric charge(q) is always equal to an integral multiple of the electron charge (e). Thus $q = ne$. Here $n=0, \pm1, \pm2, \pm3,...$ and $e=1.6\times10^{-19}C$.

---

80. How many electrons constitute 1C of charge ?

A: Given that $q = 1C$, $e = 1.6 \times 10^{-19}$ C, $n = ?$

Formula: $q = ne \Rightarrow n = \frac{q}{e} = \frac{1}{1.6\times10^{-19}} = 0.625 \times 10^{19}$ electrons

---

81. What is the importance of Gauss law in electrostatics.

A : **Importance of Gauss's Law:**

1) Gauss's law is **true for any closed surface** (of any shape and any size).
2) It gives the relation between the **electric field and the charge.**
3) It is valid for **stationary charges** as well as for **rapidly moving charges.**
4) Gauss's law is based on Coulomb's law. So Gauss' law is valid as long as Coulomb's law is valid.

---

82. Write the expression for electric intensity due to an infinitely long straight uniformly charged wire, at a radial distance r from the wire. Explain the terms in it.

A : Electric intensity due to an infinite long charged wire is $E=\frac{\lambda}{2\pi\varepsilon_0 r}$

Here λ is uniform linear charge density of the wire, r is radial distance from the wire and ε_0 is permittivity of free space.

---

83. Write the expression for electric intensity due to an infinite uniformly charged plane sheet. Explain the terms in it.

A: Electric intensity due to an infinite plane sheet of charge is $E = \frac{\sigma}{2\varepsilon_0}$

Here σ is uniform surface charge density of the sheet and ε_0 is permittivity of free space.

[☞Remaining VSAQs are In Page 91]

2 ELECTROSTATIC POTENTIAL

2 Marks

84. Can there be electric potential at a point with zero electric intensity? Give an example.

A : Yes.

Ex: Inside a charged metal spherical shell, potential is non-zero but electric intensity is zero .

---

85. Can there be electric intensity at a point with zero electric potential ? Give an example.

A : Yes.

Ex: When two charges +q and –q are separated by a distance, electric intensity at the mid point is non-zero. But the potential at the midpoint is zero [$\because$ + V–V = 0.]

---

86. The potential difference between two points is 1 V.

What is the work done to move 1 C charge between them?

A : Given potential difference V =1 Volt and charge q=1C

$\therefore$ Work done W=Vq =1×1= 1 Joule

---

87. Why is the electric field always at right angles to the equipotential surface ? Explain.

A : The potential difference between any two points on an equipotential surface dV=0.

Potential difference $dV = -\vec{E}.d\vec{r} \Rightarrow 0 = -Edr\cos\theta \Rightarrow \cos\theta = 0 \Rightarrow \theta = 90°$

$\therefore$ Electric field $\vec{E}$ is always at right angles to the equipotential surface.

---

88. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel.

a) What is the ratio of charges on each capacitor?

b) What is the ratio of potential differences across each capacitor?

A : Given that C_1 =1μF, C_2 =2μF, C_3 =3μF

In the parallel combination of capacitors we have $Q \propto C$ and V is same,

a) $Q_1 : Q_2 : Q_3 = C_1 : C_2 : C_3 =$ **1 : 2 : 3**

b) $V_1 = V_2 = V_3 =$ **1 : 1 : 1** [$\because$ **V is same**]

89. Three capacitors of capacitances 1 μF, 2μF and 3 μF are connected in series. a) What is the ratio of charges? b)What is the ratio of potential differences?

A: Given that $C_1 = 1\mu F$, $C_2 = 2\mu F$, $C_3 = 3\mu F$

In the series combination of capacitors we have **Q is same and** $V \propto \frac{1}{C}$

a) $Q_1 : Q_2 : Q_3 = 1 : 1 : 1$ [∵ **Q is same**]

b.) $V_1 : V_2 : V_3 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C_3} = \frac{1}{1} : \frac{1}{2} : \frac{1}{3} = \frac{6}{1} : \frac{6}{2} : \frac{6}{3} = 6 : 3 : 2$

---

90. What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled?

A : Capacitance $C = \frac{\varepsilon_0 A}{d} \Rightarrow C \propto A$ Here $A_2 = 2A_1$

$\therefore \frac{C_2}{C_1} = \frac{A_2}{A_1} = \frac{2A_1}{A_1} = 2 \Rightarrow C_2 = 2C_1$ ∴Capacitance becomes **double.**

---

91. What happens to the capacitance of a parallel plate capacitor if the distance between plates is doubled?

A : Capacitance $C = \frac{\varepsilon_0 A}{d} \Rightarrow C \propto \frac{1}{d}$ Here $d_2 = 2d_1$

$\therefore \frac{C_2}{C_1} = \frac{d_1}{d_2} = \frac{d_1}{2d_1} = \frac{1}{2} \Rightarrow \frac{C_2}{C_1} = \frac{1}{2} \Rightarrow C_2 = \frac{C_1}{2}$ ∴Capacitance becomes **half.**

---

92. In a hydrogen atom, the electron and proton are at a distance of 1Å . Find the dipole moment of the system. **[BMP]**

A : Given length of dipole is $2a = 1Å = 1 \times 10^{-10}$ m, we know $q = 1.6 \times 10^{-19}$ C, p = ?

Formula: Dipole moment $p = q(2a) = 1.6 \times 10^{-19} (1 \times 10^{-10}) = \mathbf{1.6 \times 10^{-29}}$ **C-m**

---

93. What is electrostatic shielding? Give an example.

A : **Electrostatic shielding** is the process of isolating a specific region from external electric fields by enclosing it within a conducting material.

Ex: Metallic car body protects passengers from thunderstorm as the electric field inside is zero.

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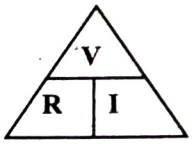
[☞Remaining VSAQs are In Page 92]

3 CURRENT ELECTRICITY

2 Marks

94. State Ohm's Law and write its mathematical form.

A : **1) Ohm's Law :** At constant temperature, the potential difference (V) between the ends of a conductor is directly proportional to the current (I) passing through it.



2) $V \propto I \Rightarrow V = RI$ (or) $I = \frac{V}{R}$ (or) $R = \frac{V}{I}$

Here R is the resistance of the conductor.

---

95. Write the vector form of Ohms law. Explain the terms in it. **[BMP]**

A : **Vector form of ohm's law:** Current density $\vec{J} = \sigma \vec{E}$.

Here σ is conductivity and $\vec{E}$ is the electric field intensity.

---

96. State Kirchhoff rules for electric network.

A: **1) Kirchoff's First Law :** The algebraic sum of currents meeting at any junction in an electric circuit is zero. Total current enters the junction is equal to total current leaving.

2) Kirchoff's Second Law : The algebraic sum of changes in potential around any closed loop involving resistors and cells is zero.

---

97. Define temperature coefficient of resistance.

A: The **temperature coefficient of resistance**(α) is the ratio of change in **resistance per unit** original resistance per degree centigrade change in temperature. $\left[\because \alpha = \frac{\Delta R}{R_0 \Delta T}\right]$

---

98. Define resistivity or specific resistance.

A: **1) Resistivity** (ρ) is the resistance of a conductor per unit length per unit area of cross section .

2) Formula: $\rho = \frac{RA}{l}$. Its SI unit is Ωm.

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99. Two wires of equal length, of copper and manganin, have the same resistance. Which wire is thicker? (Given $\rho_{manganin} > \rho_{copper}$)

A : We know that resistivity $\rho = \frac{RA}{l}$. As length l and resistance R are constant, $\rho \propto A$

Resistivity ρ is high for **manganin.**

Hence **manganin has more area and more thicker.**

---

100. If the voltage V applied across a conductor is increased to 2V, how will the drift velocity of the electrons change?

A: Drift velocity $v_d = \frac{I}{neA} = \frac{V}{neAR}$ $\left(\because I = \frac{V}{R}\right) \Rightarrow v_d \propto V$ ($\because$ n,e,A,R are constants)

$\therefore \frac{v_d^1}{v_d} = \frac{V^1}{V} = \frac{2V}{V} = 2$ $\qquad \therefore v_d^1 = 2v_d$.

Thus, the **drift velocity becomes double.**

---

101. Define mobility of charge carriers.

A: **1) Mobility(μ)of a charge carrier** is the magnitude of its drift velocity (v_d) per unit electric field strength (E).

2) Formula: $\mu = \frac{v_d}{E}$

---

102. Electrical power is transmitted from power stations to homes and factories at high voltages, why?

A : **1) Electric power** is transmitted at **high voltages** in primarily to reduce heat energy loss.

2) Power P= VI $\Rightarrow$ V and I are related inversely and also Heat loss = I^2R

$\therefore$ High voltages give low current and hence low heat energy loss.

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[☞ Remaining VSAQs are in Page 92]

4 MOVING CHARGES & MAGNETISM

2 Marks

103. What is the importance of Oersted's experiment?

A : Importance of Oersted's Experiment :

Oersted's experiment established the connection between electric current and magnetism. It showed that moving charge (current) produces magnetic field in its surrounding space.

---

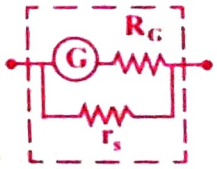
104. What is the principle of moving coil galvanometer?

A: Principle of MCG: The current carrying coil placed in a uniform magnetic field experiences a **torque.**

---

105. How do you convert a moving coil galvanometer into an ammeter?

A: A galvanometer (G) can be converted into an ammeter (A) by connecting a very **small shunt resistance** (r_s) in parallel with it.

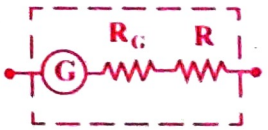


A

---

106. How do you convert a moving coil galvanometer into a voltmeter?

A : A galvanometer (G) can be converted into a voltmeter (V) by connecting a **high resistance (R)** in series with it.



V

---

107. Write the formula for the magnetic field due to infinite straight wire and explain the terms in it. **[BMP]**

A : Magnetic field due to infinite straight wire is $B = \frac{\mu_0 I}{2\pi r}$. Here μ_0 is the permeability of free space, I is electric current, r is the perpendicular distance from line.

---

108. Two parallel conducting wires, separated by a distance d carry I_1 and I_2 currents. When do they attract and repel? What is the magnitude of force per unit length acting on any wire?

A : 1) The two parallel current conducting wires attract when the currents flow in the same direction. They repel when the currents flow in the opposite direction.

2) The magnitude of the force per unit length is $f = \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$

Here μ_0 is permeability of free space and d is the distance between the wires.

[☞ Remaining VSAQs are in Page 93]

5 MAGNETISM & MATTER

2 Marks

109. Magnetic lines of force don't intersect. Why?

A: Any two magnetic field lines do not intersect each other because if they do, it means that at the point of intersection it shows two different directions which is impossible.

---

110. Magnetic lines form continuous closed loops. Why?

A: The magnetic field lines of a magnet or a solenoid form continuous closed loops because the magnetic poles N and S always exist together in pairs.

---

111. Define magnetisation of a material and write its formula.

A: **Magnetisation** (M) is defined as the **net magnetic dipole moment per unit volume** of the material. **Formula:** $M = \frac{m_{net}}{V}$

---

112. What is the magnetic moment associated with a solenoid?

A: **Magnetic moment** of a solenoid is $\bar{M} = NI\bar{A}$. Here N is the number of turns, I is the current passing through it and $\bar{A}$ is the area of cross section.

---

113. Define magnetic susceptibility and relative permeability. **[BMP]**

A: 1) **Magnetic Susceptibility** (χ) is the ratio between **magnetisation (M)** and **magnetic intensity (H) Formula:** $\chi = \frac{M}{H}$. It has no units. It determines how easily a material can be magnetized.

2) **Relative permeability** (μ_r) is the ratio of the magnetic permeability of a specific medium (μ) to the magnetic permeability of free space (μ_0) $\mu_r = \mu/\mu_0$

---

114. State Gauss law in magnetism and write it its mathematical expression.

A: 1) **Gauss law:** The net magnetic flux (ϕ) through any closed surface is equal to zero.

2) **Mathematical expression:** $\oint \bar{B}.d\bar{S} = 0$

Here $\bar{B}$ is the magnetic vector, and $d\bar{S}$ is surface vector area.

---

115. Classify the following materials with regard to magnetism.
Iron, Platinum, Copper, Magnesium.

A: 1) **Ferromagnetic material:** Iron

2) **Paramagnetic material:** Platinum, Magnesium

3) **Diamagnetic material:** Copper

6 ELECTRO MAGNETIC INDUCTION

2 Marks

116. State Faraday's law of electromagnetic induction. **[BMP]**

A : **1) Faraday's law:** The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2) Mathematical expression: Induced emf $\varepsilon = -\frac{d\phi_B}{dt}$

In the case of a coil of N turns, $\varepsilon = -N\frac{d\phi_B}{dt}$

---

117. State Lenz's law.

A : **1) Lenz's Law (Neumann's law) :** The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux.

2) Faraday's law of induction: $\varepsilon = -\frac{d\phi_B}{dt}$. The negative sign indicates Lenz's law.

---

118. Define magnetic flux.

A : **1) Magnetic Flux** is defined as the total number of magnetic field lines passing normally through the given plane.

2) Formula: $\phi_B = \vec{B}.\vec{A} = BA\cos\theta$. Here θ is the angle between $\vec{A}$ and $\vec{B}$.

Magnetic flux is a scalar. Its SI unit is Tm^2

---

119. What is the working principle of A.C. generator?

A: AC generator works on the principle of **electromagnetic induction.**

(The AC generator converts mechanical energy into electrical energy by rotating a coil with in a magnetic field, which changes the magnetic flux and induces an alternating emf.)

---

120. Write the expression for self-induced emf and explain the terms therein.

A: The expression for self induced emf is $\varepsilon = -L\frac{dI}{dt}$

Here L is the self inductance, $\frac{dI}{dt}$ is the rate of change of current.

121. What is self-induction?

A: 1) **Self induction** is the magnetic flux linked with the coil when unit current flows in it. (or) When the current in a coil changes, an emf is induced in the coil opposing the change in the current through the coil. This phenomenon is called self induction.

2) $\therefore \phi \propto I \Rightarrow \phi = LI$ (or) $\text{Emf} = -L\frac{dI}{dt}$

---

122. What is mutual induction?

A: **1) Mutual induction** is defined as the flux linked with secondary coil when unit current flows through the primary coil. (or) When the current in one coil changes, an emf is induced in another coil kept near it. This phenomenon is called mutual inductance.

2) $\phi \propto I \Rightarrow \phi = MI$ (or) $\text{Emf} = -M\frac{dI}{dt}$

---

123. Why do birds fly off a high tension wire when the current is switched 'ON'?

A: 1) When current is switched on, **induced current flows** through the body of the bird.

This induced current flows in the opposite direction through the wings of the bird.

2) Hence, the wings experience a force of mutual repulsion.

Thus, the wings spread and hence the birds fly due to electromagnetic induction.

---

124. Why is spark produced in the switch of a fan when it is switched off?

A: A fan motor contains coils of high inductance. When the fan is switched off, the current drops to zero rapidly. This sudden change induces a very high back emf (voltage) across the switch contacts. This high voltage is strong enough to ionize the air in the small gap of the switch, resulting in a **visible spark.**

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[☞ Remaining VSAQs are in Page 94]

7 ALTERNATING CURRENT

2 Marks

125. What is transformer ratio?

A : **1) Transformer Ratio** is the ratio of output emf to input emf of a transformer. (or)
It is the ratio between number of turns in the secondary coil and the number of turns in the primary coil of a transformer.

2) Formula: Transformer ratio $= \frac{N_s}{N_p} = \frac{V_s}{V_p}$

---

126. A transformer converts 200 V AC into 2000 V AC.
Calculate the number of turns in the secondary if the primary has 10 turns.

A : Given $V_p = 200$ V, $V_s = 2000$ V, $N_p = 10$, $N_s = ?$

Transformer ratio formula : $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow N_s = \frac{V_s}{V_p} N_p = \frac{2000}{200} \times 10 = 100$ turns

---

127. Write the expression for the reactance of (i) an inductor and (ii) a capacitor.

A : **(i) Reactance of an inductor** is $X_L = \omega L$
Here ω is angular frequency and L= Inductance

(ii) Reactance of a capacitor is $X_C = \frac{1}{\omega C}$
Here ω is angular frequency and C = Capacitance

---

128. A Capacitor has a capacitance of $\frac{1}{\pi}\mu F$. Find its reactance for a frequency of 50 Hz.

Sol: Given Frequency t = 50Hz , Capacitance $C = 1 \times 10^{-6}$ F **[BMP]**

Reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times \frac{1}{\pi} \times 10^{-6}} = \frac{1}{100 \times 10^{-6}} = \frac{1}{10^{-4}} = 10 k\Omega$

---

129. A pure inductor of 21mH is connected to a source of 220V.
Find the inductive reactance if the frequency of the source is 50Hz.

A: Given Inductance L = 21mH = 21×10^{-3} H, Voltage V = 220V, Frequency f = 50Hz

$\therefore$ Inductive reactance $X_L = 2\pi fL = 2 \times 3.14 \times 50 \times 21 \times 10^{-3} = 6.597\Omega = 6.6\Omega$

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130. What is the phase difference between A.C emf and current in the following: pure resistor, pure inductor and pure capacitor.

A : 1) In a **pure resistor**, the phase difference between AC voltage and AC current is **zero.**

2) In a **pure Inductor,** the **current lags** behind the voltage by $\pi/2$.

3) In a **pure capacitor**, the **current leads** ahead the voltage by $\pi/2$.

---

131. Define power factor. On which factors does the power factor depend?

A : 1) **Power Factor** is the cosine of the phase angle (ϕ) between the voltage and the current.

Formula 1: $P = V I \cos\phi$. Here $\cos\phi$ is called power factor.

Formula 2: Power factor $\cos\phi = \frac{R\,(\text{Resistance})}{Z\,(\text{Impedance})}$

2) Power factor depends on the phase difference between voltage and current or equivalently on the ratio R/Z.

---

132. What is meant by wattless component of current?

A : **Wattless current :** In a purely inductive or capacitive circuit, the power factor $\cos\phi=0$. Then no power is dissipated even though current is flowing in the circuit.

In such cases, the current is said to be **wattless current.**

---

133. Why capacitor blocks Direct Current (D.C.)?

A: D.C. has constant magnitude and direction. Once a capacitor is fully charged, it behaves like an open circuit for D.C. When current passes through the capacitor it blocks direct current D.C.

---

134. On what principle metal detector works? Explain it.

A: 1) Metal detector works on the principle of **electromagnetic induction.**

2) The primary search coil in metal detector generates an alternating magnetic field. When a metal object is nearby, due to the influence of the magnetic field, current will be induced in the metal. The induced current creates a secondary magnetic field which is detected by the metal detector's receiver coil, and it generates a sound as an alarm.

---

135. How the phenomenon of resonance in the series LCR circuit be used in tuning mechanism of a radio and TV set?

A: In the series LCR circuit, energy alternates between the inductor and capacitor, enabling the radio or TV to pick up a precise frequency. **Resonance in the circuit** enables radio or T.V. tuning by selecting a specific station's frequency while rejecting others.

8 ELECTRO MAGNETIC WAVES

2 Marks

136. Give two uses of infrared rays. **[BMP][AP 18,19,22,23]**

A : 1) Infrared rays are used in **remote control systems of TV**

2) Infrared rays are used for the **treatment of skin diseases.**

3) Infrared rays are used in **earth satellites.**

4) Infrared rays are used for taking photographs where there is no visible light due to fog or mist.

---

137. What are the applications of microwaves? **[AP 15,16,17,18,19,25]**

A: Applications of microwaves:

1) Microwaves are used in **electrical ovens** for cooking and heating.

2) Microwaves are used in **wi-fi network**, satellite communication.

3) Microwaves are used for deep **muscle pain relief.**

4) Microwaves are used in **radar systems** due to their short wavelengths .

---

138. What is the principle of production of electromagnetic waves? **[AP 19, 24]**

A: 1) Electromagnetic waves are produced by **accelerating or oscillating charged particles.**

2) When a charged particle like an electron vibrates or changes speed, it creates oscillating electric and magnetic fields that ripple outward together as self-propagative EM waves.

---

139. What is the displacement current, and what is its formula?

A: 1) Displacement current (I_d) is equal to ε_0 times the rate of change of electric flux through the surface.

2) Formula: $I_d = \varepsilon_0 \frac{d\phi}{dt}$ where $\frac{d\phi}{dt}$ is the rate of change of electric flux.

The displacement current occurs when there is change in electric flux.

---

140. The charging current for a capacitor is 0.6 A. **[AP 20]**

What is the displacement current across its plates?

A : The displacement current is 0.6 A.

Reason: The displacement current across the plates of a capacitor is always equal to charging current of the capacitor.

---

141. What is the wavelength range of X-rays?

A: The wavelength range of X-rays: **10^{-3} nm to 10 nm** (or) **0.01 Å to 100 Å**

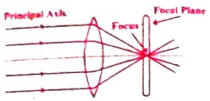
[☞ Remaining VSAQs are in Page 94]

9 RAY OPTICS

2 Marks

142. What do you understand by the focal plane? Explain with a neat ray diagram.

A: The light rays coming from an object at infinity, when pass through a lens converge (or appear to diverge) at a **point** to form a clear image. The plane passing through **that point and perpendicular** to the principal axis of the lens is called **focal plane**.



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143. Define power of a convex lens. What is its unit?

A: **1) Power of a convex Lens** is defined as the reciprocal of its focal length (f).

Power $P = \frac{1}{f}$ **SI unit :** dioptre (D)

2) Power of a lens is its ability to converge incoming light rays.

---

144. Two thin lens of power +4D and -2D are in contact. What is the focal length of the combination?

A: Given $P_1 = +4D$, $P_2 = -2D$; Equivalent power $P_{eq} = P_1 + P_2 = 4 + (-2) = +2D$

Focal length $f = \frac{1}{P_{eq}} = \frac{1}{2} = 0.5m = 50\ cm$

---

145. What is the critical angle? Give one application of total internal reflection? [BMP]

A: **1) Critical Angle (i_C):** When a light ray travels from denser medium to rarer medium, the angle of incidence for which the angle of refraction 90° is called critical angle.

2) Applications of TIR : Optical fibre, Mirage.

---

146. State the conditions under which total internal reflection takes place.

A: **Conditions for TIR:**

(i) Light must travel from a denser to rarer medium.

(ii) The angle of incidence at the interface must be greater than the critical angle. ($i > i_C$)

---

147. A small angled prism of 4° deviates a ray through 2.4°. Find the refractive index of the prism.

A: Given angle of prism A = 4°, angle of deviation $D_m = 2.4°$, refractive index $n_{21} = ?$

Formula: For a thin angled prism $D_m = (n_{21} - 1)\,A$

$$\Rightarrow 2.4 = (n_{21} - 1)\,4 \Rightarrow n_{21} - 1 = \frac{2.4}{4} = 0.6 \Rightarrow n_{21} = 1 + 0.6 = 1.6$$

[☞ Remaining VSAQs are in Page 95]

10 WAVE OPTICS

2 Marks

148. State Malus' law. Write the expression of Malus' law. **[BMP]**

A : **1) Malus' law:** The intensity of plane polarized light transmitted through an analyzer varies as the square of cosine of angle between transmission axes.

2) Expression: $I=I_0\cos^2\theta$

Here I_0= initial intensity, I = transmitted intensity, θ= angle between axes

---

149. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves?

A: **1) Polarization of light** is the phenomenon of restricting vibrations of light to one direction perpendicular to propagation.

2) The electric field vector $\vec{E}$ is used to represents the polarization of an electromagnetic wave.

---

150. Two identical coherent waves, each o f intensity I_0, are producing an interference pattern. Write the value of resultant intensity at a point of

(1) constructive interference (2) destructive interference.

A: 1) For constructive interference, (when $I_1 = I_2 = I_0$), $I_{max} = \left(\sqrt{I_0} + \sqrt{I_0}\right)^2 = 4I_0$

2) For destructive interferences, $I_{max} = \left(\sqrt{I_0} - \sqrt{I_0}\right)^2 = 0$

---

151. If a wave undergoes reflection and refraction, what will be the phase change in each case?

A: 1) In **reflection** if light reflects from a denser medium there is a phase change of **π radians.**

2) In **refraction**, there is a **no phase change.**

---

152. Compare Maxwell's Electromagnetic theory with Huygen's theory of light?

A:

Maxwell Electromagnetic Theory	Huygen's theory of light
1) Light is an **electromagnetic wave.**	1) Light is an transverse **mechanical wave.**
2) Explains electric and magnetic nature	2) Explains reflection and refraction
3) Explains polarization	3) Does not explain polarization properly
4) Does not require a medium to propagate.	4) Requires a hypothetical ether medium.

[☞ Remaining VSAQs are in Page 96]

11 DUAL NATURE OF RADIATION & MATTER

(2 Marks)

153. What are cathode rays? [AP 17,18,23]

A: 1) **Cathode Rays are streams of electrons.**
2) These rays are emitted from cathode of a discharge tube, when the gas in it is subjected to low pressure (0.001 mm of Hg) and strong electric field.

---

154. What important fact did Millikan's oil drop experiment establish?

A : 1) Millikan's experiment established the fact that electric charge is quantised.
2) Millikan found the value of charge of electron is $e = 1.602 \times 10^{-19}$ C.

---

155. What is photoelectric effect? [AP 17,18,20]

A : **Photoelectric Effect** is the emission of electrons from a metal surface when it is illuminated by electro magnetic radiation of sufficient frequency. The emitted electrons are called **photoelectrons.**

---

156. Define work function. Give an expression for it.

A: 1) **Work Function:** The minimum energy required by an electron to escape from the surface of a metal is called work function (ϕ_0) of the metal.
2) **Expression:** $\phi_0 = h\nu_0$ Here h is planck's constant and ν_0 is threshold frequency.

---

157. Give examples of photosensitive substances. Why are they called so? [AP 16]

A : 1) Alkali metals like lithium, sodium, potassium, cesium and rubidium are photosensitive substances.
2) Because, these alkali metals emit electrons when they are illuminated by light.

---

158. Write down Einstein's photoelectric equation. Explain the terms in it? [BMP]

A : **Einstein's Photoelectric Equation :** $K_{max} = h\nu - \phi_0$
Here K_{max} is maximum kinetic energy of the photoelectron, h is Planck's constant, ν is the frequency of incident light and ϕ_0 is the work function of the metal.

---

159. Write down de-Broglie's relation and explain the terms therein. [AP 16,18,22]

A : **De Broglie's relation :** Wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$
Here h = Planck's constant and p = mv is the momentum of the particle

---

160. Define 1 electron volt.

A: An **electron volt** is a tiny unit of energy, defined as the kinetic energy gained by a single electron when it accelerates through a potential difference of 1 volt in vaccum. $1\ eV = 1.6 \times 10^{-19}$ J

---

161. Which physical quantities are conserved in photon-particle collisions in photoelectric effect?

A: In a photon- particle collision, the **total energy** and **total momentum** are conserved.

12 ATOMS

2 Marks

162. Define impact parameter. **[BMP]**

A: The **impact parameter (b)** is defined as the perpendicular distance of the initial velocity vector of the α– particle from the centre of the nucleus.

---

163. Define distance of closest approach.

A: **Distance of closest approach** is the minimum distance to which a fast moving α particle can approach the nucleus when it collides head-on before its KE is completely converted into electrostatic PE.

---

164. What is the difference between α - particle and helium atom?

A: 1) Helium atom is electrically neutral whereas alpha particle is positively charged.

2) Helium atom consists of 2 protons, 2 neutrons and 2 electrons whereas alpha particle is a double ionized helium ion (He^{2+}) consisting of 2 protons and 2 neutrons but with zero electrons.

---

165. Why did most of the α - particles pass through the gold foil undeflected in Rutherford's scattering?

A: Most of the α particles passed straight through the gold foil undeflected.

Because most of the space inside an atom is empty.

Thus nucleus occupies only a small fraction of total volume of atom.

---

166. What is the angular momentum of an electron in the second orbit of Bohr's model of hydrogen atom?

A: From Bohr's model of hydrogen atom, the angular momentum of electron in n^{th} orbit is $L = \frac{nh}{2\pi}$

For n = 2, $L = \frac{2h}{2\pi} = \frac{h}{\pi}$

---

167. Express the magnitude of electrostatic force of repulsion between the α - particle and the positively charged nucleus?

A: From Coulomb's law, Force $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{r^2}$

Here r is the distance between the alpha particle and the nucleus, (2e) is the charge of the alpha particle, and (Ze) is the charge of the nucleus.

[☞ Remaining VSAQs are in Page 97]

13 NUCLEI

2 Marks

168. What are isotopes and isobars? **[BMP]**

A: 1) **Isotopes** are the nuclei having same atomic number (Z), but different mass numbers (A)

Ex: ${}_1H^1, {}_1H^2, {}_1H^3$

2) **Isobars** are the nuclei having same mass number (A), but different atomic numbers (Z)

Ex: ${}_6C^{14}$, ${}_7N^{14}$

---

169. What are isotones and isomers?

A: 1) **Isotones** are the nuclei having the same neutron number (N), but different atomic numbers (Z).

Ex: ${}_{20}Ca^{40}$, ${}_{19}K^{39}$. Here number of neutrons in each = 20

2) **Isomers** are the nuclei having same atomic number (Z) and same mass number (A) but different nuclear properties such as radioactive decay and magnetic moments.

Ex: ${}^{80}_{35}Br$ (metastable), ${}^{80}_{35}Br$ (ground state)

---

170. Why are nuclear fusion reactions called thermo nuclear reactions?

A: 1) **Nuclear fusion** reactions take place at extremely high temperature (of the order 10^7K) in order to overcome strong electrostatic repulsions between positively charged nuclei.

2) Hence, the nuclear fusion reactions are called **thermo nuclear reactions.**

---

171. Compare the radii of the nuclei of mass numbers 27 and 64.

Sol: Given $A_1 = 27, A_2 = 64, \frac{R_1}{R_2} = ?$

Nuclear radius $R = R_0 A^{1/3} \Rightarrow R \propto A^{1/3} \Rightarrow \frac{R_1}{R_2} = \frac{\sqrt[3]{A_1}}{\sqrt[3]{A_2}} = \frac{\sqrt[3]{27}}{\sqrt[3]{64}} = \frac{3}{4}$

$\therefore R_1 : R_2 = 3 : 4$

---

172. What is the significance of a mass defect in a nucleus?

A: 1) The mass defect represents the **stability of the nucleus.**

2) A large **mass defect** generally means a larger binding energy, which makes the nucleus more stable. It proves that mass and energy are interchangeable according to $E = mc^2$.

[☞ Remaining VSAQs are in Page 98]

14 SEMI CONDUCTORS

2 Marks

173. What are intrinsic and extrinsic semiconductors ?

A: **1) Intrinsic Semiconductor:**

Pure semiconductor like silicon is called intrinsic semiconductor.

In an intrinsic semiconductor, number of electrons(n_e) = number of holes(n_h).

2) Extrinsic Semiconductor: A semiconductor doped with trivalent or pentavalent impurity is called extrinsic semiconductor. Extrinsic semiconductors are of two types.

(i) n-type semiconductor and (ii) p-type semiconductor.

---

174. What is an n-type semiconductor?

What are the majority and minority charge carriers in it?

A: **1) n-type Semiconductor:**

Silicon or Germanium doped with pentavalent impurity like Arsenic is called n-type semiconductor. The pentavalent atoms add extra electrons and make it n-type semiconductor.

2) In n-type semiconductor, **majority charge carriers** are **electrons** and **minority charge carriers** are **holes.**

---

175. What is a p-type semiconductor ?

What are the majority and minority charge carriers in it?

A : **1) p-type Semiconductor:**

Silicon or Germanium doped with trivalent impurity like Indium is called p-type semiconductor. The trivalent atoms add extra holes and make it p-type semiconductor.

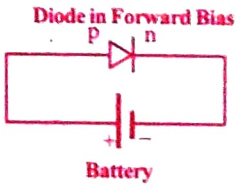
2) In p-type semiconductor, **majority** charge carriers are **holes** and **minority** charge carriers are **electrons.**

---

176. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? **[BMP]**

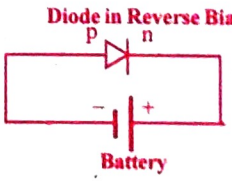
A: **1) Forward bias:**

In forward bias, a battery is connected to the diode such that its positive terminal is towards p-end of diode.



2) Reverse bias:

In reverse bias, a battery is connected to the diode such that its positive terminal is towards n-end of diode.



177. What do you mean by donor impurity and acceptor impurity?

A: 1) **Donor impurities** are atoms with five valence electrons-like Phosphorus (Pentavalent) added to semiconductors to donate free electrons forming n-type materials.

2) **Acceptor impurities** are atoms with three valence electrons- like Boron (trivalent) added to semiconductors to create vacancies (holes) forming p-type materials.

---

178. What is reverse saturation current of p-n junction diode?

A: **Reverse saturation current** is the small amount of current that flows through a p-n junction diode when it is reverse biased. It is caused by the flow of minority carriers.

---

179. Define depletion layer and what is the breakdown voltage of p-n junction diode.

A: 1) **Depletion layer:** It is the narrow region on either side of the p-n junction which is free from mobile charge carriers.

2) **Breakdown voltage:** It is the voltage in a reverse biased p-n junction at which the current suddenly rises to a large values.

---

180. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased?

A: i) In **forward bias**, the width of the depletion layer of a diode **decreases**

ii) In **reverse bias**, the width of the depletion layer of a diode **increases**.

---

181. What is rectification? Which device is used as rectifier?

A: 1) **Rectification** is the process of converting an alternating current in to direct current.

2) The device used for this rectification process is called a **rectifier.**

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[☞ Remaining VSAQs are in Page 98]

When Reading becomes Boring, BBQ makes it Caring!

AMAZING MIDDLE SECTION

SAQ

SECTION-C

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SAQ CHAPTERS

32 MARKS

★★★	1.	ELECTRIC CHARGES AND FIELDS (4 Marks)	32 - 34
★★★	2.	ELECTROSTATIC POTENTIAL (4+4 Marks)	35- 39
★★★	3.	CURRENT ELECTRICITY (4 Marks)	40 - 42
★★★	4.	MAGNETISM AND MATTER (4 Marks)	43 - 44
★★½	5.	ELECTROMAGNETIC INDUCTION (4 Marks)	45 - 46
★★	6.	ALTERNATING CURRENT (4 Marks)	47 - 48
★★½	7.	WAVE OPTICS (4 Marks)	49 - 50
★★★	8.	DUAL NATURE (4+4 Marks)	51- 54
★★★	9.	ATOMS (4 Marks)	55 - 57
★★½	10.	SEMICONDUCTOR ELECTRONICS (4 Marks)	58 - 60

Super Scoring Pages

★★★SSP★★★

Page 32, 33, 35, 36,37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55,59

DEAR DUDE!

Please note that SAQs in each chapter are Jumbled, and

Easy SAQs are brought to the beginning.

1 ELECTRIC CHARGES & FIELDS

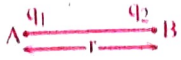
4 Marks

14. State and explain Coulomb's inverse square law in electricity. [AP 17,18,23]

A: 1) Coulomb's inverse square law:

The force of attraction or repulsion between two electric charges is directly proportional to product of their charges and is inversely proportional to the square of distance between them and acts along the line joining the charges.

2) Explanation: Consider two point charges q_1, q_2 separated by a distance r.



3) From Coulomb's law: i) $F \propto q_1q_2$ ii) $F \propto \frac{1}{r^2}$

$$\therefore F \propto \frac{q_1q_2}{r^2} \Rightarrow F = k\frac{q_1q_2}{r^2} \Rightarrow F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$$

Here ε_0 is the permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ Nm^2C^{-2}$

4) If the charges are in a medium with permittivity ε then $F_{med} = \frac{1}{4\pi\varepsilon}\frac{q_1q_2}{r^2}$

---

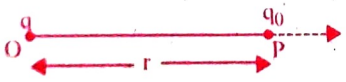
15. Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge. [AP 16]

A: 1) Intensity of Electric Field (E):

The intensity of electric field at a point in space is defined as the force (F) experienced by a unit positive test charge (q) placed at that point in the field.

2) Formula: $E = \frac{F}{q_0}$ **SI Unit:** N/C

3) Intensity due to a Point Charge :



Consider a point charge q kept at point O.

Let a test charge q_0 be kept at P, which is at a distance r from O.

4) The force between the two charges is $F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2} \Rightarrow \frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

$$\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\left(\because E = \frac{F}{q_0}\right)$$

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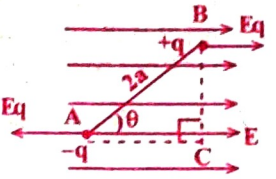
Derive an equation for the couple acting on an electric dipole in a uniform electric field. [AP 16,25]



1) Consider an electric dipole consisting of two equal and opposite charges +q & −q separated by a distance 2a.
2) Let the dipole be placed in a uniform electric field of intensity E at an angle θ to the direction of electric field.
3) The electrostatic forces acting on the charges q and −q are Eq and −Eq respectively. These two forces form a couple.
4) Couple τ = Force × Perpendicular distance between the forces

$\tau = Eq \times BC$.....(1)

5) From Δ ABC, $\sin\theta = \frac{BC}{AB} = \frac{BC}{2a} \Rightarrow BC = 2a\sin\theta$.....(2)

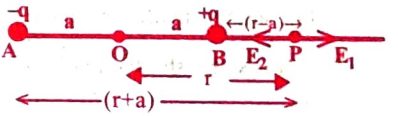
From (1) and (2), $\tau = Eq(2a\sin\theta) = (2aEq)\sin\theta = q(2a)E\sin\theta$

6) But we know dipole moment $p = q(2a)$

$\therefore \tau = pE\sin\theta$.....(3)

Derive an expression for the intensity of the electric field at a point on the axial line of a dipole. [BMP][AP 18,19,22]

1) Consider an electric dipole AB consisting of two equal and opposite charges −q & q separated by a distance 2a.



2) Let P be a point on the axial line of the dipole at a distance r from its centre O.
Then distance PB = (r − a) and PA = (r+a).
3) Intensity of electric field at P due to +q is $E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2}$.....(1)
4) Intensity of electric field at P due to −q is $E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+a)^2}$.....(2)
5) The resultant intensity of electric field due to the dipole at point P is $E = E_1 - E_2$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{(r-a)^2} - \frac{q}{(r+a)^2}\right) = \frac{q}{4\pi\varepsilon_0}\left[\frac{(r+a)^2-(r-a)^2}{(r^2-a^2)^2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{4ra}{(r^2-a^2)^2}\right] = \frac{q}{4\pi\varepsilon_0}\left[\frac{2\times 2ar}{(r^2-a^2)^2}\right] = \frac{1}{4\pi\varepsilon_0}\left[\frac{2(q2a)r}{(r^2-a^2)^2}\right] \left[\because (a+b)^2-(a-b)^2 = 4ab\right]$$

$$\therefore E = \frac{1}{4\pi\varepsilon_0}\left[\frac{2pr}{(r^2-a^2)^2}\right] \text{.....(3)} \ [\because p = q(2a)]$$

6) For shorter dipole we have (a<<<r) so that a^2 can be neglected. $\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{r^4} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$

Thus, the intensity of electric field on the axial line is $E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$

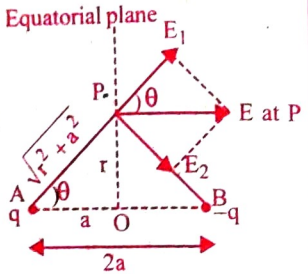
18. Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole. [AP 17,19,24]

A : 1) Consider an electric dipole AB consisting of two equal and opposite charges –q & q separated by a distance **2a.**
Let P be a point on the equatorial plane of the dipole at a distance r from its centre.

2) Electric field at P due to +q,–q is $E_1 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)$.....(1); $(\because AP^2 = r^2 + a^2)$

3) Electric field at P due to –q is $E_2 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)$.....(2) $(\because BP^2 = r^2 + a^2)$

From figure, it is clear that the y-components of E_1 and E_2 are equal and opposite. Hence they cancel each other.
But the sum of x-components gives the resultant field E at P.



4) Magnitude of resultant electric field is $E = E_1\cos\theta + E_2\cos\theta$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta + \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta = 2\times\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta$$

5) From Δ PAO, $\cos\theta = \frac{AO}{AP} = \frac{a}{\sqrt{r^2+a^2}}$

$$\therefore E = 2\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\frac{a}{\sqrt{r^2+a^2}} = \frac{2qa}{4\pi\varepsilon_0(r^2+a^2)^{3/2}}$$

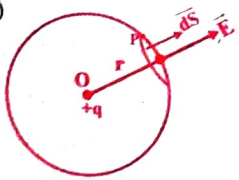
6) But we know dipole moment $p = q(2a)$ $\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2+a^2)^{3/2}}$.....(3)
For shorter dipole we have $(a \ll r)$ so that a^2 can be neglected.

7) From (3), for shorter dipole, $E = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}$

---

19. State Gauss's law in electrostatics and prove it. [AP, 20]

A: 1) **Gauss's Law:** The total electric flux (ϕ) through any closed surface is equal to $1/\varepsilon_0$ times the net charge (q) enclosed by the closed surface. Thus $\phi = \frac{1}{\varepsilon_0}(q)$



2) Proof of Gauss's Law
Suppose an isolated positive point charge q is situated at the centre O of a sphere of radius r.
Consider a point P on the sphere enclosed in a small closed surface area dS.

3) The electric field intensity at P is $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$.....(1)

4) Total electric flux over the entire surface of sphere is $\phi_E = \oint_S \vec{E}\cdot\vec{dS} = \oint_S EdS\cos\theta$
Here, $\vec{E}$ and $\vec{dS}$ are parallel. Hence $\theta = 0° \Rightarrow \cos\theta = \cos 0° = 1$

5) $\therefore \phi_E = \oint EdS = E\oint dS = E[S] = E\times 4\pi r^2$ .....(2) [$\because$ Surface area of sphere $S = 4\pi r^2$]

6) From (1) & (2), $\phi_E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\times 4\pi r^2 = \frac{q}{\varepsilon_0} = \frac{1}{\varepsilon_0}(q)$

$\therefore \phi = \frac{1}{\varepsilon_0}(q)$ Hence Gauss law is proved.

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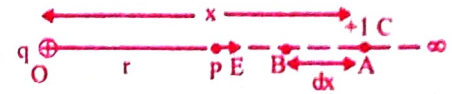
2 ELECTROSTATIC POTENTIAL

4 M+ 4M = 8 Marks

20. Derive an expression for the electric potential due to a point charge. [AP 19,23,25]

A : Potential due to a Point Charge:



1) Consider a positive point charge q at O.
   Let P be a point at a distance r from O.
2) Consider a unit positive test charge (+1 C) at A on the line OP at a distance x from O.
3) At A, electrostatic force on unit positive charge (+1 C) is $F = \frac{q \times 1}{4\pi\varepsilon_0 x^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}$ ....(1)
4) If dW is the small work done in moving the unit positive charge at A against the force through a small distance (AB) = dx, then $dW = -F\,dx$ [∵ the two positive charges repel as θ=180°]
5) From (1), $dW = -\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}dx$
6) Total work done(W) is obtained by integrating this between the limits from ∞ to r.

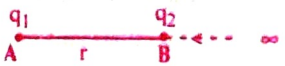
$$W = -\int_{\infty}^{r}\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2} = \frac{-q}{4\pi\varepsilon_0}\int_{\infty}^{r}\frac{1}{x^2}dx = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{x}\right]_{\infty}^{r} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r}\right)\left(\because \int\frac{1}{x^2}dx = \frac{-1}{x}\right)$$

The electric potential due to a point charge is this work done. $\therefore V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r}\right)$

---

21. Derive an expression for the electrostatic potential energy of a system of two point charges and find its relation with electric potential of a charge.

A: Potential Energy of System of Two Charges :



1) Consider a positive charge q_1 located at A.
   Let another charge q_2 be brought from infinity to the point B.
   Let r be the distance between A and B.
2) Potential at B due to charge q_1 is $V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r}$
3) The work done to bring the charge q_2 from infinity to B is $W = V_1(q_2) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$
   As this is the total work done to build the system, it gives the potential energy of the system.
4) Potential energy $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$
   The same result is obtained while bringing the charge q_1 first and q_2 later.

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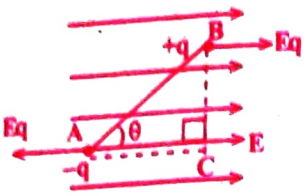
22.- Derive an expression for the potential energy of an electric dipole placed in a uniform electric field.



A: 1) Consider an electric dipole AB consisting of two equal and opposite charges –q & q separated by a distance **2a** placed in a uniform electric field E.

2) The torque on the dipole is $\vec{\tau} = \vec{p} \times \vec{E} = pE \sin\theta$

3) If an external torque τ_{ext} is applied to rotate the dipole from θ_0 to θ_1, the work done is given by $W = \int_{\theta_0}^{\theta_1} \tau_{ext} d\theta = \int_{\theta_0}^{\theta_1} pE \sin\theta d\theta = pE[-\cos\theta]_{\theta_0}^{\theta} = pE(\cos\theta_0 - \cos\theta_1)$

4) This work done is stored as the potential energy in the system.

If $\theta_0 = \pi/2$ and $\theta_1 = \theta$, the potential energy of the dipole at an angle θ, in uniform electric field is $U = pE(\cos\pi/2 - \cos\theta) = pE(0 - \cos\theta) = -pE\cos\theta = -\vec{p}.\vec{E}$

$\therefore U = -\vec{p}.\vec{E}$

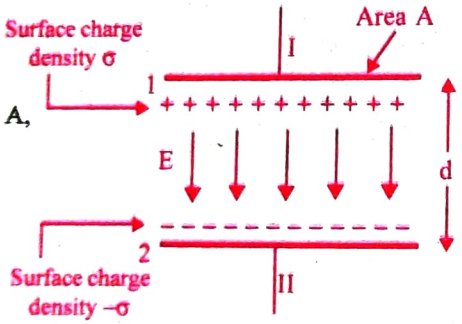
This is the expression for potential energy of dipole at an angle θ in a uniform electric field.

---

23. Derive an expression for the capacitance of a parallel plate capacitor.

[BMP][AP 16,18,22,24]

A : Capacitance of a parallel plate capacitor :



1) Consider a parallel plate capacitor consisting of two parallel plates of area A, separated by a small distance 'd'.

2) Let V be the potential difference between two plates.

3) The charges of the plates are Q and –Q.

4) The plate 1 has uniform surface charge density σ and the plate 2 has uniform surface charge density –σ.

5) The electric field due to plate 1 is $\frac{\sigma}{2\varepsilon_0}$. The electric field due to plate 2 is $-\frac{\sigma}{2\varepsilon_0}$.

6) Electric field between two charged plates is $E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = 2\left(\frac{\sigma}{2\varepsilon_0}\right) = \frac{\sigma}{\varepsilon_0}$

We know $\sigma = Q/A$

$\therefore$ The field inside the capacitor is $E = \frac{Q}{A\varepsilon_0}$ .....(1)

7) But, $E = \frac{V}{d}$ and $Q = CV$ From (1), $\frac{V}{d} = \frac{CV}{A\varepsilon_0} \Rightarrow C = \frac{\varepsilon_0 A}{d}$

$\therefore$ Capacitance of the parallel plate capacitor is given by $C = \frac{\varepsilon_0 A}{d}$

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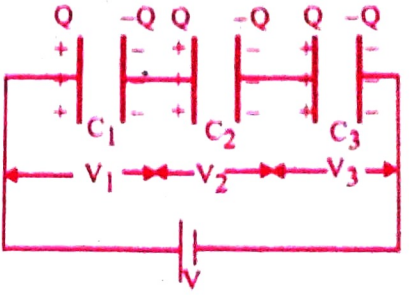
24. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series.

A : 1) Series Combination: In series combination, the second plate of first capacitor is connected to first plate of second capacitor and so on. Also, first plate of first capacitor and second plate of last capacitor are connected to opposite terminals of a battery.

2) Equivalent capacitance in series:



Let us suppose that three capacitors of capacitances C_1, C_2,C_3 are connected in series to a potential difference V.

Let Q be charge in each capacitor which remains constant in each capacitor.

3) Let V_1, V_2 , V_3 be the potential differences between the plates of the capacitors C_1, C_2,C_3 and C be the equivalent capacitance

4) Total potential differences is $V = V_1 + V_2 + V_3$ ...(1)

5) $\therefore V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$ $\quad [\because V_1 = \frac{Q}{C_1}, V_2 = \frac{Q}{C_2}$ and $V_3 = \frac{Q}{C_3}]$

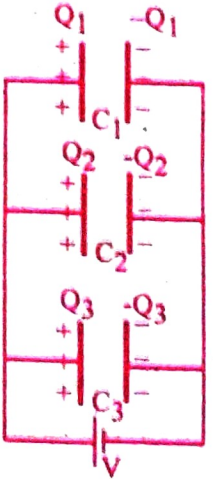
6) $\frac{V}{Q} = \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right] \Rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ .....(2)

7) For n number of capacitors in series, $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...+ \frac{1}{C_n}$

. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel. [AP 16]

Parallel Combination:

1) In parallel combination, the first plates of all the capacitors are connected to a common terminal and second plates of all the capacitors are connected to common terminal and these terminals are connected to opposite terminals of the battery.



2) Equivalent capacitance in parallel:

Consider three capacitors of capacitances C_1, C_2, C_3 connected in parallel to a potential difference V.

In parallel combination of capacitors, potential difference V remains the same on all the capacitors.

3) Let Q_1, Q_2 , Q_3 be the charges stored on the plates of capacitors C_1,C_2,C_3. In parallel combination, the total charge(Q) is equal to sum of charges stored in the capacitors.

4) $\therefore Q = Q_1 + Q_2 + Q_3$ .....(1)

$Q = C_1V + C_2V + C_3V$ $\quad [\because Q_1 = C_1V, Q_2 = C_2V, Q_3 = C_3V,]$

5) $\frac{Q}{V} = C_1 + C_2 + C_3 \Rightarrow C = C_1 + C_2 + C_3$ .....(2)

This is the equivalent capacitance of a combination.

6) For n number of capacitors in parallel, $C = C_1 + C_2 + C_3 ..... + C_n$

26. Derive an expression for the energy stored in a capacitor.

A: Energy Stored in a Capacitor :

1) When a capacitor of capacitance C is connected to a battery of potential difference V, capacitor gets charged to Q. The work done to move the charge is stored in the capacitor in form of potential energy called energy stored (U).

2) If dW is the small work done to store a charge of dQ at the potential drop V, we have

$dW = V\,dQ$.....(1)

3) The total work done to store the charge can be obtained by integrating it in the limits from 0 to

$$W = \int_0^Q V dQ \quad \text{But we know } C = \frac{Q}{V} \Rightarrow V = \frac{Q}{C}$$

4) $$\therefore W = \int_0^Q \frac{Q}{C} dQ = \frac{1}{C}\int_0^Q Q dQ = \frac{1}{C}\left(\frac{Q^2}{2}\right) = \frac{Q^2}{2C} \quad .....(2) \qquad \left(\because \int x dx = \frac{x^2}{2}\right)$$

5) This work done is stored in the form of energy U.

$$\therefore U = \frac{Q^2}{2C} = \frac{(CV)^2}{2C} = \frac{CV^2}{2} \qquad \left(\because \frac{Q}{C} = V \Rightarrow Q = CV\right)$$

$\therefore$ Energy Stored in a Capacitor is $U = \frac{1}{2}CV^2$

---

27. Explain the behaviour of dielectrics in an external field. **[AP 19]**

A : 1) Dielectrics are non-conducting substances.

The molecules of a dielectric may be polar or non-polar.

2) **Polarization:** When a dielectric with non-polar molecules is placed in an external electric field, the positive and negative charges of non-polar molecules are displaced in opposite directions. Thus induced dipole moments are developed in the dielectric by the external field.

Hence the dielectric is said to be polarised by the external field. All the induced dipole moments of the non-polar molecules add up to give a net dipole moment to the dielectric.

3) In dielectric with polar molecules, in the absence of external electric field, the different permanent dipoles are oriented randomly due to **thermal agitation.** Hence, the total dipole moment becomes zero. When an external electrical field is applied, the individual dipole moments tend to align with the field. As a result, a net dipole moment develops in the dielectric in the direction of external field. Hence the dielectric gets polarised.

4) Thus in either case, whether polar or non-polar, a dielectric develops a net dipole moment in the presence of an external field. The dipole moment per unit volume is called polarisation (P).

5) For linear isotropic dielectrics, $P = \chi_e E$

Here χ_e is electric susceptibility and E is intensity of external field.

PROBLEMS

28. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel.
a) What is the total capacitance of the combination?
b) Determine the charge on each capacitor if the combination is connected to a 100V supply. **[BMP] [AP 17] [TS 15]**

Sol: a) $C_p = 2+3+4 = 9pF$

b) For each capacitor, V = 100 Volt

$q_1 = C_1V = 2\times100 = 200pC$; $q_2 = C_2V = 3\times100 = 300pC$; $q_3 = C_3V = 4\times100 = 400pC$

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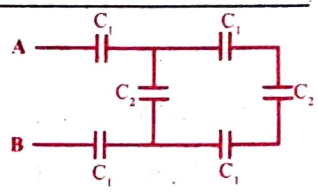
29. If $C_1 = 3$ pF and $C_2 = 2$ pF (fig.). Calculate the equivalent capacitance of the given network between points A and B.



Sol: Capacity $C_1 = 3pF$, Capacity $C_2 = 2pF$

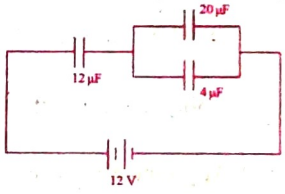
Initially C_1, C_2, C_1 are in series $\Rightarrow \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_1} = \frac{1}{3} + \frac{1}{2} + \frac{1}{3} = \frac{2+3+2}{6} \therefore C_S = \frac{6}{7}pF$

Now C_S and C_2 are in parallel.

$\therefore C_P = C_s + C_2 = \frac{6}{7} + 2 = \frac{6+14}{7} = \frac{20}{7}pF$ Then C_1, C_P, C_1 are in series.

---

30. In the circuit shown in fig. the charge on the capacitor of 4 μF is 16 μC. Calculate the energy stored in the capacitor of 12 μF capacitance.



Sol: Given that, charge on the capacitor of 4μF is of 16μC.

Voltage across 4 μF capacitor. $V_2 = \frac{q}{C} = \frac{16}{4} = 4V$

$\therefore$ Voltage across 12 μF capacitor $V_1 = V - V_2 = 12 - 4 = 8V$

$\therefore$ Energy stored in the capacitor 12μF is $U = \frac{1}{2}CV_1^2 = \frac{1}{2}(12)(8)^2 = 384\ \mu J$

$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_P} + \frac{1}{C_1} = \frac{1}{3} + \frac{7}{20} + \frac{1}{3} = \frac{20+21+20}{60} = \frac{61}{60}$ $\therefore$ Resultant capacity $C = \frac{60}{61}pF$

---

31. The electric potential at 0.9m from a point charge is +50V. What is the magnitude and sign of the charge?

Sol: Given potential V = 50V, distance r = 0.9m

Potential $V = K\frac{q}{r} \Rightarrow q = \frac{Vr}{K} = \frac{50\times0.9}{9\times10^9} = 5\times10^{-9}C$

Since the given potential is positive, the charge must also be positive.

[☞ Remaining SAQs are in Page 99]

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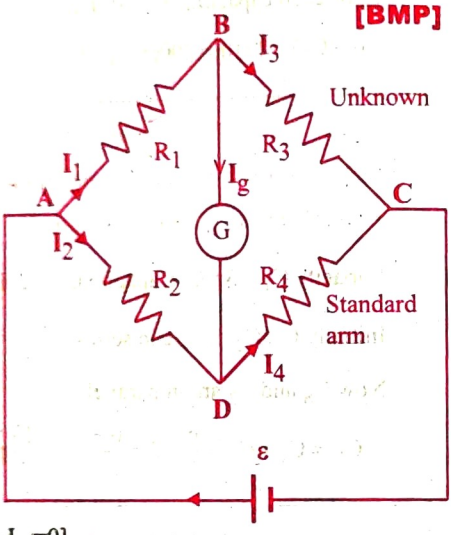
3 CURRENT ELECTRICITY

4 Marks

32. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. [BMP]



A: 1) **Wheatstone's Bridge** consists of four resistors R_1, R_2, R_3, R_4 with four junctions A,B,C, D as shown in the figure.

2) AC is battery arm, BD is galvanometer arm. When the bridge is balanced, the current flowing through the galvanometer $I_g = 0$,.

3) Applying Kirchoff's first law at B,

we have $I_1 - I_g - I_3 = 0 \Rightarrow I_1 = I_3$ .....(1) [$\because I_g = 0$]

4) Applying Kirchoff's first law at D, we have $I_2 + I_g - I_4 = 0 \Rightarrow I_2 = I_4$ .....(2) [$\because I_g = 0$]

5) Applying Kirchoff's second law to closed loop ABDA, we have $I_1 R_1 + I_g G - I_2 R_2 = 0$

$\Rightarrow I_1 R_1 = I_2 R_2 \Rightarrow \frac{I_1}{I_2} = \frac{R_2}{R_1}$ .....(3)[$\because I_g = 0$]

6) Applying Kirchoff's second law to closed loop BDCB, we have $I_g G + I_4 R_4 - I_3 R_3 = 0$

$\Rightarrow I_3 R_3 = I_4 R_4 = \Rightarrow \frac{I_3}{I_4} = \frac{R_4}{R_3}$ .....(4)

7) But from (1) & (2), $I_1 = I_3$ and $I_2 = I_4$ From (4), we have $\frac{I_1}{I_2} = \frac{R_4}{R_3}$ .....(5)

8) Now from (3) & (5) we get $\frac{R_2}{R_1} = \frac{R_4}{R_3}$ .....(6)

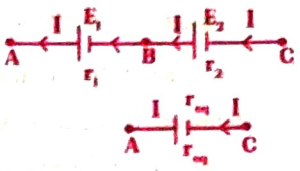
9) $\therefore \frac{R_1}{R_2} = \frac{R_3}{R4}$ (or) $\frac{R_1}{R_3} = \frac{R_2}{R4}$. This is the balance condition of Wheatstone's bridge to make $I_g = 0$.

[☞ Remaining SAQs are in Page 99]

33. Derive equivalent emf and internal resistance of two cells in series combination.

A: 1) Consider two cells of emf's E_1 and E_2 and their internal resistances r_1, r_2 connected in series.

Let V_A, V_B, V_C be the potentials at points A, B, C.



2) The potential difference between positive and negative terminals of the first cells between A and B is

$V_{AB} = V_A - V_B = E_1 - I r_1$ .....(1)

Similarly, $V_{BC} = V_B - V_C = E_2 - I r_2$ .....(2)

3) The potential difference between positive and negative terminals of A and C is

$V_{AC} = V_A - V_C = (V_A - V_B) + (V_B - V_C) = (E_1 - Ir_1) + (E_2 - Ir_2)$

$\therefore V_{AC} = (E_1 + E_2) - I[r_1 + r_2]$.....(3)

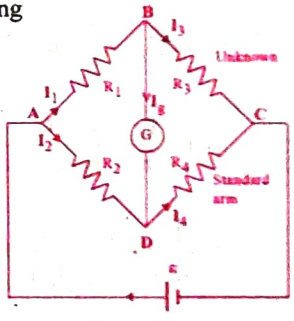
4) If we replace the combination by a single cell between A and C of emf (E_{eq}) and internal resistance (r_{eq}) then $V_{AC} = E_{eq} - Ir_{eq}$ .....(4)

5) Comparing (3) and (4), we get $E_{eq} = E_1 + E_2$ and $r_{eq} = r_1 + r_2$

---

34. State and explain Kirchhoff rules for an electric network.

A: **1) Kirchhoff's First Law :** The algebraic sum of currents meeting at any junction in an electric circuit is zero.



2) Applying **Kirchhoff's first law at B,**

we have $I_1 - I_g - I_3 = 0$

3) Kirchhoff's Second Law : The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

4) Applying **Kirchhoff's second law to closed loop ABDA,**

we have $I_1R_1 + I_gG - I_2R_2 = 0$

---

35. When the resistance connected in series with the cell is halved, the current is equal to or slightly less or slightly greater than double, Why?

A: 1) The current in a circuit with a battery of emf E, internal resistance r and external resistance R is given by $I = \frac{E}{R+r}$

2) When the resistance is halved (R/2) we have the new current $I' = \frac{E}{(R/2)+r}$

3) If r is negligible when compared to R/2 then we get $I' = \frac{2E}{R} = 2I \quad (\because \frac{E}{R} = I)$

4) Also if $r \ll (R/2)$ then the new current I' is slightly greater than 2I and if $r > R$ then the new current I' is slightly less than 2I.

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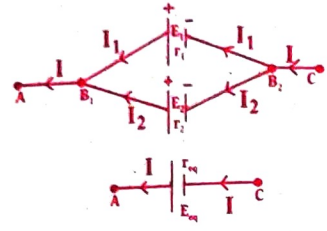
36. Derive equivalent emf and internal resistance of two cells in parallel combination.

Sol: 1) Resistors in Parallel : Consider two cells of emf's E_1 and E_2 and internal resistance r_1 and r_2 connected in parallel as shown in fig.



2) Let I_1 and I_2 be the currents leaving the positive terminals of the cells. At junction B_1 the currents I_1 and I_2 flow in whereas the current I flows out.

Total current at junction B is $I = I_1 + I_2$ .....(1)

3) Let V_{B_1} and V_{B_2} be the potentials at B_1 and B_2.

For the first cell, potential difference is $V = V_{B_1} - V_{B_2} = E_1 - I_1 r_1$

$$\Rightarrow I_1 r_1 = E_1 - V \Rightarrow I_1 = \frac{E_1 - V}{r_1} \quad .....(2)$$

4) For the second cell, potential differences is $V = V_{B_1} - V_{B_2} = E_2 - I_2 r_2$

$$\Rightarrow I_2 r_2 = E_2 - V \Rightarrow I_2 = \frac{E_2 - V}{r_2} \quad .....(3)$$

5) From (1), (2) and (3) we have $I = I_1 + I_2 = \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$

$$\Rightarrow I = \frac{E_1}{r_1} - \frac{V}{r_1} + \frac{E_2}{r_2} - \frac{V}{r_2} \Rightarrow I = \left[\frac{E_1}{r_1} + \frac{E_2}{r_2}\right] - V\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$$

$$\Rightarrow V\left[\frac{1}{r_1} + \frac{1}{r_2}\right] = \left[\frac{E_1}{r_1} + \frac{E_2}{r_2}\right] - I$$

$$\Rightarrow V\left[\frac{r_1 + r_2}{r_1 r_2}\right] = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 r_2}\right] - I = \left[\frac{(E_1 r_2 + E_2 r_1) - I r_1 r_2}{r_1 r_2}\right]$$

$$\Rightarrow V = \left[\frac{\cancel{r_1 r_2}}{r_1 + r_2}\right]\left[\frac{(E_1 r_2 + E_2 r_1) - I r_1 r_2}{\cancel{r_1 r_2}}\right] \Rightarrow V = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}\right] - I\left[\frac{r_1 r_2}{r_1 + r_2}\right] \quad .....(4)$$

6) If we replace the combination by a single cell between A and C of emf (E_{eq}) and internal resistance (r_{eq}) then $V_{AC} = E_{eq} - I r_{eq}$ .....(5)

7) Comparing (4) and (5) we get $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$ and $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$

4 MAGNETISM & MATTER

4 Marks

37. Mention the properties of magnetic field lines.

A: Properties of magnetic field lines :

1) The magnetic field lines of a magnet form continuous closed loops.
2) The magnetic line of force starts from a north pole and ends at south pole outside the magnet and will be from south pole to north pole inside the magnet.
3) At a given point of the field line, the tangent represents the direction of the field at that point.
4) The magnetic field lines don't intersect. If they intersect, magnetic field at that point will have two directions which is not possible.
5) The magnetic lines of force experience longitudinal strain. Due to this property unlike poles attract each other.
6) The magnetic lines of force exert lateral pressure. Due to this property like poles repel each other.

---

38. Explain the properties of diamagnetic materials.

A: Properties of diamagnetic substances:

1) Diamagnetic substances are weakly magnetised in opposite direction to that of magnetising field
   Ex: Cu,Ag,Au,H_2O
2) They are feebly repelled by a magnet.
3) They move from higher intensity to lower intensity in a non-uniform magnetic field.
4) They are independent of temperature.
5) Their relative permeability $\mu_r < 1$, ($\chi < 0$, χ is small and negative)
6) Resultant magnetic moment of its atom is zero, in the absence of external field.

---

39. Explain the properties of paramagnetic materials. **[BMP]**

A: Properties of Paramagnetic substances:

1) Paramagnetic substances are weakly magnetised in the direction of magnetising field.
   Ex: Al, Na, Pt.
2) They are feebly attracted by a magnet.
3) They move from lower intensity to higher intensity in a non-uniform magnetic field.
4) They are dependent on temperature.
5) Their relative permeability $\mu_r > 1.$, ($\chi > 0$, χ is small and positive)
6) Resultant magnetic moment of its atom is non-zero, due to unpaired electrons.

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40. How the existence of domains, explain the behavior of ferromagnetic material?

A: **Ferromagnetic materials** are the substances which are strongly attracted by magnets. Their atomic dipoles align spontaneously to form **strong clusters called domains.**

1) In ferromagnetic materials the atoms form innumerable small effective regions called "domains".
2) The size of domain varies from about 10^{-6} m to 10^{-3}m.
3) Each domain contains 10^{17} to 10^{21} atomic magnets aligned in the same direction.
4) In an external magnetic field, the resultant magnetism increases due to reorientation of all domains into the direction of field.
5) In the absence of external field the domains are oriented randomly, so that net magnetic moment is zero.

---

41. State Gauss' law in magnetism. Show that how this law proved the non-existence of magnetic monopoles.

A: **1) Gauss' Law for Magnetism:** Gauss' Law for magnetism states that the net magnetic flux through any closed surface is always zero.

Gauss law for magnetism is $\oint \bar{B}.d\bar{S} = 0$

2) Explanation: The above equation states that the net magnetic flux through any closed surface is zero. Magnetic filed lines always form closed continues loops. They do not begin or end at any point, unlike electric field lines.

3) Non-existence of Magnetic Monopoles: Since the net flux through a closed surface is zero there is no isolated magnetic poles (monopoles) in nature. Magnetic poles always exist in pairs (North & south)

4) Conclusion: Gauss law in Magnetism proves that magnetic monopoles do not exist.

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42. Define magnetisation of a sample, magnetic induction and susceptibility. How magnetic induction is related to magnetic intensity?

A: **1) Magnetisation (M)** of a sample is defined as the net magnetic moment per unit volume.

Formula: $M = \frac{m_{net}}{V}$ **SI unit:** $A\,m^{-1}$.

2) Magnetic induction(B) is defined as the total number of magnetic lines of force passing normally through a unit area of a substance. **SI unit:** Tesla (T)

3) Magnetic Susceptibility (χ) is defined as the ratio between magnetisation (M) and magnetic intensity (H). **Formula:** $\chi = \frac{M}{H}$. It has no units

4) Relation between (B) and (H):

In a material of a susceptibility χ_m, magnetic induction B is proportional to the magnetic intensity H Thus $B \propto H \Rightarrow B = \mu H$; Here μ= permeability of the substance. In free space, $B = \mu_0 H$

5) The total magnetic induction B = B (applied field) + B (Magnetization)

$\therefore B = \mu_0 H + \mu_0 H \chi_m = \mu_0 H(1+\chi_m) = \mu_0(1+\chi_m)H = \mu_0(\mu_r)H$ $[\because 1+\chi_m = \mu_r]$

$\therefore B = \mu H$ $[\because \mu_0(\mu_r) = \mu]$

5 ELECTRO MAGNETIC INDUCTION

4 Marks

43. Explain the Faradays law of induction and Lenz's law.

A : **1) Faraday's law of Electromagnetic Induction: The** magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2) Mathematically, the induced emf is given by $\varepsilon = -\frac{d\phi_B}{dt}$

3) In the case of a coil of N turns, $\varepsilon = -N\frac{d\phi_B}{dt}$

4) Lenz's Law(Neumann's law) : The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux.

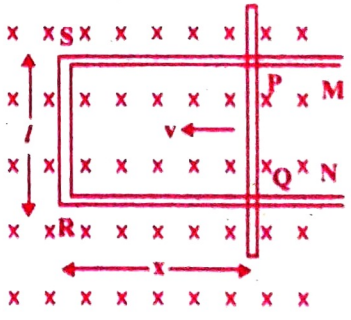
5) Faraday's law of induction is $\varepsilon = -\frac{d\phi_B}{dt}$, The negative sign in it indicates Lenz's law.

---

44. Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field.

A : Induced emf due to motion of a conductor in a magnetic field:

1) Consider a rectangular conductor PQRS suspended in a uniform magnetic field B which is perpendicular to the plane of the system.



2) The conductor PQ is free to move. Suppose, the conductor PQ is moved towards left with velocity v.

3) Let RQ = x, RS = l.

Then the magnetic flux enclosed by the loop PQRS is $\phi_B = B\,l\,x$

4) The induced emf is given by $\varepsilon = -\frac{d\phi_B}{dt} = -\frac{d}{dt}(B\,l\,x) \Rightarrow \varepsilon = -B\,l\,\frac{dx}{dt}$. But $\frac{dx}{dt} = -v$

5) Thus the induced emf in a conductor moving in a perpendicular and uniform magnetic field is given by $\varepsilon = B\,l\,v$

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[BMP]

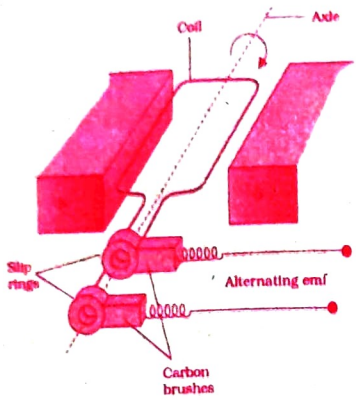
45. Explain briefly working of AC generator.

A: AC Generator : AC generator converts **mechanical energy** into **electrical energy.**

Construction:

1) AC generator consists of a rectangular coil is placed between two magnetic poles.

2) The ends of the coil are connected to slip rings and brushes.



AC Generator

Working:

3) When the coil is rotated with a constant angular speed ω, $\theta = \omega t$.
where θ is the angle between vector area A and the magnetic field B at given time t.
The flux at time t is given by

$\phi_B = BA\cos\theta = BA\cos\omega t \quad (\because \theta = \omega t)$

4) From Faraday's law, the induced emf is $\varepsilon = -N\frac{d\phi_B}{dt} = -N\frac{d}{dt}(BA\cos\omega t)$

$= -NBA\frac{d}{dt}(\cos\omega t) = -NBA\ (-\sin\omega t)(\omega) = NBA\,\omega\sin\omega t.$

5) Thus the instantaneous emf is $\varepsilon = NBA\,\omega\sin\omega t$.

Here, $NBA\omega = \varepsilon_m$ = maximum emf.

$\therefore \varepsilon = \varepsilon_m \sin\omega t \quad (\because \varepsilon_m = NBA\omega)$

Hence the voltage produced by the generator is Sinusoidal.

---

46. Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid.

A : Magnetic Energy stored in a Solenoid:

1) The energy required to build up the current in a solenoid is given by $U = \frac{1}{2}LI^2$ .....(1)
where L is the self inductance of the solenoid.

2) Self inductance of a solenoid is given by $L = \mu_0 n^2 A l$ ..... (2)

here n is number of turns per unit length, A is area of cross section, l is length of the solenoid.
Substituting eq (2) in eq (1), we get $U = \frac{1}{2}(\mu_0 n^2 A l)I^2$ ..... (3)

3) Magnetic induction inside a solenoid is $B = \mu_0 n I \Rightarrow I = B/\mu_0 n$ ..... (4)

4) Putting this in eq (3), we get $U = \frac{1}{2}(\mu_0 n^2 A l)\frac{B^2}{\mu_0^2 n^2} = \frac{1}{2\mu_0}B^2 Al$

This is the expression for magnetic energy stored in a solenoid.

[☞ Remaining SAQs are in Page 100]

6 ALTERNATING CURRENT

4 Marks

47. State the principle on which a transformer works. [BMP]

Describe the working of a transformer with necessary theory.

A : 1) **Transformer: Transformer** is a device which transforms AC electric power from one circuit to another. A transformer consists of two coils called primary coil and secondary coil wound on a continuous iron core.

2) **Principle**: Transformer is based on the principle of **mutual induction**. When current in primary coil changes, an induced current is produced in the secondary coil.

Transformer ratio $\frac{N_s}{N_p}=\frac{V_s}{V_p}$ Here N_s = number of turns in secondary coil,

N_p = number of turns in primary coil.

V_s = AC voltage in secondary, V_p = AC voltage applied to primary,

3) **Working:** When an alternating voltage V_p is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf ε_s in it. Let ϕ be the flux in each turn in the core at time t.

4) From Faraday's laws of induction, induced emf in the secondary coil is $\varepsilon_s=-N_s\frac{d\phi}{dt}$ .....(1)

Emf in the primary, $\varepsilon_p=-N_p\frac{d\phi}{dt}$ .....(2) (ϕ is alternating flux)

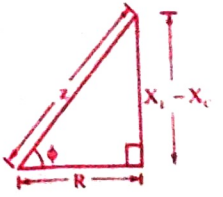
5) Here, $\varepsilon_s = V_s$ and $\varepsilon_p = V_p$ From (1) & (2) $V_s=-N_s\frac{d\phi}{dt}$ and $V_p=-N_p\frac{d\phi}{dt}$

6) Dividing them we get $\frac{V_s}{V_p}=\frac{N_s}{N_p}$. This is the formula for **transformer ratio.**

---

48. What is power factor? Derive its expression in terms of resistance R and impedance Z. Explain its significance in AC circuits.

A: 1) **Power factor** of an AC circuit is defined as the cosine of the phase angle (ϕ) between voltage and current. **Formula:** $\cos\phi=\frac{R}{Z}$



2) **Expression:** Consider impedance triangle for series LCR circuit as shown in the figure.

Here Z= hypotenuse, base =R,

perpendicular= $X_L - X_C$ and ϕ is the phase angle.

From the right angled triangle Power factor, $\cos\phi=\frac{R}{Z}$

3) **Significance:** If power factor cos ϕ = 1 the maximum power is dissipated in the circuit. If power factor cos ϕ = 0 then no power is consumed. This leads to the concept of wattless current dissipated even though current is flowing in the circuit

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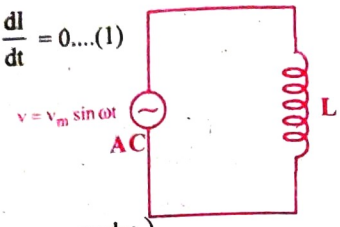
49. Obtain an expression for the current through an inductor when an AC emf is applied.

A : Current through an Inductor with AC emf :

1) Consider a pure inductor with self inductance L connected to an AC of voltage $v = v_m \sin\omega t$

2) Let I be the instantaneous current at any time t passing through the inductor.

Applying the Krichhoffs law on the circuit, we have $v - L\frac{dI}{dt} = 0$....(1)



$$\Rightarrow L\frac{dI}{dt} = v \Rightarrow \frac{dI}{dt} = \frac{v}{L} = \frac{v_m \sin\omega t}{L} \quad (\because v = v_m \sin\omega t)$$

3) Integrating it with respect to time, we get

$$\int \frac{dI}{dt}dt = \frac{v_m}{L}\int \sin(\omega t)dt \Rightarrow I = \frac{v_m}{L}\left(\frac{-\cos(\omega t)}{\omega}\right) \quad \left(\because \int \sin kx = \frac{-\cos kx}{k}\right)$$

$$\Rightarrow I = \frac{v_m}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)....(2) \quad \left[\because -\cos(\omega t) = \sin\left(\omega t - \frac{\pi}{2}\right)\right]$$

4) We know amplitude of current $I_m = \frac{v_m}{\omega L}$ $\quad \therefore (2) \Rightarrow I = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$

Current in an AC inductor circuit is $I = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$

Comparing this with $v = v_m \sin\omega t$, we claim that the current **lags** the voltage by $\pi/2$.

---

50. Obtain an expression for the current in a capacitor when an AC emf is applied.

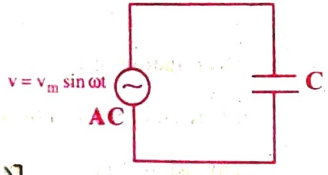
A : Current through an Capacitor with AC :

1) Consider a pure capacitor C connected to an AC voltage $v = v_m \sin\omega t$ .....(1)
Let I be the instantaneous current at any time t passing through the capacitor.

2) The potential difference across the capacitor of capacitance C is $v = \frac{q}{C}$ .....(2)

From (1) & (2), $\frac{q}{C} = v_m \sin\omega t$ $\quad (\because v = v_m \sin\omega t) \Rightarrow q = C\, v_m \sin\omega t$

3) We know $I = \frac{dq}{dt} \Rightarrow I = \frac{d}{dt}(C\, v_m \sin\omega t)$



$$\Rightarrow I = \omega C\, v_m \cos(\omega t) \quad \left(\because \frac{d}{dx}\sin kx = k\cos kx\right)$$

$$I = \omega C\, v_m \sin\left(\omega t + \frac{\pi}{2}\right)....(2) \quad \left[\because \cos(\omega t) = \sin\left(\omega t + \frac{\pi}{2}\right)\right]$$

4) We know amplitude of current $I_m = \frac{v_m}{X_C} = \frac{v_m}{1/\omega C} = v_m \omega C$

$$(2) \Rightarrow I = I_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

5) Comparing this with $v = v_m \sin\omega t$, we claim that the current **leads** the voltage by $\pi/2$.

[☞ Remaining SAQs are in Page 100]

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7 WAVE OPTICS

4 Marks

51. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. [AP 16,18]

A : Expression for Intensity at a point due Interference of Light :

1) Consider two interfering light rays coming from two sources.
The displacements of the two rays are given by $y_1 = A\cos\omega t$, $y_2 = A\cos(\omega t + \phi)$ where a is amplitude, ω is angular frequency and ϕ is the phase difference between the waves.

2) From the principle of superposition of the rays the resultant displacement is $y = y_1 + y_2$

$\Rightarrow y = A[\cos\omega t + \cos(\omega t + \phi)]$

$$\Rightarrow y = \left(2A\cos\left(\frac{\phi}{2}\right)\right)\cos\left(\omega t + \frac{\phi}{2}\right) \quad \left(\because \cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)\right)$$

Here, $2A\cos\left(\frac{\phi}{2}\right)$ represents the amplitude of the resultant wave at the point.

3) The intensity of resultant wave is $I = 4I_0\cos^2\left(\frac{\phi}{2}\right)$ where I_0 is the intensity of each wave.

4) Conditions for Maximum Intensities:

The maximum value of resultant intensity I is $I_{max}=4I_0$ when $\cos^2(\phi/2) = 1$.
Thus, the condition for maximum intensity at the point is $\phi = 0, \pm 2\pi, \pm 4\pi$ .....

5) Conditions for Zero Intensities:

The minimum value of resultant intensity I is $I_{min} = 0$ when $\cos^2(\phi/2) = 0$.
Thus, the condition for minimum intensity at the point is $\phi = \pm\pi, \pm 3\pi, \pm 5\pi$ . . . .

---

52. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [BMP] [AP 16,17,20,25]

A : 1) Yes. The principle of conservation of energy holds good for both **interference** and **diffraction.**

2) In interference and diffraction, patterns of bright and dark fringes are formed. Here the principle of conservation of energy holds good for patterns of bright and dark fringes of interference and diffraction.

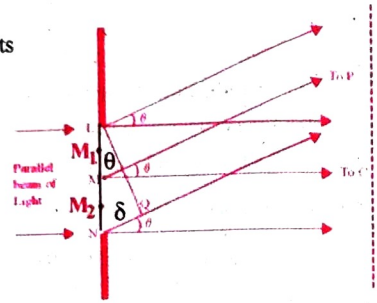
3) In interference and diffraction, light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing bright fringe. Thus there is no gain or loss of energy. Hence, total energy remains constant.

53. Explain single slit diffraction pattern with neat diagram.
State conditions for maximum intensity and minimum intensity.

A: Diffraction at Single Slit:

1) Suppose a parallel beam of light is incident normally on a single slit LN of width a.
The diffracted light goes on to meet a screen at large distance.

2) Let us consider a point P on the screen which collects the waves originating from different points of the slit at an angle θ.



3) From the diagram,

$$\sin\theta = \frac{\delta}{a} \Rightarrow \delta = a\sin\theta \Rightarrow \delta = a\theta$$

(when θ is very small we have sinθ =θ)

4) Condition for maximum intensity:

For maximum intensity $\delta = (2n+1)\frac{\lambda}{2}$ $\quad \therefore (2n+1)\frac{\lambda}{2} = a\theta \Rightarrow \theta = (2n+1)\frac{\lambda}{2a}$, n = 1,2,3,...

5) Conditions for minimum intensity:

For minimum intensity $\delta = 2n\left(\frac{\lambda}{2}\right)$ $\quad \not{2}n\left(\frac{\lambda}{\not{2}}\right) = a\theta \Rightarrow \theta = \frac{n\lambda}{a}$, n=1,2,3,....

---

54. Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids.

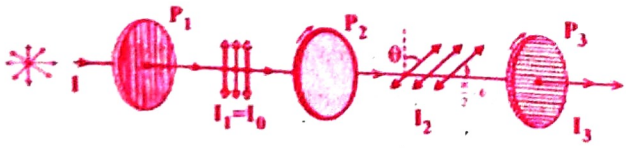
A: Intensity of Transmitted Polarised Light through a Polaroid :

1) Let I_0 be the intensity of polarised light after passing through the first polariser P_1.
Then the intensity of light after passing through second polariser P_2 will be $I_2 = I_0\cos^2\theta$
Here θ is the angle between pass axes of P_1 and P_2

2) Since P_1 and P_3 are crossed the angle between the pass axes of P_2 and P_3 will be $\left(\frac{\pi}{2} - \theta\right)$.
Hence the intensity of light emerging from P_3 will be $I_3 = I_2\cos^2(90°-\theta)$

$$I_3 = (I_0\cos^2\theta)\sin^2\theta = \frac{I_0}{4}(4\sin^2\theta\cos^2\theta) = \frac{I_0}{4}\sin^2 2\theta \qquad [\because \sin 2\theta = 2\sin\theta\cos\theta\,]$$

3) Intensity of transmitted light will be maximum when $\sin 2\theta = 1 \Rightarrow 2\theta = 90° \Rightarrow \theta = 45°$
$\therefore$ the transmitted intensity will be maximum when $\theta = \pi/4$



[☞ Remaining SAQs are in Page 101]

8 DUAL NATURE OF RADIATION & MATTER

4 +4 Marks

55. What are different types of physical processes to obtain free electrons from the metal surface. **[BMP]**

A: Types of physical processes:

1) Thermionic Emission: By suitably heating, sufficient thermal energy can be imparted to the free electrons to enable them to come out of the metal.

Ex: Vacuum tubes, CRT & X rays tubes these are the applications of this emission.

2) Field Emission: By applying a very strong electric field to a metal, electrons can be pulled out of the metal, as in a spark plug.

Ex: Spark plugs, field emission microscopes, cold cathode devices are these applications.

3) Photoelectric Emission: When light of suitable frequency illuminates a metal surface, photo electrons are emitted from the metal surface.

Ex: Photoelectric cells, solar panels and light sensors are the applications of this emission.

4) Secondary Emission: When high speed 'primary' particles like electrons strike a metal surface they transfer their kinetic energy to the electrons within the metal, causing 'secondary' electrons to be knocked out of the surface.

Ex: Photo multiplier tubes, image intensifiers are the applications of this emission.

---

56. Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential.

A: i) Threshold frequency: It is the minimum frequency of incident radiation required to liberate electrons from the metal surface. It's value depends on the nature of the metal.

ii) Work function: It is the minimum amount of energy required just to liberate an electron from the given metal surface. Work function of a metal is a physical constant.

iii) Relation between work function and threshold frequency:

The work function (ϕ) is directly proportional to the threshold frequency (ν_0) of the metal. Work function $\phi = h\nu_0$, h is planck's constant.

iv) Stopping Potential: It is the minimum negative potential given to the collector plate for which the photocurrent stops or becomes zero.

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57. What are the laws of photoelectric effect.

A: Laws of photoelectric effect:

1) **Threshold Frequency:** For a given photosensitive surface, there exists a minimum frequency called threshold frequency (ν_0), below which photoelectric emission does not take place, whatever may be the intensity of incident light.

2) **Intensity :** The photoelectric current is directly proportional to the intensity of incident light, provided the frequency is greater than threshold frequency.

3) **Kinetic Energy :** The maximum kinetic energy of emitted photoelectrons depends on the frequency of incident radiation and is independent of the intensity of light. $K_{max}=h\nu-\phi$

4) **Instantaneous Emission:** The photoelectric emission is instantaneous. There is no measurable time lag between the incidence of light and the emission of photoelectrons.

---

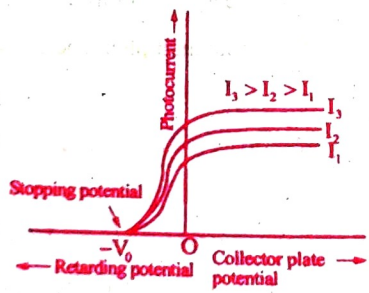
58. Explain graphically that the stopping potential is independent of intensity of incident light.

A: 1) Stopping Potential: It is the minimum negative potential given to the collector plate for which the photocurrent stops or becomes zero.

2) When light of the same frequency but different intensities falls on a photosensitive metal surface, the number of emitted photoelectrons changes then the corresponding photoelectric currents are measured.

3) Graphical Explanation:

i) We plot the photocurrent versus the collector potential (V) for light of same frequency and different intensities (I_1, I_2, I_3).



ii) From the graph it is clear that, as intensity of light increases ($I_3 > I_2 > I_1$), the saturation current also increases.

iii) The 3 curves converge at a negative potential point on the x-axis . This potential is called stopping potential V_0.

iv) From graph, the stopping potential is independent of intensity of incident light.

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59. Write Einstein's photo-electric equation.
How Millikan proved the validity of Einstein's photoelectric equation.

A: **1) Einstein's photoelectric equation:** $h\nu = \phi + \frac{1}{2}mv_{max}^2$

Here h = Planck's constant, ν= frequency of incident radiation, f = work function of the metal

$\frac{1}{2}mv_{max}^2$ = maximum kinetic energy of emitted photoelectrons

2) Since maximum kinetic energy is related to stopping potential $V_0 \Rightarrow \frac{1}{2}mv_{max}^2 = eV_0$

3) Einstein's equation can also be written as: $h\nu = \phi + eV_0 \Rightarrow V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$

4) Millikan's Verification of Einstein's Photoelectric Equation:

Millikan measured the stopping potential V_0 for different frequencies of incident light.

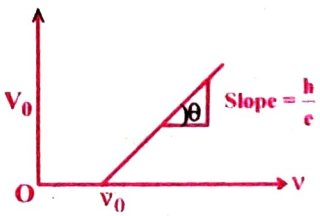
He plotted a graph between stopping potential V_0 and frequency ν

5) Observations:

i) The graph was a straight line.

ii) The slope of the graph was equal to h/e.

iii) The intercept on frequency axis gave the threshold frequency ν_0.



6) Conclusion:

The experimental results exactly matched the equation $V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$

Thus, Millikan proved the validity of Einstein's photoelectric equation.
He also determined the value of Planck's constant from the slope of the graph.

---

60. What is de Broglie's wavelength?

A: According to de Broglie's hypothesis, matter has dual nature . It states that every moving particle experiences a wave like nature.

2) de Broglie's wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$.

Here m is the mass of particle, v is the speed of particle, p is the momentum of particle.

h is the planck's constant, λ is the de Broglie wave length.

This equation is called de Broglie's relation.

3) For a microscopic particle like electron, the mass of the particle is extremely small.

4) Then the value of λ becomes very high as the mass m is in the denominator of the formula. Hence, the moving particle behaves like a wave.

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PROBLEMS

61. The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. [AP 16,24]

Sol: Given Work function $\phi_0 = 2.14\ eV = 2.14 \times 1.6\times10^{-19}J$, Stopping potential $V_0 = 0.60\ V$

Planck's constant $h = 6.63\times10^{-34}$ Js.

a) $\phi_0 = h\nu_0 \Rightarrow$ Threshold frequency $\nu_0 = \phi_0 / h = \dfrac{2.14\times1.6\times10^{-19}}{6.63\times10^{-34}} = 5.16\times10^{14}$ Hz

b) Einstein's photoelectric equation $eV_0 = h\nu - \phi_0 \Rightarrow h\nu = eV_0 + \phi_0 \Rightarrow \dfrac{hc}{\lambda} = eV_0 + \phi_0$

$$\Rightarrow \lambda = \frac{hc}{(eV_0 + \phi_0)} = \frac{(6.63\times10^{-34})(3\times10^{8})}{(0.60+2.14)1.6\times10^{-19}} = 4.54\times10^{-7}m = \mathbf{454\ nm}$$

---

62. The work function of a metal is 4.3 eV. What is the threshold wavelength?

Sol: Given work function $\phi = 4.3\ eV = 4.3\times1.6\times10^{-19} = 6.88\times10^{-19}J$

Planck's constant $h = 6.6\times10^{-34}$, Speed of light $c = 3\times10^{8}$ m/s

$\therefore$ Threshold wavelength $\lambda_0 = \dfrac{hc}{\phi} = \dfrac{12400}{4.3} = 2883.7\text{Å} = 288.37\times10^{-9}m = 288.37nm.$

---

63. A particle has a momentum of 1.5×10^{-27} kg m/s. What is its de-Broglie wavelength?

Sol: Given momentum of the particle $p = 1.5\times10^{-27}\ kg\,m/s$, Planck's constant $h = 6.626\times10^{-34}$

$\therefore$ de-Broglie wave length $\lambda = \dfrac{h}{p} = \dfrac{6.626\times10^{-34}}{1.5\times10^{-27}} = 4.4173\times10^{-7} = 441.73\times10^{-9}\ m = 441.73\ nm$

---

64. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? [BMP]

Sol: Given wavelength $\lambda = 5000\text{Å} = 5\times10^{-7}$m; Mass of electron $= 9.1'10{-}31$kg; $h = 6.63\times10^{-34}$ J/s

Kinetic energy $K = \dfrac{h^2}{2m\lambda^2}$ $\left(\because \lambda = \dfrac{h}{\sqrt{2mK}}\right)$

$$\therefore K = \frac{[6.63\times10^{-34}]^2}{2\times[9.1\times10^{-31}][5\times10^{-7}]^2} = \frac{43.9569\times10^{-68}}{18.2\times10^{-31}\times25\times10^{-14}}$$

$$= \frac{43.9569}{455}\times10^{-23} = 0.0966\times10^{-23} = 9.66\times10^{-25}J$$

$\therefore$ Kinetic energy K in eV $= \dfrac{9.66\times10^{-25}}{1.6\times10^{-19}}eV = 6.03\times10^{-6}\ eV$

[☞ Remaining SAQ are in Page 102]

9 ATOMS

4 Marks

65. State Bohr's postulates of hydrogen atom. **[BMP]**

A: Postulates of Bohr's Theory :

1) **First postulate:** Electron revolves around the positively charged nucleus in circular orbits called stationary orbits. Electrons do not radiate energy while revolving in stationary orbits.

For electron to revolve, Centripetal force= Electrostatic force of attraction

$$\therefore \frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$$

2) **Second postulate:** Angular momentum of revolving electron around the nucleus is an integral multiple of $\frac{h}{2\pi}$.

$\therefore mvr = \frac{nh}{2\pi}$, $n = 1,2,3....$ represents the principal quantum number

3) **Third postulate:** When electron jumps from higher energy level to lower energy level, the energy will be radiated in the form a light photon

$$\Delta E = E_2 - E_1 = h\nu$$

Here E_1 and E_2 are the energies of the initial and the final states,

ν is the frequency of emitted radiation and h is Planck's constant.

---

66. What are the limitations of Bohr's theory of hydrogen atom? **| AP 17,22,24|**

A : Limitations of Bohr's theory of Hydrogen atom :

1) It could not explain the **fine structure of spectral lines** in hydrogen atom.
2) It could not explain the **elliptical orbits** because Bohr assumed the circular orbits.
3) It could not explain the **wave properties** of electron.
4) It could not explain the **splitting of spectra** under **electric and magnetic field.**

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67. What are the observations of Rutherford's α - particle scattering experiment and how these findings supported the existence of atomic nucleus.

A: 1) Observations of Rutherford's α - particle scattering :

i) The majority of the α-particles passed directly through the gold foil without any deflection.

ii) A small fraction of α-particles were deflected by very small angles. (just more than 1°)

iii) Few α-particles were deflected by large angles (more than 90°) and very few (about 1 in 8000) bounced back.

2) Conclusions:

i) Most alpha particles passed straight through the gold foil. It shows that most of the space inside an atom is empty.

ii) Some positively charged alpha particles were deflected by small angles. It shows that the positive charge is concentrated at the centre of the atom.

iii) A tiny particle bounced straight back shows that it must have hit something extremely heavy and dense for which Rutherford called the nucleus of the atom.

By analyzing these findings, Rutherford concluded that an atom consists of a tiny, dense, positively charged the nucleus.

---

68. Write a short note on debroglie's explanation of Bohr's second postulate of quantization?

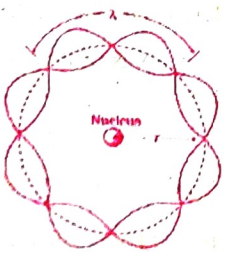
A: 1) Bohr's second postulate: The angular momentum of electron in a stationary orbit is quantised.

Thus, $m\, v_n\, r_n = \frac{nh}{2\pi}$, n=1,2,3, ...(1)

2) de -Broglie's explanation:

de- Broglie explained that the electron in the stationary orbit acts like a particle wave.

So it forms stationary waves in the orbit.



3) The distance travelled by the particle wave along circumference is an integral multiple of wavelength.

Thus, $2\pi r_n = n\lambda$ ....(2)

4) From de Broglie hypothesis, we have $\lambda = \frac{h}{p} = \frac{h}{mv_n}$

From (2), $2\pi r_n = n\left(\frac{h}{mv_n}\right) \Rightarrow m\, v_n\, r_n = \frac{nh}{2\pi}$ .....(3)

Thus, from (1) & (3) we conclude that de Broglie's explanation coincides with the second postulate of Bohr model of atom.

69. Derive expressions for potential and kinetic energies of an electron in any orbit of a hydrogen atom according to Bohr's atomic model.

A : 1) Consider an electron of H-atom of mass m, charge e.

Let it revolve in n^{th} orbit of radius r with a velocity v.

2) For electron to revolve, Centripetal force = Electrostatic force of attraction

$$\Rightarrow \frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_o}\frac{e^2}{r^2} \Rightarrow mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} \quad ...(1)$$

3) From Bohr's 2nd postulate, we have $mvr = \frac{nh}{2\pi} \Rightarrow m^2v^2r^2 = \frac{n^2h^2}{4\pi^2}$ ....(2)

$$\frac{(2)}{(1)} \Rightarrow \frac{m^2v^2r^2}{mv^2} = \frac{n^2h^2 4\pi\varepsilon_0 r}{4\pi^2.e^2} \Rightarrow r = \frac{h^2\varepsilon_0 n^2}{\pi e^2 m} \quad .....(3)$$

a) K.E of electron: We know $K.E = \frac{1}{2}mv^2$

From (1), $K = \frac{1}{2}\left(\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}\right) = \frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$ .....(4)

Substituting value of r from eq(3) in (4) we get

Kinetic Energy $K = \frac{1}{8\pi\varepsilon_0}\frac{e^2.\pi e^2 m}{h^2\varepsilon_0 n^2} = \frac{me^4}{8\varepsilon_0^2 h^2 n^2}$

b) PE of electron: P.E=Potential (V) × charge of electron.

Potential energy $U = \frac{1}{4\pi\varepsilon_0}\frac{e}{r}(-e) = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$ .....(5)

Substituting the value of radius r from (3) in eqn (5) we get

$$U = -\frac{e^2}{4\pi\varepsilon_0}\left(\frac{\pi e^2 m}{h^2\varepsilon_0 n^2}\right) = -\frac{me^4}{4\varepsilon_0^2 h^2 n^2}$$

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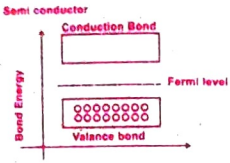
10 SEMI CONDUCTOR ELECTRONICS

4 Marks

70. Describe energy band diagram of intrinsic semiconductors at 0 K.

A: Intrinsic semiconductors:



1) Intrinsic semiconductors are made up of pure form of semi conductor. These atoms are tetravalent.

They behave like insulators at 0K.

2) At 0K, the valence band is completely filled with electrons and conduction band is empty.

Thus there are no free electrons to conduct electricity. Hence they act as insulators at 0K.

3) At 0K there is a small energy gap between the top of the valence band and the bottom of the conduction band. Since, no thermal energy is available at 0 K, no electrons can jump across this gap.

4) At higher temperatures (T > 0K) thermal energy excites some electrons from valence band to the conduction band and leaving equal number of holes there.

---

71. What are n-type and p-type semiconductors?

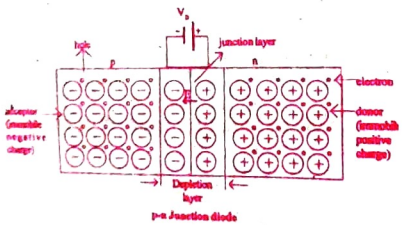
Explain the process of formation of p-n junction diode.

A: 1) n-type semiconductors are silicon or germanium doped with pentavalent impurity like arsenic.

2) p-type semiconductors are silicon or germanium doped with trivalent impurity like indium.

3) Process of formation of p-n junction diode:

When a semiconductor material such as silicon or Germanium crystal is doped such that one side of it is p-type and the other side is n-type then the interface is called p-n junction and the device is called p-n junction diode.



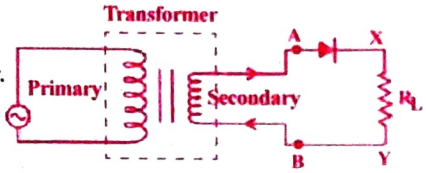
4) When p-n junction is formed, the free electrons which are in higher concentration on n-side diffuse over to p-side and combine with holes and become neutral. Similarly, hole's which are in higher concentration on p-side diffuse over to n-side and combine with electrons and become neutral.

5) This results in the formation of a narrow region on either side of the junction which is free from mobile charge carriers. This region is called **depletion layer.** This layer acts as a potential barrier between p-type and n-type material of diode.

72. Describe how a semiconductor diode is used as a half wave rectifier.

A: **1) Rectification:** The process of converting alternating current (AC) into direct current (DC) is called rectification.

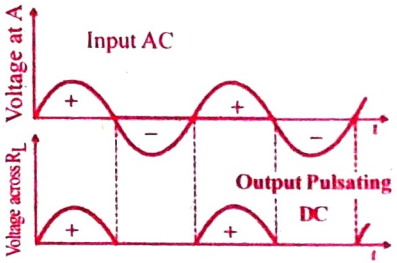
2) Half Wave Rectifier: A circuit which rectifies half of the AC wave is called Half wave rectifier.



3) Diode as a Halfwave Rectifier: A single diode is connected in series to a load resistance R_L in the secondary circuit of a transformer.

4) Working:

- During the positive half cycle, the diode is forward biased and the current flows through the diode.
- During the negative half cycle, the diode is reverse biased and the current does not flow through the diode.
- Thus, half wave is rectified using a single diode.



5) Efficiency of Halfwave rectifier $\eta = \dfrac{0.406R_L}{r_f + R_L}$

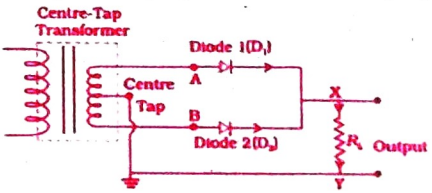
6) Maximum Efficiency of Halfwave rectifier is 40.6%

---

73. What is rectification? Explain the working of a full wave rectifier.

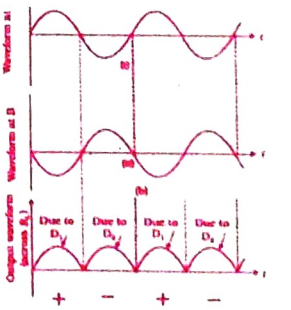
A: **1) Rectification:** The process of converting alternating current (AC) into direct current (DC) is called rectification.

2) Full Wave Rectifier: The circuit which rectifies both half cycles of an AC wave is called Full Wave Rectifier.



3) Working:

- During the positive half cycle, diode D_1 is forward biased & D_2 is reverse biased. Hence current passes through R_L.
- During the negative half cycle, the D_2 is forward biased and D_1 is reverse biased. Hence, current passes through R_L.
- Thus the full wave is rectified using two diodes.



4) Efficiency of full wave rectifier is $\eta = \dfrac{0.812R_L}{r_f + R_L}$

Here r_f = diode forward resistance, R_L = load resistance.

5) Maximum Efficiency of full-wave rectifier is 81.2%

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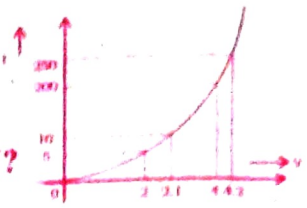
PROBLEMS

74. The I-V characteristics of a p-n junction diode in forward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V?



[BMP]

A: (i) Dynamic resistance at 2V voltage

Voltage at $V_1 = 2V$; current $I_1 = 5mA$; Voltage at $V_2 = 2.1V$; current $I_2 = 10mA$

$\therefore \Delta V = V_2 - V_1 = 2.1 - 2 = 0.1\ V$

$\Delta I = I_2 - I_1 = 10 - 5 = 5 \times 10^{-3} A$ Resistance $Rd_1 = \frac{\Delta V}{\Delta I} = \frac{0.1}{5\times10^{-3}} = 20\Omega$

(ii) Dynamic resistance at 4V voltage.

Voltage at $V_3 = 4V$, current $I_3 = 200mA$; Voltage at $V_4 = 4.2V$, current $I_4 = 250mA$

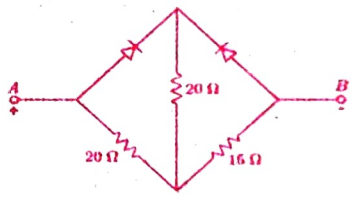
$\Delta V = V_4 - V_3 = 4.2 - 4 = 0.2V$

$\Delta I = I_4 - I_3 = 250 - 200 = 50 = 50 \times 10^{-3} A$

Resistance $R\,d_2 = \frac{\Delta V}{\Delta I} = \frac{0.2}{50\times10^{-3}} = \frac{200}{50} = 4\Omega$ $\therefore$ Ratio $= \frac{Rd_1}{Rd_2} = \frac{20}{4} = 5:1$

---

75. Two ideal diodes are connected in the network as shown in the figure. Then find the equivalent resistance between A and B?



A: 20Ω resistors are in parallel,

$\therefore \frac{1}{R_P} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10} \Rightarrow R_P = 10\Omega$

Equivalent resistance $R_{eq} = R_P + 15\Omega = 10 + 15 = 25\Omega$

---

76. Pure silicon at 300K has an equal number of electrons (n_i) and the hole (n_h) concentrations $5 \times 10^{16}\ m^{-3}$. When it was doped with the indium then the hole concentration (n_h) increases to $4.5 \times 10^{22}\ m^{-3}$ then find the concentration of electrons after doping.

A: Given $n_i = 5\times10^{16} m^{-3}$; $n_h = 4.5\times10^{22} m^{-3}$ $\therefore n_e = \frac{(n_i)^2}{n_h} = \frac{(5\times10^{16})^2}{4.5\times10^{22}} = 5.55\times10^{10} m^{-3}$

---

77. A potential barrier of 0.8V exists across the p-n junction. If the depletion region is 800 nm wide, then what is the intensity of electric field with in this region is?

A: Barrier potential V = 0.8V

Width of depletion region $d = 800\ nm = 800 \times 10^{-9}\ m = 8 \times 10^{-7}\ m$

$\therefore$ Electric field Intensity $E = \frac{V}{d} = \frac{0.8}{8\times10^{-7}} = 10^6\ V/m$

[☞ Remaining SAQs are in Page 103]

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IPE BLUE PRINT OF SR PHYSICS

CHAPTER (Total Marks)	OMQ(1M) 9×1= 9	VSAQ(2M) 14×2 = 28	SAQ(4M) 8×4 = 32	LAQ(8M) 2×8 = 16	Total 85 M
1. Electric Charges and Fields (7 M)	1	2	4	-	7
2. Electrostatic Potential (10 M)	-	2	4+4	-	10
3. Current Electricity (7 M)	1	2	4		7
4*. Moving Charges & Magnetism (10 M)	-	2	-	8	10
5. Magnetism and Matter (7 M)	1	2	4	-	7
6. Electromagnetic Induction (7 M)	1	2	4	-	7
7. Alternating Current (7 M)	1	2	4	-	7
8. Electromagnetic Waves (3 M)	1	2	-	-	3
9*. Ray Optics & Optical Instruments (10 M)	-	2	-	8	10
10. Wave Optics (7 M)	1	2	4	-	7
11. Dual Nature of Radiation & Matter (10 M)	-	2	4+4	-	10
12. Atoms (7 M)	1	2	4	-	7
13*.Nuclei (10 M)	-	2	-	8	10
14. Semiconductor Electronics (7 M)	1	2	4	-	7
Section wise Marks/ Total No.of Q's	9M/9Q	28 M/14Q	32M/12Q	16M/3Q	85 /33Q

☛ * indicates LAQ Chapters.

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3. Current Electricity		65-66	40-42		83
4. Moving Charges & Magnetism		67		16 - 20	
5. Magnetism and Matter		68	43 - 44		84
6. Electromagnetic Induction		69-70	45-46		85
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11. Dual Nature of Radiation		76	51-54		
12. Atoms		77	55-57		89
13. Nuclei		78		27-30	
14. Semiconductor Electronics		79 -80	58-60		90

☛ This chapter index helps faculty to revise lessons chapterwise.

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☛ The Chapters are **Renumbered** as per VSAQ, SAQ, LAQ division.

30 DAYS ACTION PLAN

DAY	CHAPTER	REMARKS
Day-1	1) ELECTRIC CHARGES & FIELDS (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-2	ELECTRIC CHARGES & FIELDS	SAQ
Day-3	2) ELECTRO STATIC POTENTIAL (2M + 4M + 4M = 10 M)	VSAQ+SAQ
Day-4	ELECTRO STATIC POTENTIAL	SAQ
Day-5	3) CURRENT ELECTRICITY (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-6	CURRENT ELECTRICITY	SAQ
Day-7	4) MOVING CHARGES & MAGNETISM (2M + 8M = 10 M)	LAQ
Day-8	MOVING CHARGES & MAGNETISM	VSAQ
Day-9	MOVING CHARGES & MAGNETISM Total Marks in this unit : 34 Marks	LAQ + VSAQ
Day-10	5) MAGNETISM AND MATTER (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-11	MAGNETISM AND MATTER	SAQ
Day-12	6) ELECTRO MAGNETIC INDUCTION (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-13	ELECTRO MAGNETIC INDUCTION	SAQ
Day-14	7) ALTERNATING CURRENT (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-15	ALTERNATING CURRENT	SAQ
Day-16	8) ELECTROMAGNETIC WAVES (1M + 2M = 3 M)	OMQ + VSAQ
Day-17	9) RAY OPTICS (2M + 8M = 10 M)	LAQ
Day-18	RAY OPTICS	VSAQ
Day-19	RAY OPTICS Total Marks in this unit : 34 Marks	LAQ + VSAQ
Day-20	10) WAVE OPTICS (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-21	WAVE OPTICS	SAQ
Day-22	11) DUAL NATURE OF RADIATION (2M + 4M + 4M = 10 M)	VSAQ + SAQ
Day-23	DUAL NATURE OF RADIATION	SAQ
Day-24	12) ATOMS (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-25	ATOMS	SAQ
Day-26	13) NUCLEI (2M + 8M = 10 M)	LAQ + VSAQ
Day-27	NUCLEI	LAQ + VSAQ
Day-28	NUCLEI	REVISION
Day-29	14) SEMI CONDUCTOR ELECTRONICS(1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-30	SEMI CONDUCTOR ELECTRONICS Total Marks in this unit : 41 Marks	SAQ

☞ **30 Days** x 1 Hour Reading Time (or) **3 Days** x 10 Hours Reading Time

BBQ - TIPS

LAQ SCORING CHAPTERS

Imp Chapter: MOVING CHARGES & MAGNETISM

Easy Chapter: NUCLEI

Super Scoring Pages (SSP): 16,17,21,23,27

SAQ SCORING CHAPTERS

Easy Chapters : ELECTRIC CHARGES & FIELDS, ELECTROSTATIC POTENTIAL, CURRENT ELECTRICITY, MAGNETISM AND MATTER

Imp Chapters: ELECTROMAGNETIC INDUCTION, DUAL NATURE, ATOMS

Super Scoring Pages (SSP): 32, 33, 35, 36,37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55,59

VSAQ SCORING CHAPTERS

Easy Chapters: DUAL NATURE, ALTERNATING CURRENT, ELECTRO MAGNETIC WAVES, SEMICONDUCTORS,

Imp Chapters: ELECTRIC CHARGES, ELECTROSTATIC POTENTIAL, MOVING CHARGES, ATOMS, NUCLEI

Super Scoring Pages (SSP):62, 63, 65, 67, 68, 69, 71, 73, 74, 75, 76, 77, 78, 79

BBQ HIGHLIGHTS

- **Quick Memory Bytes → (i) Answer Clarity (ii) Easy Memory .**
- Chapter concept పై Motivation కావాలంటే **Physi beats** మీ కోసం రెడీ! (on QR Code).
- **Rising Stars** కోసం **Level I** Questions, **Shining Stars** కోసం **Level II** Questions (Star Q Plus).
- **30 Days Action Plan : 30 Days** x 1 Hour Reading Time.
- **3 Days Action Plan during Public Exam : 3 Days** x 10 Hours Reading Time.
- **3 Unit Tests to complete the entire Physics syllabus in 3 units.**

☞ **SINCERE SUGGESTION:** After completing this BulletQ, Shining stars are advised to go through the Notes given by your Lecturers; follow the next Higher Level Material.

IMP FORMULAS FOR IPE PROBLEMS

ELECTROSTATIC POTENTIAL

- Potential due to a point charge is $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$
- Capacitance of a capacitor $C = \frac{Q}{V}$
- Capacitance of a parallel plate capacitor is $C = \frac{\varepsilon_0 A}{d}$
- Equivalent capacitance of 2 capacitors in parallel is $C = C_1 + C_2$
- Equivalent capacitance of 2 capacitors in series is given by $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$
- Energy stored in a capacitor is $U = \frac{Q^2}{2C}$ or $U = \frac{1}{2}QV$ or $U = \frac{1}{2}CV^2$

MOVING CHARGES & MAGNETISM

- The magnetic force on a charged particle moving in a magnetic field is $F = qvB \sin\theta$
- Magnetic field $B_a = \frac{\mu_0 I_a}{2\pi d}$
- Force $F_{ba} = B_a I_b l$
- Force per unit length $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$
- Torque = BIAN $\sin\theta$
- Maximum torque, τ_{max} = BIAN
- Shunt Resistance $r_s = \frac{I_G R_G}{I - I_G}$
- Series Resistance $R = \frac{V}{I_G} - R_G$

RAY OPTICS

- **Snell's law of refraction :**
  Refractive index $n_{21} = \frac{\sin i}{\sin r} = \text{constant}$
  (or) $n_1 \sin i = n_2 \sin r$
- **Mirror formula:** $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$
- Combined focal length (f) of thin lenses in contact: $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$
- Power (P) of a lens: $P = \frac{1}{f}$
- Refractive index of material of the prism

$$n_{21} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

For a thin prism, $D_m = (n_{21} - 1)A$

DUAL NATURE OF RADIATION

- Threshold frequency $\nu_0 = \phi_0 / h$
- Einstein's photoelectirc equation $eV_0 = h\nu - \phi$
- Wavelength $\lambda = \frac{hc}{(eV_0 + \phi_0)}$
- Threshold wavelength $\lambda_0 = \frac{hc}{\phi}$
- de-Broglie wave length $\lambda = \frac{h}{p}$
- K.E of electron $K = \frac{h^2}{2m\lambda^2}$
- Einstein's photoelectric equation is
  $K_{max} = h\nu - \phi_0$ or $\frac{1}{2}mv_{max}^2 = h(\nu - \nu_0)$

NUCLEI

- Radius of nucleus $R = R_0 A^{1/3}$
- Mass defect $\Delta M = [Zm_P + (A-Z)m_n] - M$
- Energy $E = mc^2$
- **Nuclear fission:**
  $^{235}_{92}U + ^{1}_{0}n \rightarrow ^{236}_{92}U \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\ ^{1}_{0}n$ + 200MeV
- **Nuclear fusion:**
  $4^1_1H + 4_{-1}e^0 \rightarrow ^4_2He + 2_{-1}e^0 + 2\nu + 6\gamma + 26.7MeV$

SEMI CONDUCTORS

- Half wave rectifier efficiency $\eta = \frac{0.406R_L}{r_f + R_L}$
- Full wave rectifier efficiency $\eta = \frac{0.812R_L}{r_f + R_L}$
- Current gain in a transistor is $\beta = \frac{\Delta I_C}{\Delta I_B}$

★ 8 MARKS LAQ HIT LIST

TOP 10

		Ans-Page
	★★★ MOVING CHARGES & MAGENTISM [8 MARKS] ★★★	
LAQ 1.	Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram?	[☞P 16(1)]
LAQ 2.	Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram.	[☞P 17(2)]
LAQ 3.	Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram.	[☞P 18(3)]
LAQ 4.	State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams.	[☞P 19(4)]
	★★ RAY OPTICS [8 MARKS] ★★	
LAQ 5.	Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram.	[☞P 21(5)]
LAQ 6.	a)Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index. b)Explain the working of optical fiber with a neat diagram.	[☞P 22(6)]
LAQ 7.	Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism.	[☞P 23(7)]
	★★★ NUCLEI [8 MARKS] ★★★	
LAQ 8.	Explain nuclear fission and fusion with suitable nuclear reactions.	[☞P 27(10)]
LAQ 9.	a)What is the source of stellar energy? Explain it with the help of proton-proton cycle. b)Briefly explain the formation of red giant stage of sun.	[☞P 27(11)]
LAQ 10.	Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance?	[☞P 29(13)]

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★ 4 MARKS SAQ HIT LIST

TOP 40

		Ans-Page
	★★★ELECTRIC CHARGES & FIELDS [4 MARKS] ★★★	
SAQ 1.	State and explain Coulomb's inverse square law in electricity.	[☞P 32(14)]
SAQ 2.	Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge.	[☞P 32(15)]
SAQ 3.	Derive an equation for the couple acting on an electric dipole in a uniform electric field.	[☞P 33(16)]
SAQ 4.	Derive an expression for the intensity of the electric field at a point on the axial line of a dipole.	[☞P 33(17)]
SAQ 5.	Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole.	[☞P 34(18)]
SAQ 6.	State Gauss's law in electrostatics and prove it.	[☞P 34(19)]
	★★★ELECTROSTATIC POTENTIAL [4+4 MARKS] ★★★	
SAQ 7.	Derive an expression for the electric potential due to a point charge.	[☞P 35(20)]
SAQ 8.	Derive an expression for the potential energy of an electric dipole placed in a uniform electric field.	[☞P 36(22)]
SAQ 9.	Derive an expression for the capacitance of a parallel plate capacitor.	[☞P 36(23)]
SAQ 10.	Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series.	[☞P 37(24)]
SAQ 11.	Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel.	[☞P 37(25)]
	★★★CURRENT ELECTRICITY [4 MARKS] ★★★	
SAQ 12.	Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge.	[☞P 40(32)]
SAQ 13.	Derive equivalent emf and internal resistance of two cells in series combination.	[☞P 41(33)]
SAQ 14.	Derive equivalent emf and internal resistance of two cells in parallel combination.	[☞P 42(36)]
SAQ 15.	State and explain Kirchhoff rules for an electric network.	[☞P 41(34)]
	★★★MAGNETISM & MATTER [4 MARKS] ★★★	
SAQ 16.	Mention the properties of magnetic field lines.	[☞P 43(37)]
SAQ 17.	Explain the properties of diamagnetic materials.	[☞P 43(38)]
SAQ 18.	Explain the properties of paramagnetic materials.	[☞P 43(39)]
SAQ 19.	How the existence of domains, explain the behavior of ferromagnetic material?	[☞P 44(40)]
	★★★ ELECTRO MAGNETIC INDUCTION [4 MARKS] ★★★	
SAQ 20.	Explain the Faradays law of induction and Lenz's law.	[☞P 45(43)]
SAQ 21.	Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field.	[☞P 45(44)]
SAQ 22.	Explain briefly working of AC generator.	[☞P 46(45)]
SAQ 23.	Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid.	[☞P 46(46)]

★★ ALTERNATING CURRENT[4 MARKS] ★★

SAQ 24. State the principle on which a transformer works. [☞P 47(47)]
Describe the working of a transformer with necessary theory.

SAQ 25. Obtain an expression for the current through an inductor when an AC emf is applied. [☞P 48(49)]

SAQ 26. Obtain an expression for the current in a capacitor when an AC emf is applied. [☞P 48(50)]

★★★ WAVE OPTICS [4 MARKS] ★★★

SAQ 27. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. [☞P 49(51)]

SAQ 28. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [☞P 49(52)]

SAQ 29. Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids. [☞P 50(54)]

★★★ DUAL NATURE OF RADIATION [4+4 MARKS] ★★★

SAQ 30. What are different types of physical processes to obtain free electrons from the metal surface. [☞P 51(55)]

SAQ 31. Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential. [☞P 51(56)]

SAQ 32. What are the laws of photoelectric effect. [☞P 52(57)]

SAQ 33. The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. [☞P 54(61)]

SAQ 34. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? [☞P 54(64)]

★★★ ATOMS [4 MARKS] ★★★

SAQ 35. State Bohr's postulates of hydrogen atom. [☞P 55(65)]

SAQ 36. What are the limitations of Bohr's theory of hydrogen atom? [☞P 55(66)]

SAQ 37. Write a short note on debroglie's explanation of Bohr's second postulate of quantization? [☞P 56(68)]

★★★ SEMI CONDUCTORS [4 MARKS] ★★★

SAQ 38. What are n-type and p-type semiconductors? [☞P 58(71)]
Explain the process of formation of p-n junction diode.

SAQ 39. Describe how a semiconductor diode is used as a half wave rectifier. [☞P 59(72)]

SAQ 40. What is rectification? Explain the working of a full wave rectifier. [☞P 59(73)]

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★ 2 MARKS VSAQ HIT LIST

TOP 70

		Ans-Page
	★★★ ELECTRIC CHARGES & FIELDS [2 MARKS] ★★★	
VSAQ 1.	The electric lines of force do not intersect. Why ?	[☞P 62(78)]
VSAQ 2.	What is meant by the statement charge is quantised ?	[☞P 62(79)]
VSAQ 3.	How many electrons constitute 1C of charge ?	[☞P 62(80)]
VSAQ 4.	What is the importance of Gauss law in electrostatics.	[☞P 62(81)]
VSAQ 5.	What happens to the force between two charges if the distance between them is a) halved b) doubled	[☞P 91(182)]
	★★ELECTROSTATIC POTENTIAL [2 MARKS] ★★	
VSAQ 6.	Can there be electric potential at a point with zero electric intensity? Give an example.	[☞P 63(84)]
VSAQ 7.	Can there be electric intensity at a point with zero electric potential ? Give an example.	[☞P 63(85)]
VSAQ 8.	Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel. a)What is the ratio of charges on each capacitor? b)What is the ratio of potential differences across each capacitor?	[☞P 63(88)]
VSAQ 9.	Three capacitors of capacitances 1 μF, 2μF and 3 μF are connected in series. a)What is the ratio of charges? b)What is the ratio of potential differences?	[☞P 64(89)]
VSAQ 10.	What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled?	[☞P 64(90)]
VSAQ 11.	In a hydrogen atom, the electron and proton are at a distance of 1Å . Find the dipole moment of the system.	[☞P 64(92)]
	★★★ CURRENT ELECTRICITY [2 MARKS] ★★★	
VSAQ 12.	State Ohm's Law and write its mathematical form.	[☞P 65(94)]
VSAQ 13.	Write the vector form of Ohms law. Explain the terms in it.	[☞P 65(95)]
VSAQ 14.	Define temperature coefficient of resistance.	[☞P 65(97)]
VSAQ 15.	Define resistivity or specific resistance.	[☞P 66(98)]
	★★★ MOVING CHARGES & MAGENTISM[2 MARKS] ★★★	
VSAQ 16.	What is the importance of Oersted's experiment?	[☞P 67(103)]
VSAQ 17.	What is the principle of moving coil galvanometer?	[☞P 67(104)]
VSAQ 18.	How do you convert a moving coil galvanometer into an ammeter?	[☞P 67(105)]
VSAQ 19.	How do you convert a moving coil galvanometer into a voltmeter?	[☞P 67(106)]

★★★ MAGNETISM & MATTER [2 MARKS] ★★★

VSAQ 20. Magnetic lines of force don't intersect. Why? [☞P 68(109)]

VSAQ 21. Magnetic lines form continuous closed loops. Why? [☞P 68(110)]

VSAQ 22. Define magnetisation of a material and write its formula. [☞P 68(111)]

VSAQ 23. What is the magnetic moment associated with a solenoid? [☞P 68(112)]

VSAQ 24. Define magnetic susceptibility and relative permeability. [☞P 68(113)]

VSAQ 25. Classify the following materials with regard to magnetism. [☞P 68(114)]
Iron, Platinum, Copper, Magnesium.

★★ELECTRO MAGNETIC INDUCTION [2 MARKS] ★★

VSAQ 26. State Faraday's law of electromagnetic induction. [☞P 69(116)]

VSAQ 27. State Lenz's law. [☞P 69(117)]

VSAQ 28. Define magnetic flux. [☞P 69(118)]

VSAQ 29. What is the working principle of A.C. generator? [☞P 69(119)]

VSAQ 30. What is self-induction? [☞P 70(121)]

VSAQ 31. What is mutual induction? [☞P 70(122)]

★★★ ALTERNATING CURRENT [2 MARKS] ★★★

VSAQ 32. What is transformer ratio? [☞P 71(125)]

VSAQ 33. A transformer converts 200 V AC into 2000 V AC. Calculate the number of turns in the secondary if the primary has 10 turns. [☞P 71(126)]

VSAQ 34. Write the expression for the reactance of (i) an inductor and (ii) a capacitor. [☞P 71(127)]

VSAQ 35. A Capacitor has a capacitance of $\frac{1}{\pi}\mu F$. Find its reactance for a frequency of 50 Hz. [☞P 71(128)]

VSAQ 36. A pure inductor of 21mH is connected to a source of 220V. Find the inductive reactance if the frequency of the source is 50Hz. [☞P 71(129)]

VSAQ 37. On what principle metal detector works? Explain it. [☞P 72(134)]

★★ ELECTRO MAGNETIC WAVES [2 MARKS] ★★

VSAQ 38. Give two uses of infrared rays. [☞P 73(136)]

VSAQ 39. What are the applications of microwaves? [☞P 73(137)]

VSAQ 40. What is the principle of production of electromagnetic waves? [☞P 73(138)]

VSAQ 41. The charging current for a capacitor is 0.6 A. What is the displacement current across its plates? [☞P 73(140)]

VSAQ 42. What is the wavelength range of X-rays? [☞P 73(141)]

VSAQ 43. If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon? [☞P 94(197)]

★★★ RAY OPTICS [2 MARKS] ★★★

VSAQ 44. Define power of a convex lens. What is its unit? |☞P 74(142)|
VSAQ 45. Two thin lens of power +4D and -2D are in contact. What is the focal length of the combination? |☞P 74(143)|
VSAQ 46. What is the critical angle? Give one application of total internal reflection? |☞P 74(145)|
VSAQ 47. A small angled prism of 4° deviates a ray through 2.4°. Find the refractive index of the prism. |☞P 74(147)|

★★★ WAVE OPTICS [2 MARKS] ★★★

VSAQ 48. State Malus' law. Write the expression of Malus' law. |☞P 75(148)|
VSAQ 49. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves? |☞P 75(149)|
VSAQ 50. If a wave undergoes reflection and refraction, what will be the phase change in each case? |☞P 75(151)|
VSAQ 51. Compare Maxwell's Electromagnetic theory with Huygen's theory of light? |☞P 75(152)|
VSAQ 52. Explain the statement *'light added to light can produce darkness'* |☞P 97(209)|

★★★ DUAL NATURE [2 MARKS] ★★★

VSAQ 53. What are cathode rays? |☞P 76(153)|
VSAQ 54. What is photoelectric effect? |☞P 76(155)|
VSAQ 55. Write down Einstein's photoelectric equation. Explain the terms in it? |☞P 76(158)|
VSAQ 56. Write down de-Broglie's relation and explain the terms therein. |☞P 76(159)|
VSAQ 57. Define 1 electron volt. |☞P 76(160)|

★★★ ATOMS [2 MARKS] ★★★

VSAQ 58. Define impact parameter. |☞P 77(162)|
VSAQ 59. What is the difference between α - particle and helium atom? |☞P 77(164)|
VSAQ 60. Why did most of the α - particles pass through the gold foil undeflected in Rutherford's scattering? |☞P 77(165)|

★★★ NUCLEI [2 MARKS] ★★★

VSAQ 61. What are isotopes and isobars? |☞P 78(168)|
VSAQ 62. What are isotones and isomers? |☞P 78(169)|
VSAQ 63. Why are nuclear fusion reactions called thermo nuclear reactions? |☞P 78(170)|
VSAQ 64. Compare the radii of the nuclei of mass numbers 27 and 64. |☞P 78(171)|
VSAQ 65. What is the significance of a mass defect in a nucleus? |☞P 78(172)|

★★★ SEMI CONDUCTORS [2 MARKS] ★★★

VSAQ 66. What are intrinsic and extrinsic semiconductors ? |☞P 79(173)|
VSAQ 67. What is an n-type semiconductor? What are the majority and minority charge carriers in it? |☞P 79(174)|
VSAQ 68. What is a p-type semiconductor ? What are the majority and minority charge carriers in it? |☞P 79(175)|
VSAQ 69. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? |☞P 79(176)|
VSAQ 70. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased? |☞P 80(180)|

BBQ turns, "I can't to I can".

RISING STARS' MOST FAVOURITE SECTION

LAQ

SECTION-D

Small book, Big Marks.

LAQ CHAPTERS

16 MARKS - 12 PAGES

★★★	1.	**MOVING CHARGES & MAGNETISM** (8 Marks)	16 - 20
★★	2.	**RAY OPTICS** (8 Marks)	21 - 26
★★★	3.	**NUCLEI** (8 Marks)	27 - 30

WOW TIP 1: First chapter is a 'must and should' read chapter.
It is better to practice all the 4 LAQs.

WOW TIP 2: In Ray Optics Q5, Q6 and also Q7 are easy and important.

WOW TIP 3: Q10 & Q11 are the easiest of all LAQs. Practice all the 3 tag problems of Nuclei.

☞ Practice the **diagrams** perfectly in these LAQs, as they consume little more time in the exam.

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Super Scoring Pages

★★★SSP★★★

Page 16,17,21,23,27

Total SS Pages : 5

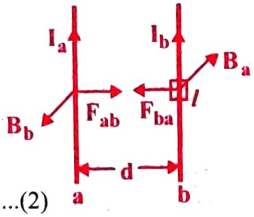
1 MOVING CHARGES & MAGNETISM

8 Marks

1. Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? [BMP]

A: (a) Force between two parallel conductors carrying Current:

1) Consider two straight parallel conductors a, b carrying currents I_a, I_b separated by a distance d .



2) Magnetic field produced by conductor a is $B_a = \frac{\mu_0 I_a}{2\pi d}$ .....(1)

3) Force on a segment l of b by magnetic field B_a is $F_{ba}=B_a I_b l$ ....(2)

From (1) & (2), $F_{ba} = \left(\frac{\mu_0 I_a}{2\pi d}\right) I_b l = \frac{\mu_0 I_a I_b l}{2\pi d}$.....(3)

The direction of this force F_{ba} is towards conductor a.

4) Similarly, the force acting on the conductor a by magnetic field B_b is $F_{ab}=\frac{\mu_0 I_a I_b l}{2\pi d}$ ....(4)

The direction of this force F_{ab} is towards conductor b.

5) From (3) & (4), we get $F_{ab}= F_{ba}=\frac{\mu_0 I_a I_b l}{2\pi d}$ ....(5)

From (5), force per unit length is $\frac{F}{l}=\frac{\mu_0 I_a I_b}{2\pi d}$

(b) Ampere: One ampere is the value of steady current when two infinitely long parallel conductors, carrying the same current separated by a distance of 1m in vacuum produces a force of 2×10^{-7} Nm^{-1} on each conductor.

QUICK MEMORY BYTES

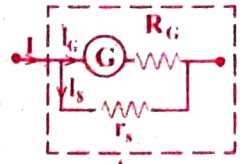
1) Imagine two friends a,b (wires) walking side by side .
2) Friend a creates a magnetic field B_a around him.
3) Friend b feels a force of attraction F_{ba} due to a.
4) Similarly, friend a also feels a force of attraction F_{ab} due to b.
5) Infact, the two forces are equal i.e $F_{ba}= F_{ab}$

Formula Ladder

- $B_a = \frac{\mu_0 I_a}{2\pi d}$
- $F_{ba}=B_a I_b l$
- $\frac{F}{l}=\frac{\mu_0 I_a I_b}{2\pi d}$

2. Explain how can a Moving coil galvanometer be converted into
(a) Ammeter (b)Voltmeter with a neat diagram.

A : (a) Conversion of Galvanometer into Ammeter:



A
(a) Ammeter

1) Galvanometer (G) measures only small currents.

2) To measure a large current (I), a galvanometer (G) is converted into an ammeter (A) by connecting a very small shunt resistance (r_s) parallel to it.

3) Then current splits and most of the current passes through r_S.

Derivation of Shunt Resistance (r_S) :

4) From the Kirchoff's 1st law, max. current $I = I_G + I_S \Rightarrow I_S = I - I_G$

5) Galvanometer and shunt are in parallel, so the potential difference across them is same:

$$\therefore I_G R_G = I_S r_S \Rightarrow I_G R_G = (I - I_G) r_S \Rightarrow r_S = \frac{I_G R_G}{I - I_G}$$

This gives the value of 'shunt resistance' required to convert galvanometer into an ammeter.

(b) Conversion of Galvanometer into Voltmeter:

V
(b) Voltmeter

1) Galvanometer (G) measures only small voltages.

2) To measure a large voltage (V), a galvanometer (G) can be converted into a voltmeter by connecting a high resistance (R) in series with it.

3) This high resistance limits the current and makes the total resistance of the voltmeter very large.

Derivation of Series Resistance (R):

4) Resistance of G is R_G and current through G is I_G (at full scale deflection)

5) **From Ohm's law:** Voltage $V = I_G(R_G + R) \Rightarrow R_G + R = \frac{V}{I_G} \Rightarrow R = \frac{V}{I_G} - R_G$,

This gives the value of 'high resistance' required to convert galvanometer into an voltmeter.

QUICK MEMORY BYTES

GA→ SR P

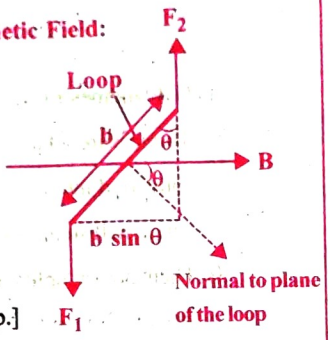
Galvanometer to Ammeter → Small shunt Resistance in Parallel

GV→ HR S

Galvanometer to Voltmeter → High Resistance in Series

3. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram.

A: (a) Torque on a current carrying Loop in a Uniform Magnetic Field:



1) Consider a rectangular current loop of length l, breadth b is suspended in a magnetic field B .
2) The magnetic forces on the arms are given by $F_1 = F_2 = Bil$
3) The two forces constitute a couple.
4) The perpendicular distance between the forces is $b\sin\theta$. [θ is the angle between the field and the normal to the loop.]
5) Torque = Force × Perpendicular distance

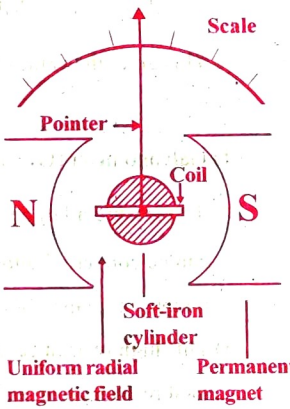
$\tau = Bil \times b \sin\theta = B\,i\,A\sin\theta$ [$\because$ Area of the loop $lb = A$]

(b) Construction and working of MCG:

The current-carrying coil placed in uniform magnetic field experience a torque.

Construction:

1) MCG consists of a rectangular coil with a pivot arrangement in a strong radial magnetic field NS.
2) A 'soft iron cylinder' is arranged at the 'centre of the coil' to produce the uniform magnetic field.
3) A spring is arranged to the lower end of the coil.



Moving coil galvanometer

Working: If the current I to be measured is passed through the MCG, the coil gets a deflection θ.

Deflecting torque acting on the coil is $\tau = BIAN$ ....(1)

Restoring torque developed in the suspension is $\tau_r = C\theta$....(2)

Here C is the torsional torque.

At equilibrium position, the deflecting magnetic torque = restoring torque due to spring

From (1) & (2), $BIAN = C\theta \Rightarrow I = \frac{C\theta}{BAN}$. Put $\frac{C}{BAN} = K$

Hence, $I = K\theta \Rightarrow I \propto \theta$ Here K is called the constant of the galvanometer.

The relation $I = K\theta$ explains the principle of MCG.

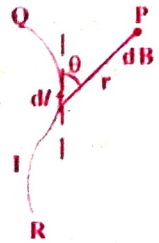
By determining the values of K and θ, the current through MCG can be calculated.

FORMULA BOX

- $F_1 = F_2 = Bil$, $\tau = B\,i\,A\sin\theta$
- $BIAN = C\theta \Rightarrow I = \frac{C\theta}{BAN}$; $\frac{C}{BAN} = K \Rightarrow I = K\theta$

4. State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams.

A: Biot-Savart Law: The magnitude of magnetic field induction due to small element of current carrying conductor is (i) directly proportional to the strength of the current (ii) length of the element (iii) sine of the angle between position vector and the element and (iv) inversely proportional to the square of the distance of the point from the element.



Explanation: Suppose RQ is a conductor carrying a current I.

dl is a small length of element on the conductor.

The distance of a point P from dl is r.

Let θ be the angle between $\overline{dl}$ and radius vector $\overline{r}$.

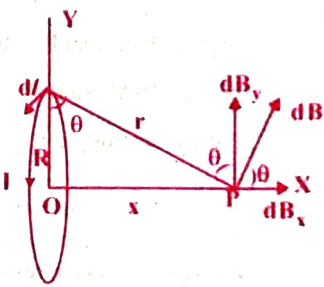
The magnetic induction due to dl is dB.

From Biot-savart law, we have

(i) $dB \propto I$ (ii) $dB \propto dl$ (iii) $dB \propto \sin\theta$ (iv) $dB \propto \frac{1}{r^2}$

Thus, $dB \propto \frac{Idl\sin\theta}{r^2}$ (or) $dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$ Here $\frac{\mu_0}{4\pi} = 10^{-7}\ Hm^{-1}$

Magnetic Induction on the axis a coil:



1) Consider a circular loop of radius R carrying a current I.

2) Let P be a point on the OX- axis at a distance x from its centre O.

3) Consider a small element dl carrying the current I on the coil.

4) The distance of P from the element dl is r.

5) From the Biot-Savart law, $dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$, [dB is the magnetic induction at P due to dl]

$\therefore dB = = \frac{\mu_0}{4\pi}\frac{Idl}{r^2}$ [∵ Angle between dl and r is 90°.]

$\Rightarrow dB = \frac{\mu_0}{4\pi}\frac{Idl}{(x^2+R^2)}$ [From the figure, $r^2 = x^2 + R^2$]

$\Rightarrow B = \int_0^{2\pi R} dB\cos\theta = \frac{\mu_0}{4\pi}\frac{I}{(x^2+R^2)}\cos\theta\int_0^{2\pi R} dl$ [B is the resultant M.I at 'P' due to circular loop]

6) From the figure, $\cos\theta = \frac{R}{r} = \frac{R}{(x^2+R^2)^{1/2}}$ & for the circular coil $\int_0^{2\pi R} dl = 2\pi R$

$\therefore B = \frac{\mu_0}{4\pi}\frac{I}{(x^2+R^2)}\frac{R}{(x^2+R^2)^{1/2}}\times 2\pi R \Rightarrow B = \frac{\mu_0 IR^2}{2(x^2+R^2)^{3/2}}$

If the coil has N turns, magnetic filed induction $B = \frac{\mu_0 NIR^2}{2(x^2+R^2)^{3/2}}$

TAG PROBLEMS

P 1. A current of 5A flows through each of two parallel long wires, the wires are 2.5 cms apart. Calculate the force acting per unit length of each wire. [BMP]

Sol: Current in each wire $I_1 = I_2 = 5A$

Distance between the wires r = 2.5 cm = 0.025 m. Also $\mu_0 = 4\pi\times10^{-7}$

Force per unit length $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r} = \frac{[4\pi\times10^{-7}][5][5]}{2\pi(0.025)} = 2\times10^{-4}\ Nm^{-1}$

P 2. A coil of area $10^{-2}m^2$ and contains 10^6 turns is used in a galvanometer. What is the maximum torque on it if the galvanometer carrying a current of 10^{-4} A is placed in a magnetic field of 10^2 T?

Sol: Given area of the coil $A = 10^{-2}m^2$, Number of turns $N = 10^6$, Current $I = 10^{-4}$ A

Magnetic induction $B = 10^2$ T, Torque will be maximum when $\theta = 90° \Rightarrow \sin 90° = 1$

Torque = BIAN sinθ

$\therefore$ Maximum torque, τ_{max} = BIAN = $10^2 \times 10^{-4} \times 10^{-2} \times 10^6$ = 100Nm

---

P 3. A Moving coil galvanometer, having a resistance of 1 ohm can measure a current 10^{-6} A. What is the resistance of shunt required to measure a current 1A ?

Sol: Given galvanometer, resistance $R_G = 1\Omega$, Current through galvanometer $I_G = 10^{-6}$ A,

Current to be measured, I= 1A, r_s = ?

Shunt Resistance $r_s = \frac{I_G R_G}{I - I_G} = \frac{10^{-6}\times1}{1-10^{-6}} \approx 10^{-6}\Omega$ $(\because 1-10^{-6} \approx 1)$

☞ Few more **Tag Problems** suggested by your **Faculty.**

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2 RAY OPTICS

8 Marks

5. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram.

A: Derivation of Mirror Equation :

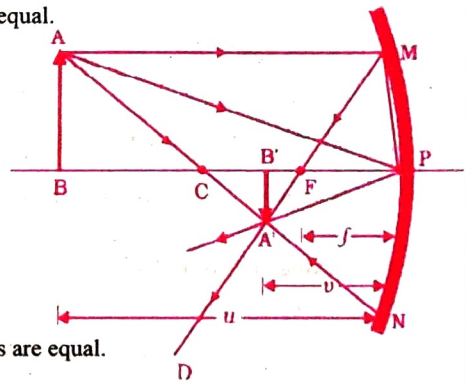
1) Consider a concave mirror with pole P, focus F, and centre of curvature C.

2) AB is an object in front of the mirror. A'B' is its image formed by the mirror.

PF = f = focal length of concave mirror; PB' = v = image distance; PB = u = object distance.

3) In the figure, Δ FB'A', ΔMFP are similar triangles.
So, the ratio of their corresponding sides are equal.

$$\therefore \frac{B'A'}{MP} = \frac{B'F}{FP}$$



4) But from the figure, MP = AB

$$\therefore \frac{B'A'}{AB} = \frac{B'F}{FP} \quad \text{.....(1)}$$

5) Δ A'B'P, ΔABP are also similar triangles.

Hence, the ratio of their corresponding sides are equal.

$$\therefore \frac{B'A'}{AB} = \frac{B'P}{BP} \quad \text{.....(2)}$$

6) From (1) and (2), $\frac{B'F}{FP} = \frac{B'P}{BP}$

7) But from the figure, B'F = B'P–FP $\quad \therefore \frac{B'P - FP}{FP} = \frac{B'P}{BP}$ .....(3)

8) From sign convention, distances from P against the direction of incident light are negative.
Here B'P = –v, FP = –f, BP = –u

9) From (3), we get $\frac{-v-(-f)}{-f} = \frac{-v}{-u} \Rightarrow \frac{v-f}{f} = \frac{v}{u} \Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

This is the **mirror equation.**

10) **Linear Magnification (m) :**
It is the ratio of height of the image (h') to height of the object (h) $\therefore m = \frac{h'}{h}$

In the case of a mirror with proper sign convention, we have $m = -\frac{v}{u}$

Hence, we conclude that a **real inverted image** is formed by a concave mirror.

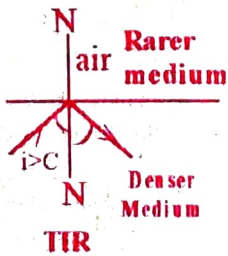
6. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index.

b) Explain the working of optical fiber with a neat diagram.

A: a) Total Internal Reflection (TIR):

1) When a light ray is passing from denser medium to rarer medium and if the angle of incidence is greater than its critical angle ($i > i_C$), it gets reflected into the same denser medium without any refraction. This is called total internal reflection.

Relation between critical angle and refractive index:



2) Let n_2 be the refractive index of the denser medium
n_1 be the refractive index of the rarer medium .

3) From Snell's law, $\frac{n_2}{n_1} = \frac{\sin r}{\sin i_C} = \frac{1}{\sin i_C}$ $(r = 90° \Rightarrow \sin 90° = 1)$

4) Equation for critical angle is $n_{21} = \frac{1}{\sin i_C}$ (or) $\sin i_C = \frac{1}{n_{21}}$

where n_{21} = refractive index of the denser medium with respect to rarer medium.

b) Working of optical fiber:

1) An optical fibre works on the principle of **TIR.**

2) Optical fibre consists of a **core and cladding.** The refractive index of the material of the core is higher than that of the cladding.

Optical fibre

3) When a light signal is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end.

4) Since light undergoes TIR at each stage, there is no appreciable loss in the intensity of the light signal.

5) Optical fibres are fabricated such that light reflected at one side of inner surface strikes the other at an angle larger than the critical angle. Even if the fibre is bent, then also light can easily travel along its length.

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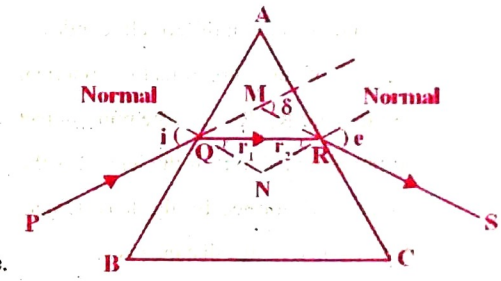
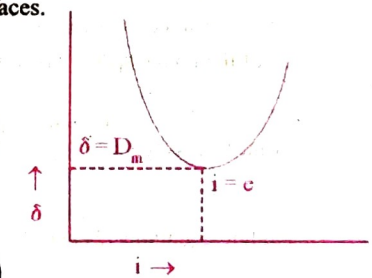
7. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. [BMP]

A: (a) Angle of incidence (i) , Angle of minimum deviation (D_m):



1) Consider a triangular glass prism ABC.
   A = Refracting angle of the prism.
   n_{21}= Refractive index of glass w.r.to air.
   PQ = Incident ray.
   QR = Refracted ray in the prism.
   RS = Emergent ray
2) The normals to AB and AC meet at N.
   i = Angle of incidence at the first surface.
   r_1 = Angle of refraction at the first surface.
   r_2 = Angle of refraction at the second surface
   e = Angle of emergence.
   δ = Angle of deviation (angle between PQ and RS).
3) In the quadrilateral AQNR , $A + 90° + N + 90° = 360° \Rightarrow A + N = 180°$ .....(1)
4) In Δ QNR, $r_1 + r_2 + N = 180°$ .....(2)
5) Equating eq. (1) and (2), $r_1 + r_2 + N = A+N \Rightarrow r_1 + r_2 = A$ .....(3)
6) The total deviation δ = sum of deviations at the two faces.
   $\therefore \delta = (i - r_1) + (e - r_2)$
   $\Rightarrow \delta = i + e - (r_1 + r_2) = i + e - A$ [From (3)]
   $\Rightarrow \delta + A = i + e$ .....(4)
7) A graph is drawn between angle of incidence i on the x-axis and angle of deviation δ on the y-axis.
8) From graph, we get $i = e$, $r_1 = r_2 = r$, $\delta = D_m$



From (3) $r_1 + r_2 = A \Rightarrow r + r = A \Rightarrow 2r = A \Rightarrow r = \left(\frac{A}{2}\right)$ .....(5)

From(4) $\delta + A = i + e \Rightarrow D_m + A = i + i \Rightarrow 2i = A + D_m \Rightarrow i = \left(\frac{A + D_m}{2}\right)$.....(6)

This is the equation for angle of incidence (i) and also angle of minimum deviation (D_m).

(b) Prism formula for refractive index (n_{21})of prism:

When the light ray is passing from medium 1 (air) to medium 2 (glass),

the refractive index (n_{21}) of medium 2 with respect to medium 1 is given by $n_{21} = \frac{\sin i}{\sin r}$ .....(7)

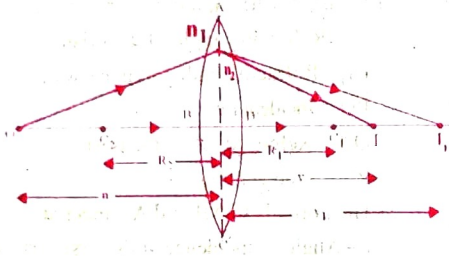
From (5) & (6), we get $n_{21} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

8. Derive the lens makers formula with suitable diagram and thereby derive the lens formula.

A: Lens makers formula & Thin lens formula:

1) Consider a thin double convex lens of material of refractive index n_2. placed in a medium of refractive index n_1.

2) The radii of curvature of its two spherical surfaces are R_1, R_2. O is the point object on the principal axis of the lens . I is the image of O formed by the lens. Object distance is u Image distance is v.



3) The general formula of refraction for curved surface is $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ .....(1)

4) **For first surface:** Object distance =u, image distance =v, RI of the object space =n_1, RI of the image space =n_2, Radius of curvature = R_1

From (1), we get $\frac{n_2}{v_1}-\frac{n_1}{u}=\frac{n_2-n_1}{R_1}$ .....(2).

5) **For second surface:**Object distance =v_1, image distance =v, RI of the object space =n_2, RI of the image space =n_1, Radius of curvature = R_2

From (1), we get $\frac{n_1}{v}-\frac{n_2}{v_1}=\frac{n_2-n_1}{-R_2}$ .....(3)

Adding equations (2) and (3), we get $\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$.....(4)

6) If the object is at infinity (u = ∞), its image is formed at focus (v = f).

Putting these in equ (4), we get $\frac{n_1}{f}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ .....(5)

Dividing it by n_1, we get $\frac{1}{f}=(n_{21}-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ ---(5) where $n_{21}=\frac{n_2}{n_1}$

This is the **Lens maker's formula.**

By comparing (4) and (5), we get $\frac{n_1}{v}-\frac{n_1}{u}=\frac{n_1}{f} \Rightarrow \frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ .....(6)

This is the **Thin lens formula.**

9. With a neat labelled diagram explain the formation of image in a simple microscope. Derive the expression for magnification when the image is formed at

i) near point ii) at infinity.

A: Simple Microscope:

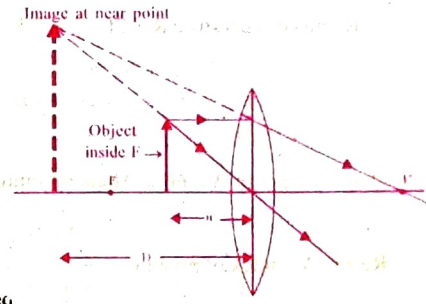
1) A simple microscope or magnifying glass is a convex lens of small focal length.

2) In this, the object is placed between the principal focus F and the optic centre of the convex lens.

3) The light ray coming from the object parallel to principal axis refracts at lens and passes through second principal focus F'.

4) Another light ray from the object passes through the optic centre of the lens undeviated.

5) These two rays meet on the back side of the object to form the image.



Simple microscope image at near point

Hence, virtual, erect and magnified image is formed on the same side of object at D = 25 cm.

6) Expression for Magnification : It is the ratio of distance of image to the distance of object.

$$m = \frac{\text{Distance of image}}{\text{Distance of object}} = \frac{D}{u}$$

7) From Len's formula, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{f} = -\frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{u} = \frac{1}{v} + \frac{1}{f}$

$$\therefore m = D\left(\frac{1}{v} + \frac{1}{f}\right) = \frac{D}{v} + \frac{D}{f}$$

i) At near point, v=D

$$\therefore m = \frac{D}{v} + \frac{D}{f} = \frac{D}{D} + \frac{D}{f} = 1 + \frac{D}{f}$$

ii) At far point , v=∞

$$\therefore m = \frac{D}{v} + \frac{D}{f} = \frac{D}{\infty} + \frac{D}{f} = \frac{D}{f}$$

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TAG PROBLEMS

P1. An object is placed at a distance of 15 cm, from a convex mirror and image is formed at a distance of 5 cm from the mirror. Calculate the radius of curvature of the mirror.

Sol: Here, object distance u = –15cm, Image distance v = +5cm

From mirror formula $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}=\frac{1}{5}+\frac{1}{(-15)}=\frac{2}{15}\Rightarrow f=\frac{15}{2}=7.5\,\text{cm};$

$\therefore$ Radius of curvature R = 2f = 2 × 7.5 = 15 cm

P2. Calculate the refractive index of material of an equilateral prism for which angle of minimum deviation is 60°.

Sol: Angle of prism $A=60^{\circ}$, Angle of minimum deviation $D_m=60^{\circ}$

Refractive index of prism $\mu=\frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}\Rightarrow\mu=\frac{\sin\left(\frac{60^{\circ}+60^{\circ}}{2}\right)}{\sin\left(\frac{60^{\circ}}{2}\right)}=\frac{\sin 60^{\circ}}{\sin 30^{\circ}}=\frac{\sqrt{3}}{2}\times\frac{2}{1}=\sqrt{3}=1.732$

P 3. A convex lens has 20 cm focal length in air. What is its focal length in water? (Refractive index of air-water = 1.33 refractive index for air-glass = 1.5.)

Sol: **Lens maker's formula :** $\frac{1}{f}=(n_{21}-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

(i) When the lens is in air, f = +20 cm, $n_{21}=\frac{n_2}{n_1}=1.5,$

$\therefore\frac{1}{20}=(1.5-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=(0.5)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.....(1)

(ii) When the lens is put in water, $n_{21}=\frac{n_2}{n_1}=\frac{1.5}{1.33}$, f = ?

$\therefore\frac{1}{f}=\left(\frac{1.5}{1.33}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.....(2)

Dividing equ (1) by equ (2), we get $\frac{f}{20}=\frac{0.5}{\left(\frac{1.5}{1.33}-1\right)}\Rightarrow f=\frac{20\times0.5\times1.33}{1.5-1.33}=\frac{13.3}{0.17}=78.2$ cm

Thus, when a lens is immersed in water, its focal length becomes roughly 4 times to its focal length in air.

☞ Few more **Tag Problems** suggested by your **Faculty.**

3 NUCLEI

8 Marks

10. Explain nuclear fission and fusion with suitable nuclear reactions. **[BMP]**

A : **I) Nuclear fission** is the splitting of a heavy nucleus into two or more fragments with the release of large energy .

1) It is the source of energy in nuclear reactors.
2) It takes place at ordinary temperature.
3) This principle is used in atom bomb.
4) The energy produced in the fission is low when compared to fusion.

Ex: The fission of Uranium-235 : $_{92}U^{235} + {}_0n^1 \rightarrow {}_{92}U^{236} \rightarrow {}_{56}Ba^{144} + {}_{36}Kr^{89} + 3\,{}_0n^1 + 200MeV$

II) Nuclear fusion is the fusing of two or more light nuclei to form a single nucleus with the release of extremely large energy.

1) It is the source of energy in the stars.
2) It takes place at very high temperature of the order 10^7 K.
3) This principle is used in hydrogen bomb.
4) The energy produced in the fusion is 7 times that of fission.

Ex : In the Sun 4 hydrogen nuclei fuse together to form a helium nucleus.

$4\,{}_1H^1 + 4\,{}_{-1}e^0 \rightarrow {}_2He^4 + 2\,{}_{-1}e^0 + 2\nu + 6\gamma + 26.7MeV$

---

11. a) What is the source of stellar energy? Explain it with the help of proton-proton cycle.
b) Briefly explain the formation of red giant stage of sun.

A: **a) Source of stellar Energy:** The energy from the sun and other stars is called stellar energy. The source of stellar energy is thermo nuclear fusion.

Nuclear fusion in Sun, stars takes place in multi steps in which the hydrogen is burn to form helium.

Proton-Proton Cycle: The nuclear reactions of proton-proton cycle are as follows.

$${}_1H^1 + {}_1H^1 \rightarrow {}_1H^2 + {}_{+1}e^0 + \bar{\upsilon} + Q_1$$

$${}_1H^1 + {}_1H^1 \rightarrow {}_1H^2 + {}_{+1}e^0 + \bar{\upsilon} + Q_2$$

$${}_1H^2 + {}_1H^1 \rightarrow {}_2He^3 + Q_3$$

$${}_1H^2 + {}_1H^1 \rightarrow {}_2He^3 + Q_4$$

$${}_2He^3 + {}_2He^3 \rightarrow {}_2He^4 + {}_1H^1 + {}_1H^1$$

$$4\,{}_1H^1 \rightarrow {}_2He^4 + 2({}_{+1}e^0) + 2\bar{\upsilon} + 26.7\,MeV$$

Thus 4 hydrogen atoms combine to form a helium atom with a release of 26.7 MeV energy.

b) Red giant stage of the Sun:

After about **5 billion years**, most of the **hydrogen fuel** in the Sun's core gets exhausted.

- Then nuclear fusion in the core **slows down**. The **core contracts** and becomes **hotter**.
- Hydrogen starts fusing in a **shell around the core**.
- This releases a lot of energy, causing the **outer layers to expand**.
- As the Sun expands, its **surface cools** and looks **red**.

Thus, the Sun becomes a **Red Giant**—a **very large, cool, and bright (luminous) star**.

12. a) Explain the composition of nucleus and express the atomic masses of nucleons in terms of atomic mass units.

b) Draw and explain the potential energy curve of pair of nucleons.

Show that the density of nucleus doesn't depend on mass number A.

A: a) **Composition of Nucleus:**

- Nucleus consists of positively charged protons and neutrons.
- Charge of proton = $+1.6\times10^{-19}$ C
- The number of protons in the nucleus is called the **atomic number** (Z)
- The no.of protons and neutrons in the nucleus is called **mass number A.**
- **Atomic Masses of Nucleons in atomic mass unit (u):**

  Mass of Proton: m_p=1.007276 u

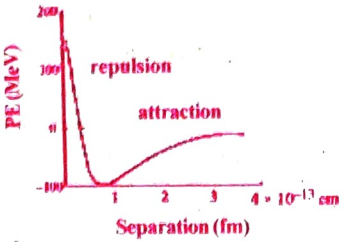
  Mass of Neutron: m_n=1.008665 u

b) **Potential energy curve of nucleons:**

The PE curve shows how potential energy between two nucleons changes with distance between them.

X-axis →Separation between two nucleons in fm.

Y-axis→ Potential energy in MeV.



The graph shows the following facts :

i) **Very small distances - Repulsive region (<0.5fm):**

Nuclear force is strongly repulsive. Potential energy is large and positive.

ii) **Intermediate distances - Attractive region (0.5 to 2 fm):**

Nuclear force is attractive. Potential energy is negative.

Point of stable equilibrium is attained in this region.

iii) **Large distances (2 to 3 fm):**

Nuclear force becomes very n :gligible. Potential energy becomes almost zero.

- **Derivation of density (ρ):**

Mass of 'A' nucleons M = mA, where m= mass of each nucleon

Volume of the nucleus is $V=\frac{4}{3}\pi R^3=\frac{4}{3}\pi R_0^3 A \quad [\because R=R_0A^{1/3}]$

$\therefore$ **Density of Nucleus is** $\rho=\frac{M}{V}=\frac{mA}{\frac{4\pi}{3}R_0^3A}=\frac{3m}{4\pi R_0^3}$

Thus, the **density of nucleus** doesn't depend on mass number A.

13. Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance?

A : **1) Mass Defect (ΔM)** is the difference between the total mass of all the nucleons of a nucleus and the actual mass of a nucleus (M) .

Formula: $\Delta M = [Zm_p + (A-Z)m_n] - M$ where Zm_p = total mass of protons,

$(A-Z)m_n$ = total mass of neutrons, M = mass of nucleus.

2) Binding Energy is the energy required to bind the nucleons in the nucleus or the energy required to break the nucleus into its constituent protons and neutrons.

From $E = mc^2$ we can write Binding energy $B.E = \Delta M\, c^2$

Here ΔM is the mass defect and c is the velocity of light. Also 1 amu = 931.5 MeV

$\therefore$ $B.E = \Delta M \times 931.5$ MeV

3) Binding Energy per Nucleon ($\overline{B.E}$) is the binding energy of a nucleus (B.E) divided by its mass number (A) . **Formula** $\overline{B.E} = B.E/A$

4) Variation of binding energy per nucleon with mass number :

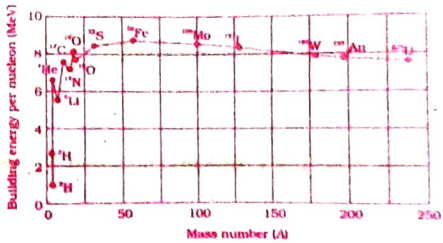
The graph shown in the figure explains how the binding energy per nucleon ($\overline{B.E}$) varies with mass number (A).

5) The graph shows the following facts :

i) $\overline{B.E}$ is constant from A = 28 to A = 138

ii) $\overline{B.E}$ =8.75 MeV is maximum for Fe-56 and $\overline{B.E}$ =7.6 MeV lowest for U-238.

iii) $\overline{B.E}$ is low for both light nuclei (A < 28) and heavy nuclei (A > 138).



6) Significance: Greater binding energy per nucleon means greater stability to the nucleus.

The light nuclei of A < 28 and heavy nuclei of A > 138 are unstable because their $\overline{B.E}$ is low. That is why, properties like radioactivity, fission, fusion exist in this region only.

The mid nuclei are stable because their $\overline{B.E}$ is high.

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TAG PROBLEMS

P1. Find the radius of Aluminum nucleus $_{13}Al^{27}$. (Given that $R_0 = 1.2 \times 10^{-15}$m).

A: Given constant $R_0 = 1.2 \times 10^{-15}$m

From $_{13}Al^{27}$, we have mass number of Aluminium is A=27

Radius of Aluminium is $R = R_0A^{1/3} = 1.2 \times 10^{-15}(27)^{1/3} = 3.6 \times 10^{-15}$ m

---

P2. If one microgram of $_{92}U^{235}$ is completely destroyed in an atom bomb, how much energy will be released?

A: Mass of uranium destroyed, m =1 microgram = 10^{-6}g=10^{-9}kg, c=3×10^{8}ms^{-1}, E = ?

Formula: Energy $E = mc^2 = 10^{-9}\times(3 \times 10^{8})^2 = 9 \times 10^{-9+8+8}$ $\therefore E = 9 \times 10^{7}$ J

P3. 200MeV energy is released when one nucleus of $_{92}U^{235}$ undergoes fission. Find the number of fissions per second required for producing a power of 1 megaw att. **[BMP]**

A: Required power to be produced P = 1 mega watt = 10^6 J/s

Energy released for per fission in E= 200 meV = $200 \times 10^6 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-11}$J

We know power $P = \frac{NE}{t} \Rightarrow \frac{P}{E} = \frac{N}{t}$

Number of fissions per second $= \frac{P}{E} = \frac{10^6}{3.2\times10^{-11}} = 3.125\times10^{16}$

☛ Few more **Tag Problems** suggested by your **Faculty.**

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When Reading becomes Boring, BBQ makes it Caring!

AMAZING MIDDLE SECTION

SAQ

SECTION-C

When Exam comes near, BBQ removes Fear!

SAQ CHAPTERS

32 MARKS

★★★	1.	ELECTRIC CHARGES AND FIELDS (4 Marks)	32 - 34
★★★	2.	ELECTROSTATIC POTENTIAL (4+4 Marks)	35- 39
★★★	3.	CURRENT ELECTRICITY (4 Marks)	40 - 42
★★★	4.	MAGNETISM AND MATTER (4 Marks)	43 - 44
★★½	5.	ELECTROMAGNETIC INDUCTION (4 Marks)	45 - 46
★★	6.	ALTERNATING CURRENT (4 Marks)	47 - 48
★★½	7.	WAVE OPTICS (4 Marks)	49 - 50
★★★	8.	DUAL NATURE (4+4 Marks)	51- 54
★★★	9.	ATOMS (4 Marks)	55 - 57
★★½	10.	SEMICONDUCTOR ELECTRONICS (4 Marks)	58 - 60

Super Scoring Pages

★★★SSP★★★

Page 32, 33, 35, 36,37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55,59

DEAR DUDE!

Please note that SAQs in each chapter are Jumbled, and

Easy SAQs are brought to the beginning.

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ONE MARK QUESTIONS

OMQ

SECTION-A

Be Bold with BBQ!

When Memory becomes Weak, BBQ makes it Peak!

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TARGET MARKS:≥5

OMQ CHAPTERS

9 MARKS

1.	ELECTRIC CHARGES & FIELDS	82
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4.	ELECTROMAGNETIC INDUCTION	85
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6.	ELECTROMAGNETIC WAVES	87
7.	WAVE OPTICS	88
8.	ATOMS	89
9.	SEMICONDUCTOR ELECTRONICS	90

☛ Being Bullet Q Book, to make it short only 5 selective OMQ[5 MCQ, 5 FIB, 5 OWQ] are taken.

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OMQ – Hints & Solutions

1. ELECTRIC CHARGES & FIELDS

I. MULTIPLE CHOICE Q'S(MCQ)

1. By quantization, charge q of a body is given by (where n is integer) [4]
   1) $q=n/2e$ 2) $q=n(e/3)$ 3) $q=n(2e/3)$ ✓4) $q=ne$
2. The value of permittivity of free space ε_0 is: [3]
   1) $9\times10^9\ N\,m^2/C^2$ 2) $8.85\times10^{-12}\frac{Nm^2}{C^2}$ ✓3) $8.85\times10^{-12}\frac{C^2}{Nm^2}$ 4) $9\times10^9\frac{C^2}{Nm^2}$
3. Two charges +2C and +6C are repelling each other with a force of 12N. If each charge is added with -2C, then the value of the force will be [4]
   1) 4 (Attractive) 2) 4 (Repulsive) 3) 8 (Repulsive) ✓4) Zero
4. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests **[BMP]** [3]
   1) $E_A > E_B > E_C$ 2) $E_A = E_B = E_C$ ✓3) $E_A = E_C > E_B$ 4) $E_A = E_C < E_B$
5. The angle between the electric dipole moment $\bar{p}$ and the direction of electric field strength due to it on the equatorial line is: [3]
   1) 0° 2) 90° ✓3) 180° 4) 60°

II. FILL IN THE BLANKS (FIB)

6. A simple apparatus to detect charge on a body is the **Gold – Leaf electroscope**.
7. A soap bubble is given a negative charge then its radius **increases**.
8. The value of basic unit of charge is **1.6×10^{-19} C**.
9. The SI unit of electric field is **N/C (or) Volt/m**.
10. SI unit of volume charge density is **C/m^3**.

III. ONE WORD ANSWER Q'S (OWQ)

11. What is the nature of force between two like charges?
    A: The nature of the force between two like charges is **repulsive.**
12. How many number of electrons are in 1 coulomb charge?
    A: Given that $q = 1C$, $e = 1.6\times10^{-19}$ C, $n = ?$
    Formula: $q = ne \Rightarrow n = \frac{q}{e} \Rightarrow n = \frac{1}{1.6\times10^{-19}} = 0.625\times10^{19}$ **electrons**
13. What is the direction of electric field due to a single positive charge?
    A: The direction of electric field due to a single positive charge is **radially outward in all directions.**
14. What is the SI unit of electric flux?
    A: SI unit of electric flux is the **Volt-meter (V-m).** It is also equal to Nm^2/C.
15. What is the SI unit of electric dipole moment?
    A: SI unit of dipole moment is **coulomb -meter (C-m)**

2. CURRENT ELECTRICITY

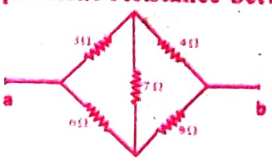
I. MCQ

1. **Which of the following material has the largest resistivity?** [3]
   1) Silicon 2) Germanium ✓3) **Glass** 4) Silver

2. **An electric bulb is rated 60 W, 220 V. The resistance of the filament is close to** [1]
   ✓**1) 807 Ω** 2) 870 Ω 3) 708 Ω 4) 780 Ω

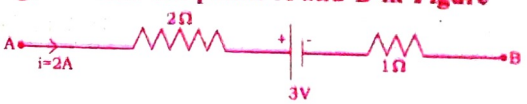
3. **The internal resistance of a 2.1 V cell which gives a current of 0.2 A through the resistance of 10 Ω is** [4]
   1) 0.8 Ω 2) 1.0 Ω 3) 0.2 Ω ✓**4) 0.5 Ω**

4. **A bridge circuit is shown in figure . The equivalent resistance between points a and b is** [4]



   1) 21 Ω 2) 7 Ω
   3) $\frac{252}{85}\Omega$ ✓**4)** $\frac{14}{3}\Omega$

5. **The potential difference ($V_A - V_B$) between the points A and B in Figure** [4]
   1) −3V 2) +3V
   3) +6V ✓**4) +9V**



II. FIB

6. The electric current in an electrolytic solution is due to the flow of **ions**.
7. The SI unit of resistivity is **Ωm** .
8. The SI unit of current density is **A/m^2** .
9. The reciprocal of resistivity is **conductivity**.
10. In an open circuit, the potential difference between the ends of a battery is equal to its **electromotive force (emf)**. **[BMP]**

III. OWQ

11. **What is the SI unit of electric current?**
    A: SI unit of electric current is the **ampere (A).**

12. **What is the SI unit of Resistance?**
    A: SI unit of electric resistance is the **ohm (Ω).**

13. **What is conductivity?**
    A: **Conductivity** is the ability of a material **to allow the flow of electric charge.**

14. **Give the relation between emf and potential difference in open circuit.**
    A: In an open circuit, **EMF is equal to the potential difference** across the terminals (E = V).

15. **What are the dimensions of Resistivity?**
    A: The dimensional formula of resistivity is $[ML^3T^{-3}A^{-2}]$

3. MAGNETISM & MATTER

I. MCQ

1. The magnetic field lines of the bar magnet looks similar to the magnetic field lines of _____ [3]

1) a current carrying circular coil 2) a straight long current carrying wire
✓3) **a long solenoid** 4) a plane current sheet

2. The universal character of every material is **[BMP]** [1]

✓1) **diamagnetic** 2) paramagnetic 3) ferromagnetic 4) ferrimagnetic

3. Which of the following materials exhibits paramagnetic property? [3]

1) Iron 2) Cobalt ✓3) **Aluminium** 4) Nickel

4. A freely suspended magnet align in _____ direction. [3]

1) South-West 2) East-West ✓3) **North-South** 4) North-East

5. Which one among the following materials exhibits higher magnetic susceptibility? [1]

✓1) **Ferromagnetic** 2) Paramagnetic
3) Diamagnetic 4) Both Paramagnetic and Diamagnetic

II. FIB

6. The SI unit of magnetisation (M) is **ampere/metre**.
7. The SI unit of magnetic field intensity (H) is **ampere/metre**.
8. The magnetic material whose susceptibility is positive but small is **paramagnetic**.
9. Domains are the characteristic feature of **ferro** magnetic materials
10. The net magnetic flux passing through any closed surface around bar magnet is **zero**.

III. OWQ

11. What is the magnetic field induction at a distance of r from the center of the bar magnet of magnetic moment m on its bisector line (equatorial line)?

A: Along the equatorial line, for short magnet $\mathbf{B_{eq}} = \frac{\mu_0}{4\pi}\frac{\mathbf{M}}{r^3}$. M = magnetic moment, r = distance

---

12. What is the analogy to permittivity ϵ_0 of free space in magnetic field?

A: **Permeability of free space (μ_0).** [$\mu_0 = 4\pi \times 10^{-7}$ H/m.]

---

13. Write a relation between susceptibility and relative permeability.

A: $\mu_r = 1 + \chi$ is a relation between relative permeability (μ_r) and magnetic susceptibility (χ).

---

14. Which law confirms that no mono magnetic poles exist?

A: **Gauss's law for magnetism** is the law that confirms that no mono magnetic poles exist.

---

15. Write down the relation between magnetic induction (B) and magnetic Intensity (H).

A: $B = \mu H$ (or) $B = \mu_0\mu_r H$
This is the relation between magnetic induction (B) and magnetic field intensity (H).

4. ELECTRO MAGENTIC INDUCTION

I. MCQ

1. **Lenz's law is a consequence of which principle?** [1]
   ✓**1) Conservation of energy** 2) Newton's first law
   3) Coulomb's law 4) Ohm's law
2. **The SI unit of self inductance is:** [4]
   1) Tesla 2) Weber 3) Ampere ✓**4) Henry**
3. **Mutual induction between two coils does not depend on:** [4]
   1) number of turns in both coils 2) cross sectional area of coils
   3) magnetic permeability of medium between them ✓**4) current flowing through the coils**
4. **A change in magnetic flux of 0.3 Wb in 0.1 s produces an average emf of:** [2]
   1) 0.03 V ✓**2) 3 V** 3) 30 V 4) 300 V
5. **If the flux linkage in a coil changes from 2 Wb to 8 Wb in 0.5 s, the average emf induced is:** **[BMP]** [2]
   1) 10 V ✓**2) 12 V** 3) 6 V 4) 4 V

II. FIB

6. According to Faraday's law, the induced emf in a coil is directly proportional to the **rate of change** of the magnetic flux.
7. The SI unit of magnetic flux is **Weber**.
8. The back emf in an inductor is always in the direction opposite to the **original emf**.
9. The flux linkage through a coil of N-turns is proportional to the **current** passing through the coil.
10. The direction of induced current is determined by **Lenz** law.

III. OWQ

11. **Which law states that the induced current opposes the change in flux that produces it?**
A: **Lenz's law**
It states that the direction of an induced current is always such as to oppose the change that causes it.

---

12. **Give the expression for the emf induced between the ends of a metal conductor moving perpendicular to a uniform magnetic field.**
A: **$E = Blv$.** Here l is the length, v is the velocity, B is the uniform magnetic field

---

13. **What is the working principle of a transformer?**
A: Transformer works on the principle of **mutual inductance.**

---

14. **Why are resistance coils double wounded in opposite direction?**
A: To **minimize/ avoid self inductance.**

---

15. **What is the unit of mutual inductance?**
A: SI unit of mutual inductance is **Henry** (H)

5. ALTERNATING CURRENT

I. MCQ

1. **The frequency of AC mains in India is:** [2]
   1) 25 Hz ✓**2) 50 Hz** 3) 100 Hz 4) 60 Hz
2. **In a series LCR circuit at resonance, the current is:** [2]
   1) minimum ✓**2) maximum** 3) zero 4) infinite
3. **The reactance of a capacitor decreases with:** [2]
   1) decrease in capacitance ✓**2) increase in frequency**
   3) both (1) and (2) 4) decrease in frequency
4. **The power factor of a series LCR circuit at resonance is:** [3]
   1) 0 2) 0.5 ✓**3) 1** 4) −1
5. **A transformer is designed to:** [1]
   ✓**1) increase or decrease AC voltage** 2) convert AC into DC
   3) increase the frequency of AC 4) store electric charge

II. FIB

6. The r.m.s. value of an AC of peak value I_0 is $\underline{I_0 / \sqrt{2}}$.
7. The phase difference between current and voltage in a purely resistive AC circuit is $\underline{0°}$.
8. In an LCR circuit at resonance, the impedance is equal to **resistance (R)**.
9. In a transformer, the core is laminated to reduce **eddy current** losses.
10. The power factor of an AC circuit is given by $\underline{\cos\phi}$.

III. OWQ

11. What is the r.m.s value of an AC whose peak value is 141 V?

A: $V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{141}{1.414} = 99.7V = \mathbf{100V}$ [∵peak value $V_0 = 141V$]

---

12. What is the time period of AC mains in India? **[BMP]**

A: The time period (T) of AC mains in India is **0.02 seconds.**

∴Time period $T = \frac{1}{\text{frequency}} = \frac{1}{50} = 0.02s = 20$ milli seconds [∵standard AC frequency in India is 50Hz.]

---

13. What is the condition for resonance in a series LCR circuit?

A: Condition for resonance: $X_L = X_C$ i.e., Inductive reactance = Capacitive reactance.

---

14. Which device is used to reduce AC without much loss of energy?

A: **Step down transformer** is the device used to reduce AC without much loss of energy.

---

15. Write the expression for power factor in an AC circuit.

A: Power factor $\cos\phi = \frac{R}{Z}$; R= Resistance, Z= Impedance, ϕ is phase angle.

6. ELECTRO MAGNETIC WAVES

I. MCQ

1. **Which colour of the light has the longest wavelength?** [3]
   1) Green 2) Blue ✓**3) Red** 4) Violet
2. **If the frequency of an electromagnetic wave is 60 MHz and it travels in free space, the wavelength of the wave (m) is:** [2]
   1) 2.5 ✓**2) 5** 3) 10 4) 2
3. **Green House effect is due to** [4]
   1) absorption of Ultra-Violet radiation by Ozone layer
   2) transmission of visible light by the atmosphere
   3) good weather
   ✓**4) reflection of infrared rays emitted by earth's atmosphere**
4. **Ozone Layer absorbs** [2]
   1) Visible light ✓**2) UV radiation** 3) Infrared radiation 4) Microwaves
5. **Which of the following waves are also called the "Heat waves"?** [3]
   1) Visible light 2) X-rays ✓**3) Infrared rays** 4) Radio waves

II.FIB

6. The spectrum of electromagnetic waves stretching in order of increasing wavelength from **Gamma** rays to long **Radio** waves.
7. In the charging capacitor both conduction current and **displacement** current may be present.
8. Faraday's law of induction states that there is an induced emf equal to the rate of change of **magnetic flux**.
9. **Accelerated** charges radiate electromagnetic waves. **[BMP]**
10. Jagadish Chandra Bose succeeded in producing electromagnetic waves of **shorter** wavelength.

III. OWQ

11. **In cellular phones which band of radio waves are used to transmit voice communication?**
A: Cellular phones use UHF band **(Ultra High Frequency)** .
The frequency range is 300 MHz to 3 GHz.

---

12. **Earth surface absorbs the incoming visible radiation from the sun and what type of waves it re-radiate?**
A: Earth re-radiates **long infrared waves.**

---

13. **What type of rays can snakes detect in addition to visible rays?**
A: Snakes can detect **infrared rays in** addition to visible rays.

---

14. **When high energy electrons bombard a metal target, what type of waves are produced?**
A: **X-rays** are produced when high energy electrons bombard a metal target.

---

15. **Which type of electromagnetic waves are used in the treatment of cancer?**
A: **Gamma rays** and **X-rays** are used in radiation therapy to destroy cancer cells.

7. WAVE OPTICS

I. MCQ

1. When three lights of red, violet and green coloured are used successively in young double slit experiment, the fringe width will be maximum for ..... [1]

✓1) Red 2) Violet 3) Green 4) Same for all

2. The phase difference between two points 'A' and 'B' on a wave front separated by distance λ is [2]

1) π/4 ✓2) 2π 3) π 4) π/2

3. The penetration of light into the region of geometrical shadow is called [3]

1) polarization 2) interference ✓3) diffraction 4) Refraction

4. Which one of the following phenomenon confirms that transverse nature of light? [4]

1) Interference 2) Diffraction 3) Dispersion ✓4) Polarization

5. Which of the following waves cannot be polarized? [3]

1) X- rays 2) Radio waves ✓3) Sound waves 4) Light waves

II. FIB

6. A wave front is the continuous locus of all particles vibrating with **same phase**.
7. The secondary wave fronts emanating from a primary wave front are also called as **wavelets**.
8. In interference pattern, all bright and dark fringes are of **same** width.
9. For constructive interference, the path difference between two waves is equal to **nλ** .
10. Due to **diffraction** phenomenon of light, colours on CD are formed.

III.OWQ

11. State Huygen's principle.

A: Every point on a wave front acts as a source of secondary spherical wavelets.

---

12. Which principle is used to find the position of a wave front after some time? **[BMP]**

A: **Huygen's principle** is used to find the position of a wave front.

---

13. What is interference of light?

A: Interference is the phenomenon where two or more light waves overlap to form a resultant wave of greater, lower or the same amplitude.

---

14. What is the main condition to produce interference of light?

A: The primary condition is that the **light sources must be coherent.**

---

15. What do you mean by the polarisation of light?

A: Polarization is the process of restricting the vibrations of light to a single plane perpendicular to the direction of propagation.

8. ATOMS

I. MCQ

1. **1eV is equivalent to** [3]
   1) 6.67×10^{-34} J 2) 1.6×10^{-27} J ✓3) 1.6×10^{-19} J 4) 6.24×10^{-22}J

2. **The total energy of an electron in the stationary states of the hydrogen atom is** [2]
   1) $E_n = -\frac{13.6}{n}$eV ✓2) $E_n = -\frac{13.6}{n^2}$eV 3) $E_n = \frac{13.6}{n}$eV 4) $E_n = \frac{13.6}{n^2}$eV

3. **Bohr's atomic model is not valid for** [4]
   1) hydrogen atom 2) singly ionized helium atom (He^+)
   3) deuterium atom ✓4) **singly ionized neon atom (Ne^+)**

4. **The angular momentum of an electron revolving in n^{th} orbit is given by** [1]
   ✓1) **$mvr = nh/2\pi$** 2) $mvr = n^2h/2\pi$ 3) $mvr = h\pi n^3$ 4) $mvr = n^2h^2/2\pi$

5. **According to Bohr atomic model, in which of the following transition will the frequency be maximum?** [2]
   1) n = 4 to n = 3 ✓2) **n = 2 to n = 1** 3) n = 5 to n = 4 4) n = 3 to n = 2

II. FIB

6. A He^+ ion is in its first excited state, its ionization energy is **13.6 eV**
7. The change in angular momentum of an electron when it jumps from 4^{th} orbit to 5^{th} orbit in hydrogen atom is **$h/2\pi$**.
8. The process of raising or transferring the electron from lower energy state to higher energy state is called **excitation**. **[BMP]**
9. The process of raising the atom from the normal state to the ionized state is called **ionization**.
10. 1 fermi is equal to **10^{-15}** meter.

III. OWQ

11. **What kind of charge is spread throughout the atom in Thomson's model?**
    A: In Thomson's model, **positive charge** is spread throughout the atom.

---

12. **What is the SI unit of the impact parameter?**
    A: The SI unit of impact parameter is **metre (m).**

---

13. **The energy of a hydrogen atom in the ground state is –13.6eV. What will be the energy of He^+ ion in the first excited state?**
    A: For He^+ ion we have Z=2 and for the first excited state n=2
    Energy $E_n = \frac{-13.6Z^2}{n^2} \Rightarrow E_2 = \frac{-13.6(2)^2}{(2)^2} = -13.6$eV.

---

14. **An electron in a hydrogen atom revolving in its second excited state having the radius of 4.65Å. What is its de-Broglie wavelength?**
    A: According to Bohr's postulate, $2\pi r = n\lambda \Rightarrow \lambda = \frac{2\pi r}{n} = \frac{2\times3.14\times4.65}{3} = 9.734$Å

---

15. **What is the path of an electron around the nucleus as per Rutherford's atomic model?**
    A: According to Rutherford's atomic model, the path of an electron around the nucleus is a **circular orbit.**

9. SEMI CONDUCTORS

I. MCQ

1. **Which of the following is used as pentavalent impurities for doping process?** [1]
   ✓**1) Arsenic** 2) Boron 3) Indium 4) Aluminium

2. **The energy band gaps of silicon and germanium are** [3]
   1) 0.7eV, 1.1eV 2) 1eV, 0.7eV ✓**3) 1.1eV,0.7eV** 4) 2eV, 1.5eV

3. **In case of full wave rectification, if the input frequency is 60 Hz, then the output frequency would be** [2]
   1) 60 Hz ✓**2) 120 Hz** 3) 100 Hz 4) 150 Hz

4. **For a p-n junction diode, the current in reverse bias is in the order of** [3]
   1) Few mA 2) Few amperes ✓**3) Few μA** 4) Few kilo amperes

5. **The increase in the width of the depletion region in a p-n junction diode is due to** [1]
   ✓**1) reverse bias only** 2) both forward bias and reverse bias
   3) high doping concentrations 4) forward bias only

II. FIB

6. In a vacuum tube, the electrons are supplied by a **cathode**.
7. In vacuum tubes, electrons flow from the cathode to the anode in **only one** direction.
8. The gap between the top of the valence band and bottom of the conduction band is called the **energy band gap/forbidden energy gap**. **[BMP]**
9. The pentavalent dopant donates one extra electron for conduction and hence this impurity is called **donor**.
10. In the p-n junction diode symbol, the direction of arrow indicates **direction of conventional current**.

III. OWQ

11. **At what temperature the intrinsic semiconductor will behave like an insulator?**

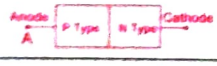
A: **At absolute zero temperature (0 K)** intrinsic semi conductor behaves like an insulator

---

12. **Draw the circuit symbol of the p-n junction diode.**

A: **Circuit symbol of the p-n junction diode:**

p-n junction diode:



---

13. **What is the forbidden energy gap in Ge semiconductor crystal in terms of Joules?**

A: **Forbidden energy gap in Ge** at room temperature $Eg=0.72\times1.6\times10^{-19}$ J $=\mathbf{1.15\times10^{-19}}$ **J**

---

14. **What is the name of the current due to motion of charge carriers when external electric field is applied.**

A: The current produced by the motion of charge carries under the influence of an external electric field is known as **drift current.**

---

15. **What is the name of the process "adding impurities to the intrinsic semiconductor to make extrinsic semiconductor"?**

A: **Doping** is the process of adding impurities to an intrinsic semiconductor to make extrinsic semi conductor.

SHINING STARS
STAR QUESTIONS PLUS

ELECTRIC CHARGES & FIELDS
VSAQ

182. What happens to the force between two charges if the distance between them is a) halved b) doubled

A : From Coulomb's law, we have $F \propto \frac{1}{r^2} \Rightarrow \frac{F_2}{F_1} = \frac{r_1^2}{r_2^2}$

a) When distance is halved $r_2 = \frac{r_1}{2}$; New force $F_2 = ?$

$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} = \frac{r_1^2}{(r_1/2)^2} = \frac{r_1^2}{r_1^2/4} = 4 \Rightarrow F_2 = 4F_1$. Thus, force **increases to 4 times.**

b) When distance is doubled $r_2 = 2r_1$, $F_2 = ?$

$\frac{F_2}{F_1} = \frac{r_1^2}{r_2^2} = \frac{r_1^2}{(2r_1)^2} = \frac{r_1^2}{4r_1^2} = \frac{1}{4} \Rightarrow F_2 = F_1/4$. Thus, the force **decreases to 1/4.**

---

183. When will be the electric flux negative and when is it positive?

A : **Electric flux is negative** due to a **negative charge** and **positive** due to a **positive charge.**

Reason: Electric flux $\Delta\phi = \vec{E} \cdot \Delta\vec{S} = E\Delta S\cos\theta$

Electric flux $\Delta\phi$ is negative when cos θ is negative i.e,, when θ belongs to Q_2 or Q_3.

Electric flux $\Delta\phi$ is positive when cos θ is positive i.e,, when θ belongs to Q_1 or Q_4.

---

184. Write the expression for electric intensity due to a charged conducting spherical shell at points outside and inside the shell. Explain the terms in it.

A : 1) Electric intensity outside a charged conducting spherical shell is $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

Here q is the total charge on the spherical shell, r is the distance of the point outside the shell and ε_0 is permittivity of free space.

2) Electric intensity inside the shell is zero. i.e., E = 0

---

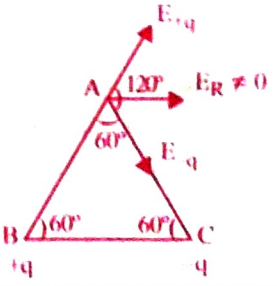
185. Consider two charges +q and -q placed at B and C of an equilateral triangle ABC. For this system, the total charge is zero. But the electric field (intensity) at A which is equidistant from B and C is not zero. Why ?

A : At point A, the electric field vectors due to +q and −q are not in opposite direction. (actually 120°).

So they do not cancel each other.

Hence the resultant vector is non-zero.

So, intensity of electric field is not zero at A.



ELECTROSTATIC POTENTIAL

186. What will be the shapes of equipotential surfaces in case of a uniform electric field and a point charge?

Sol: In a uniform electric field, the equipotential surfaces take the shape of **equidistant parallel planes** perpendicular to direction of the electric field lines.

At a point charge, equipotential surfaces are **concentric spheres** centered on the point charge.

---

187. The dielectric strength of air is $3\times10^6 Vm^{-1}$ at certain pressure. A parallel plate capacitor with air in between the plates has a plate separated by 1 cm. Can you charge the capacitor to 3×10^6 V . Give reason.

A : No.

Since air is the dielectric, the **maximum possible** potential of the capacitor is V=Ed

$$V= 3\times10^6 \times10^{-2}=3\times10^4 \text{ V}$$

But required potential is 3×10^6 V. Required potential > Possible potential .

So it is not possible for the capacitor to charge potential 3×10^6 V .

CURRENT ELECTRICITY

188. If a wire is stretched to double its original length without the loss of mass, how will the resistivity of the wire be influenced?

A : **1) Resistivity** is the property of material of a wire and it does not change with the change in dimensions (length) of the wire.

2) When a wire is stretched to double its length, its resistivity does not change, but its **resistance becomes 4 times**. Because for a given mass, volume is constant . Here $R \propto l^2$

---

189. The electron drift speed in metals is small (~mm/s) and the charge of the electron is also very small ($\sim10^{-19}$C), but we can still obtain a large amount of current in a metal. Why?

A : Current in a conductor is $I = nAev_d$

Though the values of e and v_d are very small, the value of number of free electrons per unit volume n is very very large($\sim10^{28}$) for metals. Hence large currents are possible in metals.

MOVING CHARGES & MAGNETISM

190. Express a relation among the quantities the speed of light in vacuum, the permeability of vacuum and the permittivity of vacuum.

A : The speed of light in vacuum $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Here μ_0 is permeability and ε_0 is the permittivity of vaccum.

---

191. What is the magnitude of magnetic force on a charged particle of strength q moving through a uniform magnetic field B at a velocity v which makes an angle θ with magnetic field? When does this force become maximum?

A : 1) The magnitude of magnetic force (F) on a charge (q), moving with velocity (v) at an angle θ to a magnetic field (B) is given by **F = Bqv sinθ.**

2) This force becomes maximum when θ=90°. Then sinθ = sin 90° = 1. ∴ F_{max} = Bqv

---

192. What is the magnitude of magnetic force on a current (i) carrying wire of length 'l' placed in a uniform magnetic field B if it makes an angle θ with magnetic field? When does this force become maximum?

A : 1) The magnitude of magnetic force (F) on a current (i) carrying wire of length (*l*) placed in a magnetic field (B) which makes an angle (θ), is given by **F = Bi*l* sinθ**

2) This force becomes maximum when θ=90°. Then sinθ = sin 90° = 1. ∴ F_{max} =Bi*l*

---

193. What is the work done by the magnetic field on moving charge? Give reason.

A: 1) The work done by a magnetic field on a moving charge is **zero.**

2) **Reason:** Magnetic field $\vec{F}$ is always perpendicular to the velocity $\vec{v}$ of the charge.

Work done $W = \int \vec{F}.\vec{ds} = \int \vec{F}.\vec{v}\, dt$ (dot product becomes zero)

---

194. Write expression for the magnetic field induction at a point on the axis of a circular current carrying coil. Obtain an expression for the magnetic field induction at its center.

A 1) The magnetic induction on the axis of a circular coil at a distance x from its centre is

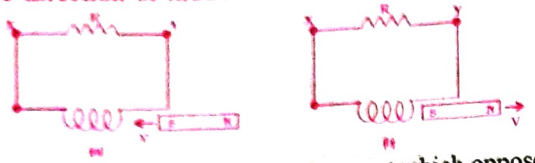
$B_x = \frac{\mu_0 NIR^2}{2(x^2 + R^2)^{3/2}}$ Here N is number of the turns, I is the current passing through the coil, R is radius of the coil, μ_0 is permeability of free space.

2) Magnetic Induction at the centre of the circular coil:

Putting x = 0 in the above equation, we get magnetic induction the at centre of the coil $B_0 = \frac{\mu_0 NI}{2R}$

ELECTRO MAGNETIC INDUCTION

195. (a) & (b). State the direction of induced current in the resistor 'R'.



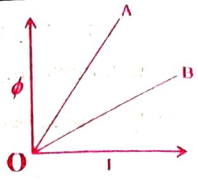
A: According to Lenz's law, the direction of induced current which opposes the cause of it.

a) If the magnet moves **towards** the coil, **flux increases.**
Hence the induced current in the resistor R moves in anti-clockwise direction from Y to X.

b) If the magnet moves **away** from the coil, **flux decreases.**
Hence the induced current in the resistor R moves in clockwise direction from X to Y.

---

196. A plot of magnetic flux (ϕ) versus current (I) is shown in figure for two inductors A and B. Which of the two has larger value of self induction. State the reason.



A: 1) In the given graph, slope of the line represents self inductance (L).

$$\text{Slope} = \frac{\phi}{I} = L$$

2) In the graph, slope of line OA is greater than the slope of the line OB.

Hence, the self inductance of A is greater than the self inductance of B. Thus $L_A > L_B$

ELECTRO MAGNETIC WAVES

197. If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon?

A: Energy of photon $E = \frac{hc}{\lambda} \Rightarrow E\alpha\frac{1}{\lambda}$ Let $\lambda_2 = 2\lambda_1$

$\therefore \frac{E_2}{E_1} = \frac{\lambda_1}{\lambda_2} = \frac{\lambda_1}{2\lambda_1} = \frac{1}{2} \Rightarrow E_2 = \frac{E_1}{2}$. Thus, energy of **photon becomes half.**

---

198. What is the ratio of speed of infrared rays and ultraviolet rays in vacuum?

A: Ratio of speed of infrared rays and ultraviolet rays in vacuum is **1:1.**
Because they both travel with same speed of light in vacuum

---

199. What is the relation between the amplitudes of the electric and magnetic fields in free space for an electromagnetic wave?

A: Velocity of light $c = \frac{E_0}{B_0}$

Here E_0 is amplitude of electric field, B_0 is amplitude of magnetic field in free space.

---

200. The energy of the EM waves is of the order of 15 KeV. To which part of the spectrum does it belong?

A: The range of x-ray part in EM waves is 0.1 keV to 100 keV. So,15 keV belongs to **x-rays.**

201. A plane electromagnetic wave of frequency 20 MHz travels in free space along the + X direction. At a particular point in space and time, the electric field vector of the wave is E_y = 9.3 V/m. Then find the magnetic field vector of the wave at that point?

A: Given Electric field $E_y = 9.3 V/m$ in the direction of y-axis.

Velocity of light $c = 3 \times 10^8$ m/s. Waves travel along +ve x-axis

Magnetic field $\vec{B} = \frac{E_y}{c} = \frac{9.3}{3\times10^8} = 3.1\times10^{-8} T = 3.1\times10^{-8}\hat{k}T$

$\hat{k}$ refers to positive z-axis as $\vec{E}, \vec{B}$ and propagation of EM waves are mutually perpendicular.

---

202. A beam of light travelling along X-axis is described by the electric field $E_y = 900 \sin\omega\left(t - \frac{x}{c}\right)$, then find the ratio of electric force to magnetic force on a charge moving along Y-axis with a speed of 3×10^8 m/s. (Given speed of light is 3×10^8 m/s).

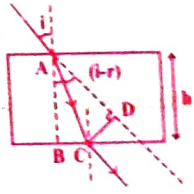
A: Force due to electric field $F_E = E_0 q$; Force due to magnetic field $F_B = B_0 q v$

$$\therefore \frac{F_E}{F_B} = \frac{E_0 \cancel{q}}{B_0 \cancel{q} v} = \frac{E_0}{B_0 v} = \frac{c\cancel{B_0}}{\cancel{B_0} v} = \frac{c}{v} = \frac{3\times10^8}{3\times10^8} = \frac{1}{1} = 1:1 \qquad (\because E_0 = cB_0)$$

RAY OPTICS

203. Discuss the lateral shift of a ray refracted through a parallel side glass slab with neat ray diagram.

A: **Lateral shift** is the perpendicular distance between the incident ray's original path and the emergent ray after passing through a parallel sided glass slab. In the diagram the **lateral shift is CD.**

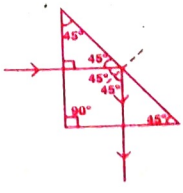


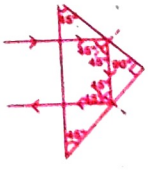
Lateral shift of a ray

---

204. Draw the ray diagrams of crown glass prism to bend rays by 90° and 180° to form an inverted image with the use of total internal reflection.

A:

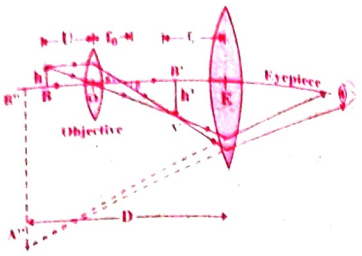




205. Draw a labelled ray diagram to show the formation of an image by a compound microscope. Write the expression for its magnifying power.

A: **Formula:** At **near point**, magnifying power

$$m = m_0 m_e = \left[\frac{v_0}{u_0}\right]\left[1 + \frac{D}{f_e}\right]$$



At **infinity**, magnifying power $m = m_0 m_e = \left[\frac{v_0}{u_0}\right]\left[\frac{D}{f_e}\right]$

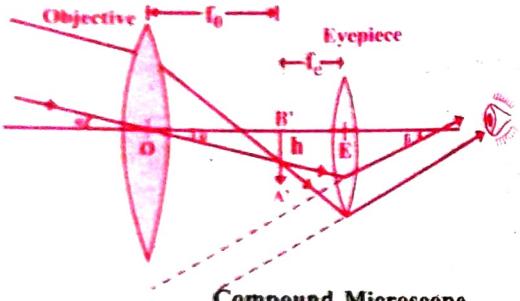
m_0 = Magnification of the object; m_e = Magnification of the eye piece, f_e = Focal length of eye piece

D = least distance of distinct of vision; u_0 = object distance; v_0 = image distance

---

206. Draw a ray diagram of an astronomical telescope (refracting telescope) in normal adjustment position (image formed at infinity). State any one drawback of this type of telescope.

A:



Compound Microscope

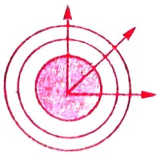
Drawback: Astronomical telescopes suffer from chromatic aberrations, blurred edges.

WAVE OPTICS

207. What is the wave front. Draw the geometrical shape of the wavefront produced by light diverging from a point source.

A: 1) A **wave front** is the locus of all points in a medium that vibrate in the **same phase.**

2) For light diverging from a point source, the wave front is **spherical.**



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208. If light bend around obstacles, then why can't we see around a building?

A: 1) Light bends around obstacles due to diffraction, but diffraction is appreciable only when obstacle size is comparable to wavelength.

2) Since wavelength of light is very small, bending is negligible around buildings.

209. Explain the statement *'light added to light can produce darkness'*

A: 1) This statement describes the phenomenon of **destructive interference**.

When two coherent light waves travel in the same direction and meet at a point out of phase, the crest of one wave falls on the trough of the other.

2) Then their amplitudes cancel out, resulting in a region of zero intensity (darkness).

---

210. Two independent light sources cannot act as coherent sources. Then how Young performed the interference pattern on the screen?

A: Young achieved a coherent interference pattern by using a single light source to illuminate two narrow slits. This ensured that the two secondary sources were perfectly synchronized in phase.

---

211. Diffraction is common in sound, but not common in light waves. Why?

A: 1) Sound waves have large wavelengths, so they diffract easily around obstacles.

2) Light waves have very small wavelengths, so diffraction is very small.

ATOMS

212. What happens when an electron jumps from higher energy level to lower energy level according to Bohr's model?

A: 1) When an electron jumps from higher energy level to lower energy level, it emits a photon of electromagnetic radiation.

2) The energy of the photon is equal to the difference between the higher energy level and the lower energy level is given by $\Delta E = h\nu$.

---

213. What is the physical meaning of negative energy of an electron?

A : 1) The physical meaning of the negative energy of an electron in an orbit means that the electron is bound to the positive nucleus due to attractive force.

2) The energy of electron in first orbit of hydrogen is –13.6 eV. It means that the energy required to take the electron out of the hydrogen atom from first orbit is 13.6 eV.

---

214. Name the physical quantity whose dimensions are the same as those of angular momentum?

A: Planck's constant (h) has same dimensions as **angular momentum.**

$$E = h\nu \Rightarrow h = \frac{E}{\nu} = \frac{ML^2T^{-2}}{T^{-1}} = ML^2T^{-1}$$

NUCLEI

215. A fusion reaction releases energy of 1.6×10^{-12}J. Find the equivalent mass defect in kg.

A: Given $E=1.6\times10^{-12}$ J. We know $c=3\times10^{8}$ m/s . We have $E=mc^2$

$$\therefore \text{Mass defect } \Delta m=\frac{E}{c^2}=\frac{1.6\times10^{-12}}{(3\times10^{8})^2}=1.78\times10^{-29}\text{kg}$$

---

216. The radius of germanium (Ge) nuclide is measured to be twice the radius of $_4Be^9$. The number of nucleons in Ge are

A: Nuclear radius $R=R_0A^{1/3}\Rightarrow R\propto A^{1/3}$, Given $R_{Ge}=2R_{Be}$ and also $A_{Be}=9$

$$\text{Now } \frac{^{R}G_e}{^{R}B_e}=\left[\frac{^{A}G_e}{^{A}B_e}\right]^{1/3}\Rightarrow\frac{2\cancel{R_{Be}}}{\cancel{R_{Be}}}=\left(\frac{A_{Ge}}{9}\right)^{1/3}\Rightarrow(2)^3=\left(\frac{A_{Ge}}{9}\right)\Rightarrow 8=\frac{^{A}G_e}{9}$$

$$\therefore {}^{A}G_e=8\times9=72$$

---

217. An unstable heavy nucleus at rest breaks into two nuclei move away with velocities in the ratio of 8:27. What is the ratio of radii of the nuclei (assumed to be spherical)?

A: From the law of conservation of momentum $m_1v_1=m_2v_2\Rightarrow\frac{m_1}{m_2}=\frac{v_2}{v_1}$.....(1)

Mass(m) is proportional to mass number (A) $\therefore\frac{m_1}{m_2}=\frac{A_1}{A_2}$.....(2)

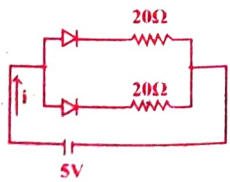
From (1) and (2); $\frac{A_1}{A_2}=\frac{v_2}{v_1}=\frac{27}{8}$

Nuclear radius $R\propto A^{1/3}\Rightarrow\frac{R_1}{R_2}=\left[\frac{A_1}{A_2}\right]^{1/3}=\left[\frac{27}{8}\right]^{1/3}=\left[\frac{3^3}{2^3}\right]^{1/3}=\frac{3}{2}$

$\therefore$ Ratio of radii of nuclei is 3 : 2.

SEMI CONDUCTORS

218. What is the current through the battery in the circuit as shown in figure, here diodes are ideal?



A: Both the diodes are in forward bias and both are ideal.

R_1 and R_2 are parallel. $(\because R_1=20\Omega;\ R_2=20\Omega)$

$$\therefore R_{eq}=\frac{R_1R_2}{R_1+R_2}=\frac{20\times20}{20+20}=\frac{400}{40}=10 \qquad \text{Hence, } R_{eq}=10\Omega$$

$$\therefore \text{Current } I=\frac{V}{R}=\frac{5}{10}=\frac{1}{2}=0.5A$$

SAQ

ELECTROSTATIC POTENTIAL

219. Define electric field intensity and electric potential and then derive the relation between them.

Sol: **Electric field intensity (E)** at a point in an electric field is defined as the force experienced by unit positive test charge (q_0) kept at that point. **Formula:** $E=\frac{F}{q_0}$

Electric potential (V) is defined as the work done per unit positive charge in bringing it from infinity to that point against the electric field.

Relation between E and V:

Consider two points separated by a distance 'd' in a uniform electric field of intensity E.

Let V be the potential difference between these points. The amount of work done in moving a charge q is W=Vq.....(1) The force acting on the charge q is F=Eq.

Work done W =Fd=(Eq)d.....(2) From (1) & (2), Vq=Eqd

$\therefore V=Ed \Rightarrow E=\frac{V}{d}$. This is the required relation between E and V.

CURRENT ELECTRICITY

220. Define electric resistance and write its SI unit. How does the resistance of a conductor vary if a)conductor is stretched to 4 times of its length?

b) temperature of the conductor is increased?

A: **Electric Resistance** of a conductor is the ratio of the potential difference (V) across the ends of a conductor to the electric current (i) passing through it.

Formula: $R=\frac{V}{i}$. **SI unit:** ohm (Ω).

Also resistance of a conductor is given by $R=\rho\frac{l}{A}$

where ρ is resistivity of material, l is length and A is area of cross section of the conductor.

a) Conductor stretched 4 times of its Length : When a conductor is stretched, its length increases but its area of cross section decreases so that its volume (Al) remains constant.

$R=\rho\frac{l}{A}=\rho\frac{l^2}{Al}$ When ρ and Al are constant then $R \propto l^2$

Thus if length l stretched 4 times, its resistance R becomes 4^2 = **16 times.**

b) When temperature of a conductor is increased, its resistance R increases.

Resistivity of a conductor $\rho=\frac{m}{ne^2\tau}$. Here m is mass of electron, n is number of electrons per unit volume, e is charge of electron and τ is average time of collisions.

When temperature is increased, the number of collisions increases and the average time of collisions τ decreases. The term in the denominator decreases.

Thus, its resistivity and hence **resistance increases.**

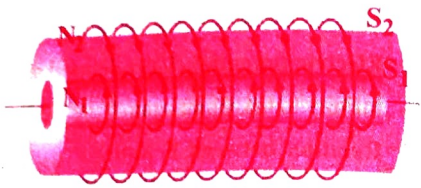
ELECTROMAGNETIC INDUCTION

221. Obtain an expression for the mutual inductance of two long co-axial solenoids.

A: 1) **Mutual Inductance** is the phenomenon where a changing current in one coil induces an EMF in a nearby secondary coil.

Mutual induced emf $\varepsilon = -M\frac{di}{dt}$, here M is mutual inductance of the coils.

2) Expression for mutual induction:



Consider two co-axial solenoids S_1, S_2 each of length l. Let A be the common cross-section area of solenoids.

Let N_1 and N_2 be the total number of turns in the solenoids S_1 and S_2.

3) When current I_1 is passing through solenoid S_1, Magnetic field produced is $B_1 = \mu_0 \frac{N_1}{l} I_1$ .....(1)

4) Magnetic flux linked with each turn of solenoid S_2 is $B_1A = \left(\mu_0 \frac{N_1}{l} I_1\right)A$.

5) $\therefore$ Total magnetic flux linked with N_2 turns of solenoid S_2 is, $\phi_2 = (B_1A)N_2$

$\Rightarrow \phi_2 = \left(\mu_0 \frac{N_1}{l} I_1 A\right)N_2 = \frac{\mu_0 N_1 N_2 I_1 A}{l}$ .....(2)

6) Now total magnetic flux linked with N_2 turns of solenoid S_2 is $\phi_2 = M_{12}I_1$ .....(3) where M_{12} is mutual inductance of S_1 with respect to S_2

7) From eq (2) and eq (3) we get, $M_{12}I_1 = \frac{\mu_0 N_1 N_2 I_1 A}{l} \Rightarrow M_{12} = \frac{\mu_0 N_1 N_2 A}{l}$ .....(4)

8) Let n_1, n_2 be the number of turns per unit length of the solenoids S_1, S_2

$\therefore M_{12} = \mu_0 n_1 n_2 lA$.....(5) $\left(\because n_1 = \frac{N_1}{l}, n_2 = \frac{N_2}{l}\right)$

9) Similarly, mutual inductance of solenoid S_2 with respect to S_1 is $M_{21} = \mu_0 n_1 n_2 lA$ .....(6)

Comparing eq (5) and eq (6), we get $M_{21} = M_{12}$

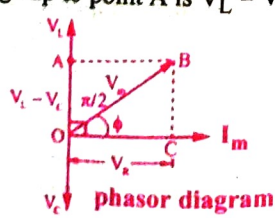
10) $\therefore$ Mutual inductance of two long coaxial solenoids, $M = \mu_0 n_1 n_2 lA$

ALTERNATING CURRENT

222. Derive the expression for the impedance of a series LCR circuit by using phasor diagrams technique.

A: The **Phasor diagram** for series LCR circuit is as shown in figure. Here, voltage V_L is in upward direction and voltage V_C is in downward direction. So, net voltage up to point A is $V_L - V_C$.

Maximum voltage is denoted by V_m.



From the phasor diagram, $OB = \sqrt{(OC)^2 + (CB)^2}$

$\therefore V_m = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(I_m R)^2 + (I_m X_L - I_m X_C)^2}$

$$\Rightarrow V_m = I_m\sqrt{R^2+(X_L-X_C)^2}$$

$$\frac{V_m}{I_m} = Z = \sqrt{R^2+(X_L-X_C)^2} \text{ for } X_L > X_C$$

This is the expression for impedance of series LCR circuit.

223. Define resonance in a series LCR circuit. Derive the condition for resonance and explain what happens to current and impedance at resonance.

A: IN RESONANCE LCR CIRCUIT :

An LCR circuit that allows the maximum current at a specific frequency of AC is known as a **series resonance (LCR) circuit.** Consider an LCR circuit of given frequency n.

At resonance, capacitive reactance (X_C)is equal to the inductive reactance (X_L).

Thus at resonance $X_L = X_C$

Impedance $Z = \sqrt{R^2+(X_L-X_C)^2} = \sqrt{R^2+0} = R \Rightarrow Z = R$

Again, at resonance $X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

We know $\omega = 2\pi n \Rightarrow 2\pi n = \frac{1}{\sqrt{LC}} \Rightarrow n = \frac{1}{2\pi\sqrt{LC}}$ Here n is resonant frequency.

Current and Impedance at Resonance: i) We know $I_0 = \frac{E_0}{R}$

$\therefore$ At resonance if R is minimum then current in the circuit becomes maximum.

ii) At resonance, $X_L = X_C \Rightarrow Z = R$.

Hence if the impedance of the circuit is minimum then resistance also becomes minimum.

WAVE OPTICS

224. Compare and contrast the double slit interference pattern with single slit diffraction pattern.

A:

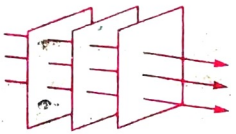
Feature	Double slit Interference	Single slit Diffraction
source	Produced by the superposition of two separate wave fronts from two coherent sources.	Produced by the superposition of secondary wavelets originating from different parts of the same wave front.
Fringe width	All fringes are generally of equal width.	The central bright fringe is twice as wide as the secondary maxima.
Intensity	All bright fringes have the same intensity.	The intensity decreased rapidly as we move away from the central maximum.
Constant	There is usually a good constant	There is poor constant dark fringes are not perfectly dark.

225. Explain any two types of wave fronts. Draw the ray diagram to show the behavior of plane wave fronts as they pass through a) a thin prism b) a thin convex lens.

A: A wave front is defined as the locus of all points is a medium that are overlapping in the same phase. There are two types of wave fronts.

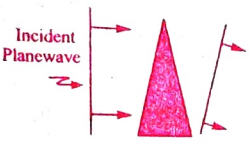
i) **Spherical wave front:** It is produced by a point source of light. As the light travels outward in all directions, the surface of constant phase is a sphere.

ii) **Plane wave front:** It is produced by a source at an infinite distance or a point source placed at the focus of a convex lens. The rays are traveled and the wave front is a plane.

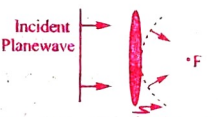


Behaviour of plane wave front:

a) **Through a thin prism:** When a plane wave front passes through a prism, the lower part of the wave front travels through more glass than the upper part. Since light travels slower in glass, the lower part is delayed, causting the emerging wave front to be tilted.



b) **Through a thin convex lens:** A plane wave front incident on a convex lens is refracted such that the central part of the wave front travels through the thickest part of the lens and is delayed the most. This causes the emerging wave front to become spherical and converge towards the focal point.



Spherical wavefront of radius f

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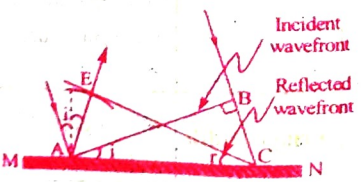
226. State Huygen's principle. Deduce the law of reflection on the basis of Huygen's wave theory.

A : 1) **Huygens' Principle:** Each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave. These wavelets are called secondary wavelets and a common tangent drawn to these wavelets gives the position of wavefront at a later time.

2) **Deduction of law of reflection:** We consider a plane wave AB incident at an angle on a reflecting surface MN.



3) If 'υ' represents the speed of the wave in the medium and if 't' represents the time taken by the wave front to advance from the point B to C. Then the distance BC = υt.

4) In order to construct the reflected wave front we draw a sphere of radius υt from the point A. Let CE represent the tangent plane drawn from point C to this sphere. Then, AE = BC = υt. The triangle EAC and BAC are congruent, the angle i and r would be equal. This is the law of reflection

DUAL NATURE OF RADIATION

227. Explain why macroscopic objects do not show significant wave-like properties when compared to sub-atomic particles.

A: Macroscopic objects do not show significant wave properties, because their mass is incredibly large making their associated de Broglie wavelength is very small.

De Broglie wavelength $\lambda = \frac{h}{P}$

Every moving particle has a wave nature, with a wave length λ calculated by the equation

$\lambda = \frac{h}{mv}$

Here h is planck's constant, m is mass of the particle, v is the velocity of the object.

Macroscopic objects fail to show wave properties because, the numerator (h) is small, the macroscopic mass (m) is larger than the wavelength close to zero.

SEMI CONDUCTOR ELECTRONICS

228. Draw and explain the current-voltage (I -V) characteristic curves of a junction diode in forward and reverse bias.

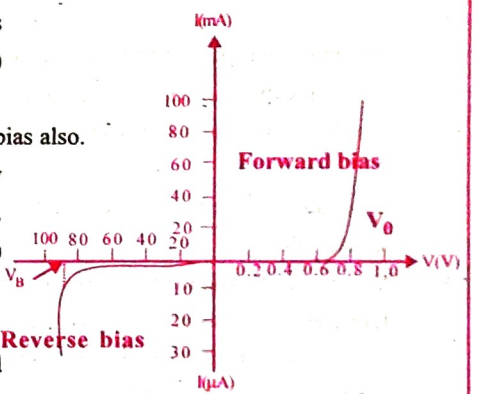
A: **Voltage - Current Characteristics of Diode:**

A diode is connected in forward bias and readings are noted with applied potential differences (PDs) and obtained currents.

The same process should be repeated in reverse bias also.

With the measured values a graph is drawn by taking **voltage** on **x-axis** and **current** on **y-axis.**

The graph obtained is called characteristic (I-V) curve of the diode.



Forward bias - Threshold voltage (V_0)

Threshold voltage is the voltage in a forward biased p-n junction at which the current suddenly rises to a large values.

The curve shows that the diode conducts electric current during the forward bias but can not conduct during the reverse bias. Due to this property, diode is used as a rectifier.

Reverse bias - Breakdown voltage (V_B)

Breakdown voltage: It is the voltage in a reverse biased p-n junction at which the current suddenly rises to a larger values.

CHAPTERS: 1,2,3,4

Q 1 UNIT TEST-1

Time : 1 Hour

SR. PHYSICS

Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. By quantization, charge q of a body is given by (where n is integer) [☞P 82(1)]
   1) q=n/2e 2)q=n(e/3) 3)q=n(2e/3) 4) q=ne
2. The SI unit of electric field is _____ [☞P 82(9)]
3. What is the SI unit of electric current? [☞P 83(11)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. What is the importance of Gauss law in electrostatics. [☞P 62(81)]
5. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel. [☞P 63(88)]
   a) What is the ratio of charges on each capacitor?
   b) What is the ratio of potential differences across each capacitor?
6. State Ohm's Law and write its mathematical form. [☞P 65(94)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. State and explain Coulomb's inverse square law in electricity. [☞P 32(14)]
8. Derive an expression for the capacitance of a parallel plate capacitor. [☞P 36(23)]
9. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. [☞P 40(32)]

(or)

10. Derive an equation for the couple acting on a electric dipole in a uniform electric field. [☞P 33(16)]
11. Derive an expression for the electric potential due to a point charge. [☞P 35(20)]
12. Define electric resistance and write its SI unit. How does the resistance of a conductor vary if a)conductor is stretched to 4 times of its length? b) temperature of the conductor is increased? [☞P 99(220)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? [☞P 16(1)]
14. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. [☞P 17(2)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 5,6,7,8,9

Q 2 UNIT TEST-2

Time : 1 Hour

SR. PHYSICS

Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. The universal character of every material is
   1) diamagnetic 2) paramagnetic
   3) ferromagnetic 4) ferrimagnetic [☞P 84(2)]
2. The direction of induced current is determined by _____ law. [☞P 85(10)]
3. What is the time period of AC mains in India? [☞P 88(12)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. Magnetic lines form continuous closed loops. Why? [☞P 68(110)]
5. State Faraday's law of electromagnetic induction. [☞P 69(116)]
6. What is transformer ratio? [☞P 71(125)]

SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Explain the properties of paramagnetic materials. [☞P 43(39)]
8. Explain briefly working of AC generator. [☞P 46(45)]
9. State the principle on which a transformer works. [☞P 47(47)]
   Describe the working of a transformer with necessary theory.

(or)

10. Obtain an expression for the mutual inductance of two long co-axial solenoids. [☞P 99(220)]
11. How the existence of domains, explain the behavior of ferromagnetic material? [☞P 44(40)]
12. Obtain an expression for the current through an inductor when an AC emf is applied. [☞P 48(49)]

SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram. [☞P 21(5)]
14. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. [☞P 23(7)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

CHAPTERS: 10,11,12,13,14

Q 3 UNIT TEST-3

Time : 1 Hour **SR. PHYSICS** Max.Marks : 25

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. $3 \times 1 = 3$

1. Which of the following waves cannot be polarized? [☞P 88(5)]
   1) X- rays 2) Radio waves 3) Sound waves 4) Light waves
2. The process of raising or transferring the electron from lower energy state to higher energy state is called _____ [☞P 89(8)]
3. The gap between the top of the valence band and bottom of the conduction band is called the _____ [☞P 90(8)]

SECTION-B

II. Answer ALL the following questions $3 \times 2 = 6$

4. State Malus' law. Write the expression of Malus' law. [☞P 75(148)]
5. What is photoelectric effect? [☞P 76(155)]
6. Define impact parameter. [☞P 77(162)]

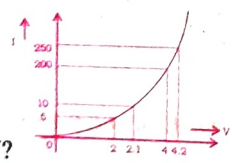
SECTION-C

III. Answer ANY TWO of the following questions. $2 \times 4 = 8$

7. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. [☞P 49(51)]
8. State Bohr's postulates of hydrogen atom. [☞P 55(65)]
9. What are different types of physical processes to obtain free electrons from the metal surface. [☞P 51(55)]

(or)

10. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [☞P 49(52)]
11. What are the laws of photoelectric effect. [☞P 52(57)]
12. The I-V characteristics of a p-n junction diode in forward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V? [☞P 60(74)]



SECTION-D

IV. Answer ANY ONE of the following questions. $1 \times 8 = 8$

13. Explain nuclear fission and fusion with suitable nuclear reactions. [☞P 27(10)]
14. Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance? [☞P 29(13)]

☞ **'Extra choice' is given in Section-C just to cover all the IMP questions**

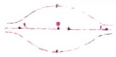
Q4 BOARD MODEL PAPER

Time : 3 Hours **SR. PHYSICS** Max.Marks : 85

SECTION-A

Ans-Page Index

I. Answer ALL the following questions. $9 \times 1 = 9$

1. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests 

   1) $E_A > E_B > E_C$ 2) $E_A = E_B = E_C$ 3) $E_A = E_C > E_B$ 4) $E_A = E_C < E_B$ [☞P 82(4)]
2. The universal character of every material is [☞P 84(2)]
   1) diamagnetic 2) paramagnetic 3) ferromagnetic 4) ferrimagnetic
3. If the flux linkage in a coil changes from 2 Wb to 8 Wb in 0.5 s, the average emf induced is: [☞P 85(5)]
   1) 10 V 2) 12 V 3) 6 V 4) 4 V
4. In an open circuit, the potential difference between the ends of a battery is equal to its _____ [☞P 83(10)]
5. What is the time period of AC mains in India? [☞P 86(12)]
6. _____ charges radiate electromagnetic waves. [☞P 87(9)]
7. Which principle is used to find the position of a wave front after some time? [☞P 88(12)]
8. The process of raising or transferring the electron from lower energy state to higher energy state is called_____ [☞P 89(8)]
9. The gap between the top of the valence band and bottom of the conduction band is called the_____ [☞P 90(8)]

SECTION-B

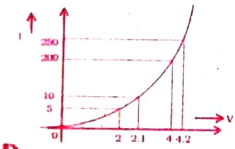
II. Answer ALL the following questions $14 \times 2 = 28$

10. The electric lines of force do not intersect. Why ? [☞P 62(78)]
11. In a hydrogen atom, the electron and proton are at a distance of 1Å . Find the dipole moment of the system. [☞P 64(92)]
12. Write the vector form of Ohms law. Explain the terms in it. [☞P 65(95)]
13. Write the formula for the magnetic field due to infinite straight wire and explain the terms in it. [☞P 67(107)]
14. Define magnetic susceptibility and relative permeability. [☞P 68(113)]
15. State Faraday's law of electromagnetic induction. [☞P 69(116)]
16. A Capacitor has a capacitance of $\frac{1}{\pi}\mu F$. Find its reactance for a frequency of 50 Hz. [☞P 71(128)]
17. Give two uses of infrared rays. [☞P 73(136)]
18. What is the critical angle? Give one application of total internal reflection? [☞P 74(145)]
19. State Malus' law. Write the expression of Malus' law. [☞P 75(148)]
20. Write down Einstein's photoelectric equation. Explain the terms in it? [☞P 76(158)]
21. Define impact parameter. [☞P 77(162)]
22. What are isotopes and isobars? [☞P 78(168)]
23. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? [☞P 79(176)]

SECTION-C

III. Answer ANY EIGHT of the following questions. 8 × 4 = 32

24. Derive an expression for the intensity of the electric field at a point on the axial line of a dipole. |☞P 33(17)|
25. Derive an expression for the capacitance of a parallel plate capacitor. |☞P 36(23)|
26. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel. |☞P 39(28)|
    a) What is the total capacitance of the combination?
    b) Determine the charge on each capacitor if the combination is connected to a 100V supply.
27. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. |☞P 40(32)|
28. Explain the properties of paramagnetic materials. |☞P 43(39)|
29. Explain briefly working of AC generator. |☞P 46(45)|
30. State the principle on which a transformer works. Describe the working of a transformer with necessary theory. |☞P 47(47)|
31. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. |☞P 49(52)|
32. What are different types of physical processes to obtain free electrons from the metal surface. |☞P 51(55)|
33. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? |☞P 54(64)|
34. State Bohr's postulates of hydrogen atom. |☞P 55(65)|
35. The I-V characteristics of a p-n junction diode in for ward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V? |☞P 60(74)|



SECTION-D

IV. Answer ANY TWO of the following questions. 2 × 8 = 16

36. (a) Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? |☞P 16(1)|
    (b)A current of 5A flows through each of two parallel long wires, the wires are 2.5 cms apart. |☞P 20(P1)|
    Calculate the force acting per unit length of each wire.
37. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. |☞P 23(7)|
38. (a) Explain nuclear fission and fusion with suitable nuclear reactions. |☞P 27(10)|
    (b) 200MeV energy is released when one nucleus of $_{92}U^{235}$ undergoes fission. Find the number of fissions per second required for producing a power of 1 megawatt. |☞P 30(P3)|

Ans-Page Index

GUESS PAPERS

Q 5 GUESS PAPER -1

SR. PHYSICS

Time : 3 Hours Max.Marks : 85

Ans-Page Index

SECTION-A

I. Answer ALL the following questions. 9 × 1 = 9

1. The value of permittivity of free space ε_0 is: [☞P 82(2)]

   1) $9\times10^{9}\ N\,m^2/C^2$ 2) $8.85\times10^{-12}\ \frac{Nm^2}{C^2}$

   3) $8.85\times10^{-12}\ \frac{C^2}{Nm^2}$ 4) $9\times10^{9}\ \frac{C^2}{Nm^2}$

2. The SI unit of magnetic field intensity (H) is _____ [☞P 84(7)]
3. What is the unit of mutual inductance? [☞P 85(15)]
4. Which of the following material has the largest resistivity? [☞P 83(1)]
   1) Silicon 2) Germanium 3) Glass 4) Silver
5. According to Faraday's law, the induced emf in a coil is directly proportional to the _____ of the magnetic flux. [☞P 85(6)]
6. Which type of electromagnetic waves are used in the treatment of cancer? [☞P 87(15)]
7. The penetration of light into the region of geometrical shadow is called [☞P 88(3)]
   1) polarization 2) interference 3) diffraction 4) Refraction
8. The process of raising or transferring the electron from lower energy state to higher energy state is called _____. [☞P 89(9)]
9. In a vacuum tube, the electrons are supplied by a _____ [☞P 90(6)]

SECTION-B

II. Answer ALL the following questions 14 × 2 = 28

10. The electric lines of force do not intersect. Why ? [☞P 62(80)]
11. What happens to the capacitance of a parallel plate capacitor if the distance between plates is doubled? [☞P 64(90)]
12. State Ohm's Law and write its mathematical form. [☞P 65(94)]
13. What is the importance of Oersted's experiment? [☞P 67(103)]
14. Magnetic lines of force don't intersect. Why? [☞P 68(109)]
15. State Lenz's law. [☞P 69(117)]
16. What is transformer ratio? [☞P 71(125)]
17. What is the wavelength range of X-rays? [☞P 73(141)]

18. Define power of a convex lens. What is its unit? |☞P 74(143)|
19. Compare Maxwell's Electromagnetic theory with Huygen's theory of light? |☞P 75(152)|
20. What is photoelectric effect? |☞P 76(155)|
21. Define impact parameter. |☞P 77(162)|
22. What are isotopes and isobars? |☞P 78(168)|
23. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased? |☞P 80(180)|

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Derive an equation for the couple acting on an electric dipole in a uniform electric field. |☞P 33(16)||
25. Derive an expression for the electric potential due to a point charge. |☞P 35(20)|
26. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel. |☞P 37(25)|
27. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. |☞P 40(32)|
28. Explain the properties of diamagnetic materials. |☞P 43(38)|
29. Explain the Faradays law of induction and Lenz's law. |☞P 45(43)|
30. Obtain an expression for the current in a capacitor when an AC emf is applied. |☞P 48(50)|
31. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. |☞P 49(51)|
32. Explain graphically that the stopping potential is independent of intensity of incident light. |☞P 52(58)|
33. Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential. |☞P 51(56)|
34. Write a short note on debroglie's explanation of Bohr's second postulate of quantization? |☞P 56(68)|
35. Describe how a semiconductor diode is used as a half wave rectifier. |☞P 59(72)|

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. |☞P 17(2)|
37. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram. |☞P 21(5)|
38. a) What is the source of stellar energy? Explain it with the help of proton-proton cycle. |☞P 27(11)|
    b) Briefly explain the formation of red giant stage of sun.

Q 6

GUESS PAPERS

GUESS PAPER -2

Time : 3 Hours

SR. PHYSICS

Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. What is the SI unit of electric dipole moment? [☛P 82(14)]
2. A freely suspended magnet align in _____ direction. [☛P 84(4)]
   1) South-West 2) East-West 3) North-South 4) North-East
3. The direction of induced current is determined by _____law. [☛P 85(10)]
4. What is the SI unit of electric current? [☛P 83(11)]
5. In a series LCR circuit at resonance, the current is: [☛P 86(2)]
   1) minimum 2) maximum 3) zero 4) infinite
6. The spectrum of electromagnetic waves stretching in order of increasing wavelength from ____ rays to long ____waves. [☛P 87(6)]
7. Due to _____phenomenon of light, colours on CD are formed. [☛P 88(10)]
8. What is the SI unit of the impact parameter? [☛P 89(12)]
9. The energy band gaps of silicon and germanium are [☛P 90(2)]
   1) 0.7eV, 1.1eV 2) 1eV, 0.7eV 3) 1.1eV,0.7eV 4) 2eV, 1.5eV

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What is the importance of Gauss law in electrostatics. [☛P 62(81)]
11. Can there be electric potential at a point with zero electric intensity? Give an example. [☛P 63(84)]
12. Define temperature coefficient of resistance. [☛P 65(97)]
13. How do you convert a moving coil galvanometer into an ammeter? [☛P 67(105)]
14. Magnetic lines form continuous closed loops. Why? [☛P 68(110)]
15. Define magnetic flux. [☛P 69(118)]
16. A transformer converts 200 V AC into 2000 V AC. Calculate the number of turns in the secondary if the primary has 10 turns. [☛P 71(126)]
17. What are the applications of microwaves? [☛P 73(137)]
18. What do you understand by the focal plane? Explain with a neat ray diagram. [☛P 74(142)]
19. Two identical coherent waves, each o f intensity I_0, are producing an interference pattern. Write the value of resultant intensity at a point of (1) constructive interference (2) destructive interference. [☛P 75(150)]

Ans-Page Index

20. What are cathode rays? [☛P 76(153)]
21. Define distance of closest approach. [☛P 77(163)]
22. What are isotones and isomers? [☛P 78(169)]
23. What are intrinsic and extrinsic semiconductors ? [☛P 79(173)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. State and explain Coulomb's inverse square law in electricity. [☛P 32(14)]
25. Derive an expression for the capacitance of a parallel plate capacitor. [☛P 36(23)]
26. Derive an expression for the potential energy of an electric dipole placed in a uniform electric field. [☛P 36(22)]
27. State and explain Kirchhoff rules for an electric network. [☛P 41(34)]
28. Mention the properties of magnetic field lines. [☛P 43(37)]
29. Obtain an expression for the mutual inductance of two long co-axial solenoids. [☛P 100(221)]
30. Obtain an expression for the current through an inductor when an AC emf is applied. [☛P 48(49)]
31. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [☛P 49(52)]
32. What are the laws of photoelectric effect. [☛P 52(57)]
33. Write Einstein's photo-electric equation. [☛P 53(59)]
    How Millikan proved the validity of Einstein's photoelectric equation.
34. What are the limitations of Bohr's theory of hydrogen atom? [☛P 55(66)]
35. What is rectification? Explain the working of a full wave rectifier. [☛P 59(73)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram. [☛P 18(3)]
37. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index. [☛P 22(6)]
    b) Explain the working of optical fiber with a neat diagram.
38. Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance? [☛P 29(13)]

GUESS PAPERS

Q 7 GUESS PAPER -3

Time : 3 Hours

SR. PHYSICS

Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. 9 × 1 = 9

1. SI unit of volume charge density is _____
2. Write a relation between susceptibility and relative permeability.
3. The SI unit of self inductance is:
   1) Tesla 2) Weber 3) Ampere 4) Henry
4. What are the dimensions of Resistivity?
5. The r.m.s. value of an AC of peak value I_0 is _____
6. Which colour of the light has the longest wavelength?
   1) Green 2) Blue 3) Red 4) Violet
7. State Huygen's principle.
8. 1eV is equivalent to
   1) 6.67×10^{-34} J 2) 1.6×10^{-27} J 3) 1.6×10^{-19} J 4) 6.24×10^{-22}J
9. The pentavalent dopant donates one extra electron for conduction and hence this impurity is called _____

SECTION-B

II. Answer ALL the following questions 14 × 2 = 28

10. What is meant by the statement charge is quantised ?
11. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel.
    a) What is the ratio of charges on each capacitor?
    b) What is the ratio of potential differences across each capacitor?
12. Define resistivity or specific resistance.
13. How do you convert a moving coil galvanometer into a voltmeter?
14. What is the magnetic moment associated with a solenoid?
15. What is self-induction?
16. Define power factor. On which factors does the power factor depend?
17. What is the principle of production of electromagnetic waves?
18. A small angled prism of 4° deviates a ray through 2.4°. Find the refractive index of the prism.
19. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves?

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[☞P 75(149)]

20. Write down de-Broglie's relation and explain the terms therein. [☞P 76(159)]
21. What is the angular momentum of an electron in the second orbit of Bohr's model of hydrogen atom? [☞P 77(166)]
22. Compare the radii of the nuclei of mass numbers 27 and 64. [☞P 78(171)]
23. What is an n-type semiconductor? [☞P 79(174)]
What are the majority and minority charge carriers in it?

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. State Gauss's law in electrostatics and prove it. [☞P 34(19)]
25. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel. [☞P 39(28)]
a) What is the total capacitance of the combination?
b) Determine the charge on each capacitor if the combination is connected to a 100V
26. Explain the behaviour of dielectrics in an external field. [☞P 38(27)]
27. Derive equivalent emf and internal resistance of two cells in series combination. [☞P 41(33)]
28. How the existence of domains, explain the behavior of ferromagnetic material? [☞P 44(40)]
29. Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field. [☞P 45(44)]
30. State the principle on which a transformer works. [☞P 47(47)]
Describe the working of a transformer with necessary theory.
31. Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids. [☞P 50(54)]
32. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? [☞P 54(64)]
33. The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. [☞P 54(61)]
34. State Bohr's postulates of hydrogen atom. [☞P 55(65)]
35. What are n-type and p-type semiconductors? [☞P 58(71)]
Explain the process of formation of p-n junction diode.

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams [☞P 19(4)]
37 With a neat labelled diagram explain the formation of image in a simple microscope. Derive the expression for magnification when the image is formed at i) near point ii) at infinity. [☞P 25(9)]
38. a) Explain the composition of nucleus and express the atomic masses of nucleons in terms of atomic mass units. [☞P 28(12)]
b) Draw and explain the potential energy curve of pair of nucleons. Show that the density of nucleus doesn't depend on mass number A.

GUESS PAPERS

Q 8 GUESS PAPER -4

Time : 3 Hours

SR. PHYSICS

Max.Marks : 85

SECTION-A

Ans-Page Index

I. Answer ALL the following questions. $9 \times 1 = 9$

1. The angle between the electric dipole moment $\vec{p}$ and the direction of electric field strength due to it on the equatorial line is:
   1) 0° 2) 90° 3) 180° 4) 60° [☞P 82(5)]
2. The magnetic material whose susceptibility is positive but small is _____ [☞P 84(8)]
3. What is the working principle of a transformer? [☞P 85(13)]
4. The SI unit of resistivity is _____ [☞P 83(7)]
5. What is the condition for resonance in a series LCR circuit? [☞P 86(13)]
6. Which of the following waves are also called the "Heat waves"? [☞P 87(5)]
   1) Visible light 2) X-rays 3) Infrared rays 4) Radio waves
7. What is the main condition to produce interference of light? [☞P 88(14)]
8. 1 fermi is equal to _____ meter. [☞P 8910)]
9. For a p-n junction diode, the current in reverse bias is in the order of [☞P 90(4)]
   1) Few mA 2) Few amperes 3) Few μA 4) Few kilo amperes

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. Write the expression for electric intensity due to an infinitely long straight uniformly charged wire, at a radial distance r from the wire. Explain the terms in it. [☞P 91(184)]
11. Three capacitors of capacitances 1 μF, 2μF and 3 μF are connected in series. [☞P 64(89)]
    a) What is the ratio of charges? b)What is the ratio of potential differences?
12. Define mobility of charge carriers. [☞P 66(101)]
13. What is the principle of moving coil galvanometer? [☞P 67(104)]
14. Classify the following materials with regard to magnetism. [☞P 68(115)]
    Iron, Platinum, Copper, Magnesium.
15. Write the expression for self-induced emf and explain the terms therein. [☞P 69(120)]
16. Write the expression for the reactance of (i) an inductor and (ii) a capacitor. [☞P 71(127)]
17. The charging current for a capacitor is 0.6 A. [☞P 73(140)]
    What is the displacement current across its plates?
18. State the conditions under which total internal reflection takes place. [☞P 74(146)]
19. If a wave undergoes reflection and refraction, what will be the phase change in each case? [☞P 75(151)]

20. Define work function. Give an expression for it. [☞P 76(156)]
21. What is the difference between α - particle and helium atom? [☞P 77(164)]
22. What is the significance of a mass defect in a nucleus? [☞P 78(172)]
23. What is a p-type semiconductor ? What are the majority and minority charge carriers in it? [☞P 79(175)]

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge. [☞P 32(15)]
25. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series. [☞P 37(24)]
26. Derive an expression for the energy stored in a capacitor. [☞P 38(26)]
27. Derive equivalent emf and internal resistance of two cells in parallel combination. [☞P 42(36)]
28. State Gauss' law in magnetism. Show that how this law proved the non-existence of magnetic monopoles. [☞P 44(41)]
29. Explain briefly working of AC generator. [☞P 46(45)]
30. What is power factor? Derive its expression in terms of resistance R and impedance Z. Explain its significance in AC circuits. [☞P 47(48)]
31. Explain single slit diffraction pattern with neat diagram. State conditions for maximum intensity and minimum intensity. [☞P 50(53)]
32. What are different types of physical processes to obtain free electrons from the metal surface. [☞P 51(55)]
33. What is de Broglie's wavelength? [☞P 53(60)]
34. Derive expressions for potential and kinetic energies of an electron in any orbit of a hydrogen atom according to Bohr's atomic model. [☞P 57(69)]
35. Describe energy band diagram of intrinsic semiconductors at 0 K. [☞P 58(70)]

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. [☞P 17(2)]
37. Derive the lens makers formula with suitable diagram and thereby derive the lens formula. [☞P 24(8)]
38. a)What is the source of stellar energy? Explain it with the help of proton-proton cycle. b)Briefly explain the formation of red giant stage of sun. [☞P 27(11)]

GUESS PAPERS

GUESS PAPER -5

SR. PHYSICS

Time : 3 Hours Max.Marks : 85

SECTION-A

I. Answer ALL the following questions. $9 \times 1 = 9$

1. How many number of electrons are in 1 coulomb charge?
2. Which of the following materials exhibits paramagnetic property?
   1) Iron 2) Cobalt 3) Aluminium 4) Nickel
3. The SI unit of magnetic flux is _____
4. The reciprocal of resistivity is _____
5. The power factor of a series LCR circuit at resonance is:
   1) 0 2) 0.5 3) 1 4) –1
6. What type of rays can snakes detect in addition to visible rays?
7. The phase difference between two points 'A' and 'B' on a wave front separated by distance λ is
   1) $\pi/4$ 2) 2π 3) π 4)$\pi/2$
8. The process of raising the atom from the normal state to the ionized state is called_____
9. Draw the circuit symbol of the p-n junction diode.

SECTION-B

II. Answer ALL the following questions $14 \times 2 = 28$

10. What happens to the force between two charges if the distance between them is a) halved b) doubled
11. What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled?
12. If a wire is stretched to double its original length without the loss of mass, how will the resistivity of the wire be influenced?
13. What is the magnitude of magnetic force on a charged particle of strength q moving through a uniform magnetic field B at a velocity v which makes an angle θ with magnetic field? When does this force become maximum?
14. Define magnetisation of a material and write its formula.
15. What is the working principle of A.C. generator?
16. What is the phase difference between A.C emf and current in the following: pure resistor, pure inductor and pure capacitor.
17. If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon?

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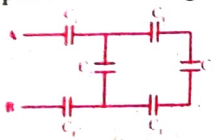
18. Two thin lens of power +4D and -2D are in contact. What is the focal length of the combination? |☞P 74(144)|
19. Diffraction is common in sound, but not common in light waves. Why? |☞P 97(211)|
20. Give examples of photosensitive substances. Why are they called so? |☞P 76(157)|
21. Name the physical quantity whose dimensions are the same as those of angular momentum? |☞P 97(214)|
22. Why are nuclear fusion reactions called thermo nuclear reactions? |☞P 78(170)|
23. What is reverse saturation current of p-n junction diode? |☞P 80(178)|

SECTION-C

III. Answer ANY EIGHT of the following questions. $8 \times 4 = 32$

24. Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole. |☞P 34(18)|
25. Derive an expression for the electrostatic potential energy of a system of two point charges and find its relation with electric potential of a charge. |☞P 35(21)|
26. If $C_1 = 3$ pF and $C_2 = 2$ pF (fig.). Calculate the equivalent capacitance of the given network between points A and B. |☞P 39(29)|



27. When the resistance connected in series with the cell is halved, the current is equal to or slightly less or slightly greater than double, Why? |☞P 41(35)|
28. Define magnetisation of a sample, magnetic induction and susceptibility. How magnetic induction is related to magnetic intensity? |☞P 44(42)|
29. Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid. |☞P 46(46)|
30. State the principle on which a transformer works. Describe the working of a transformer with necessary theory. |☞P 47(47)|
31. State Huygen's principle. Deduce the law of reflection on the basis of Huygen's wave theory. |☞P 102(226)|
32. The work function of a metal is 4.3 eV. What is the threshold wavelength? |☞P 54(62)|
33. A particle has a momentum of 1.5×10^{-27} kg m/s. What is its de-Broglie wavelength? |☞P 54(63)|
34. What are the observations of Rutherford's α - particle scattering experiment and how these findings supported the existence of atomic nucleus. |☞P 56(67)|
35. Describe how a semiconductor diode is used as a half wave rectifier. |☞P 59(72)|

SECTION-D

IV. Answer ANY TWO of the following questions. $2 \times 8 = 16$

36. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram. |☞P 18(3)|
37. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index. |☞P 22(6)|
    b) Explain the working of optical fiber with a neat diagram.
38. Explain nuclear fission and fusion with suitable nuclear reactions. |☞P 27(10)|

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5. Magnetism and Matter

6. Electromagnetic Induction

7. Alternating Current

8. Electromagnetic Waves

9. Ray Optics

10. Wave Optics

11. Dual Nature of Radiation

12. Atoms

13. Nuclei

14. Semiconductor Electronics

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A.P.J. ABDUL KALAM

Dream Beyond Your Limits

From a small town Rameswaram with limited means, he dreamed big, worked hard and became the People's President.

LESSON: Your background may be small. Your dream need not be.

YOUR TURN: Write one dream that you refuse to give up.

YUVRAJ SINGH

The Fighter Who Came Back

After winning the World Cup, he was diagnosed with cancer. He fought bravely, completed treatment and returned to cricket.

LESSON: Life can knock you down. Don't stay down.

YOUR TURN: Don't give up because of one setback.

AMITABH BACHCHAN

The Phoenix

His company failed and he faced huge losses and pressure. Instead of giving up, he started again and made a great comeback.

LESSON: A failure can be a chapter, not the whole story.

YOUR TURN: When you fail, ask – "What can I start again?"

SINDHUTAI SAPKAL

From Pain to Purpose

She faced extreme hardship in life. She turned her pain into a mission and became 'Maai', mother of thousands of orphans.

LESSON: You can turn your pain into someone else's hope.

YOUR TURN: Help one person who needs you.

MARY KOM

I Will Try Again

From a small village in Manipur, she faced many defeats, challenges and motherhood. But she kept coming back.

LESSON: A setback can stop you for a while. It need not stop you forever.

YOUR TURN: Try once more at something you recently gave up.

SUDHA MURTY

Simplicity is Strength

She was once rejected in a job interview. She didn't lose hope and went on to become the first woman engineer in that company.

LESSON: Don't accept "You can't" without asking "Why not?"

YOUR TURN: Speak respectfully when something is unfair.

MS DHONI

From Railway Job to World Champion

He worked as a ticket examiner in the railway. He kept playing cricket. Hard work changed his life.

LESSON: Your present situation does not decide your final destination.

YOUR TURN: Keep working towards your goal daily.

SRINIVASA RAMANUJAN

Talent Can Come From Anywhere

He struggled in formal education, but his love for mathematics was extraordinary. His ideas amazed the world.

LESSON: One weakness does not cancel all your strengths.

YOUR TURN: Find the subject or skill in which YOU are good.

DR. VERGHESE KURIEN

One Idea Can Change Millions

He was not from a farming family, but helped start the White Revolution (Amul) and changed millions of farmers' lives.

LESSON: Don't only think of your success, think of others too.

YOUR TURN: Do one thing that helps your whole group.

RATAN TATA

Success With Humility

He became a top industrial leader but stayed humble and used his success to help many people.

LESSON: Grow in success. Stay small in ego.

YOUR TURN: Help someone without telling everyone.

GOODNESS ACTIONS (NO MONEY NEEDED!)

- Plant a Sapling
- Make Someone Smile
- Read & Share Good Books
- Save Water
- Help Someone
- Keep It Clean
- Listen Patiently
- Encourage Others
- Give Without Expecting
- Respect Your Nation

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కొందరి కోసమే తయారైన బేబి బుల్లెట్ ఎందరికో ఎన్నో రకాలుగా...

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